

Bones, Joints, and Soft Tissue Tumors

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BONE

BASIC STRUCTURE AND FUNCTION OF BONE

The adult human skeleton is composed of 206 bones accounting for ~12% of body weight. Bone functions include mechanical support, force transmission, internal organ protection, and mineral homeostasis, and it also serves as the major site of hematopoiesis during postnatal life. Bone consists of extracellular matrix and several cell types.

Matrix

The extracellular component of bone, the matrix, is composed of **osteoid (35%) and minerals (65%)**. The latter, primarily hydroxyapatite [$\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$], gives bone its hardness and serves as a repository for 99% of the calcium and 85% of the phosphorus in the body.

Osteoid consists of type I collagen and smaller amounts of glycosaminoglycans and other proteins. One of these proteins, osteocalcin, is produced by osteoblasts and contributes to the regulation of bone formation, mineralization, and calcium homeostasis; it may also have additional metabolic effects. Serum osteocalcin levels are used as a specific marker of osteoblast activity. Bone maturation and metabolism are also sensitive to cytokines and growth factors and are thus regulated by diverse inputs, which include locally and systemically produced factors, as well as mechanical force.

Bone matrix can be *woven* or *lamellar* (Fig. 26.1). Woven bone is produced rapidly (e.g., during fetal development or fracture repair), but the haphazard arrangement of collagen fibers imparts less structural integrity than the parallel collagen fibers of lamellar bone. Woven bone is always abnormal in adults, but its presence is not specific for any particular disease. Long bones are composed of a dense outer cortex and a central medulla. The latter is supported by bony trabeculae interspersed with marrow; the latter may be fatty (white) or hematopoietic (red).

Cells

The cellular components of bone include **osteoblasts, osteocytes, and osteoclasts**.

- *Osteoblasts* on the surface of the osteoid matrix synthesize, transport, and assemble matrix and regulate mineralization (Fig. 26.2A). Osteoblast activity is tightly regulated by hormonal and local mediators. Quiescent osteoblasts, which can be recognized by their small volume of cytoplasm, may remain on the trabecular surface or become embedded within the matrix as osteocytes.

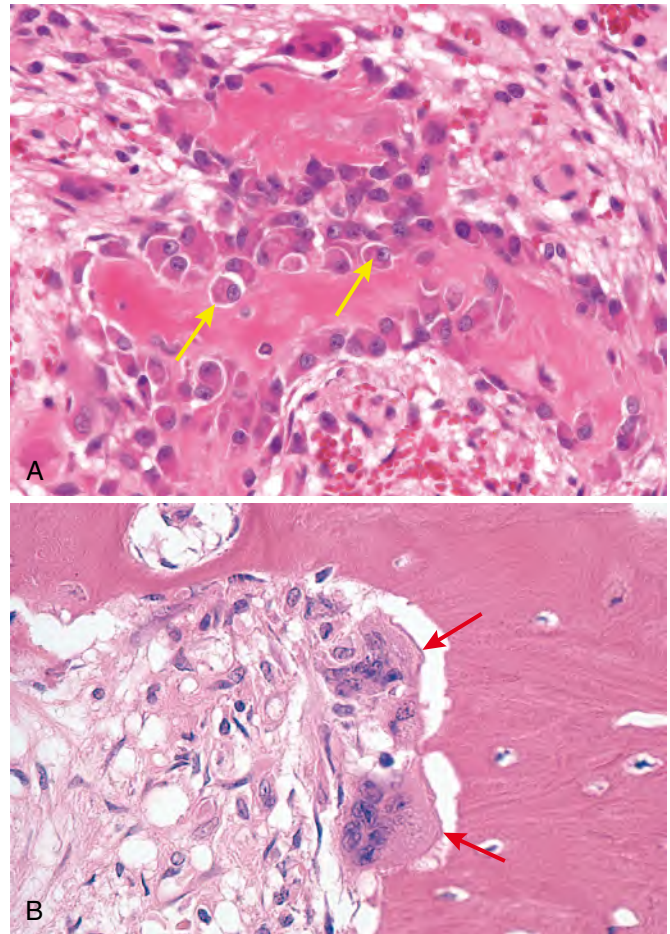


Fig. 26.2 (A) Active osteoblasts (yellow arrows) synthesizing bone matrix. The surrounding spindle cells represent osteoprogenitor cells. (B) Two osteoclasts (red arrows) resorbing bone.

- *Osteocytes* are interconnected by an intricate network of dendritic cytoplasmic processes through tunnels (canaliculi) within the matrix. Osteocytes control calcium and phosphate levels in the micro-environment, detect mechanical forces, and translate those forces into biologic activity—a process called *mechanotransduction*.

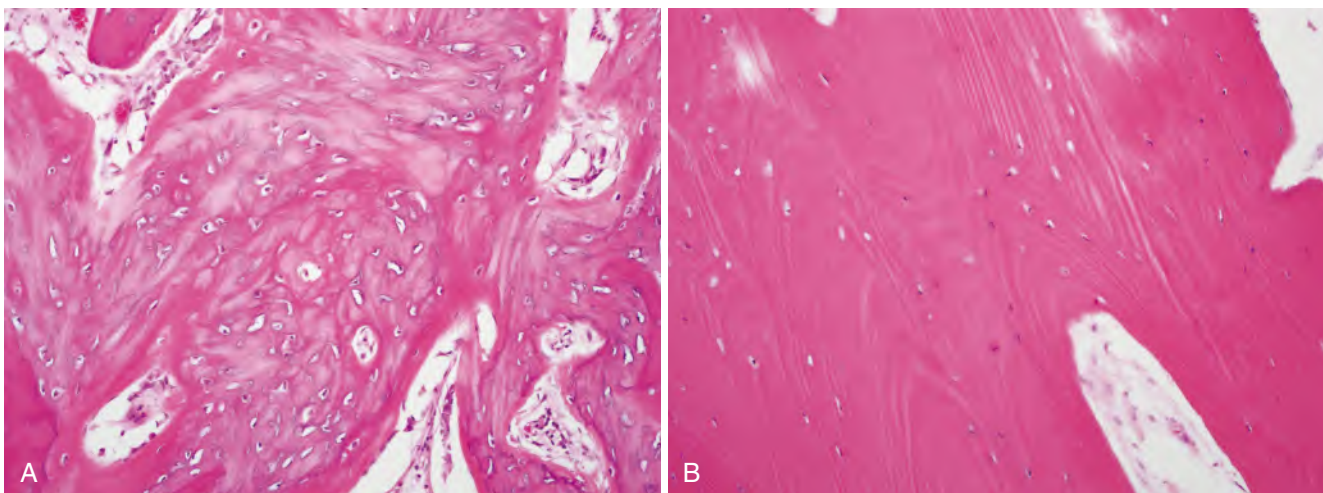


Fig. 26.1 (A) Woven bone. (B) Lamellar bone contrasts with the more cellular, disorganized appearance of woven bone.

- *Osteoclasts* are specialized multinucleated cells derived from circulating monocytes that are responsible for bone resorption (Fig. 26.2B). Surface integrin proteins allow osteoclasts to attach to the matrix and create a sealed extracellular trench (resorption pit). Secretion of acid and neutral proteases, predominantly *matrix metalloproteases* (MMPs), into the pit results in dissolution of inorganic and organic bone components.

Development

Most bones that form during embryogenesis develop from a cartilage mold via endochondral ossification. The cartilage mold (anlagen) is synthesized by mesenchymal precursor cells. A central medullary canal within the anlagen is created by chondroblasts at approximately 8 weeks of gestation. Simultaneously, osteoblasts begin to deposit the cortex beneath the nascent periosteum of the midshaft (diaphysis). This forms a primary center of ossification, resulting in radial bone growth. At the longitudinal ends (epiphysis), endochondral ossification forms secondary centers of ossification. Eventually, plates of cartilage anlage become entrapped between the expanding secondary centers of ossifications to form the growth plate (physis) (Fig. 26.3). Chondrocytes in the growth plates undergo sequential proliferation, hypertrophy, and apoptosis. The cartilage matrix mineralizes during apoptosis and is invaded by capillaries, providing the nutrients for activation of osteoblasts and osteoid synthesis. Most calcified cartilage matrix is ultimately resorbed, leaving only strut-shaped remnants that serve as scaffolding for bone deposits known as primary spongiosa, the earliest bone trabeculae (see Fig. 26.3). Over time, this process produces longitudinal bone growth.

Flat bones, for example the cranium, are formed by *intramembranous ossification*, in which a dense layer of mesenchyme is directly ossified by osteoblasts without a cartilage anlage. Flat bones enlarge by deposition of new osteoid on a preexisting surface, a process called appositional growth.

Normal bone development depends on the complex interplay of multiple factors, including:

- *Growth hormone* (GH) secreted by the anterior pituitary gland induces and maintains chondrocyte proliferation.
- *Thyroid hormone* secreted by the thyroid gland acts on proliferating chondrocytes to induce hypertrophy.
- *Indian hedgehog* is secreted locally by prehypertrophic chondrocytes and coordinates chondrocyte proliferation and differentiation with osteoblast proliferation.
- *Parathyroid hormone–related protein* (PTHrP) produced by perichondrial stromal cells and early proliferating chondrocytes activates the PTH receptor to maintain chondrocyte proliferation.
- *Wnt growth factors* are expressed in the growth plate–proliferating zone and, via Frizzled and LRP5/6 receptors, activate β -catenin to promote chondrocyte proliferation and maturation.
- *SOX9* is a transcription factor expressed by proliferating, but not hypertrophic, chondrocytes that is essential for differentiation of chondrocyte precursors.
- *RUNX2* is a transcription factor expressed in early hypertrophic chondrocytes and immature mesenchymal cells that controls terminal chondrocyte and osteoblast differentiation.
- *SATB2*, a transcription factor, regulates skeletal development by interacting with matrix attachment regions of DNA. SATB2 is also a marker of osteoblastic differentiation.
- *Fibroblast growth factors* (FGFs) are secreted by a variety of mesenchymal cells. FGFs (most notably FGF-3) act on hypertrophic chondrocytes to inhibit proliferation and promote differentiation.
- *Bone morphogenetic proteins* (BMPs), members of the TGF- β family, are expressed at various stages of chondrocyte development and have diverse effects on chondrocyte proliferation and hypertrophy at the growth plate.

Homeostasis and Remodeling

The adult skeleton appears static but actually undergoes continuous change via a tightly regulated process known as remodeling. Remodeling turns over approximately 10% of the skeleton each year, repairs damage, and may change the shape of bones in response to mechanical forces. Remodeling occurs within the *bone (or basic) multicellular unit* (BMU), which consists of a locally coupled “unit” of osteoblast and osteoclast activity on the bone surface. Osteoclast attachment, bone resorption, osteoblast attachment and proliferation, and, finally, matrix synthesis occur sequentially within the BMU.

Events within the BMU are regulated by cell-to-cell interactions, cytokines, and several signaling pathways (Fig. 26.4). One pathway involves three factors: (1) the transmembrane receptor activator for NF- κ B (RANK), which is expressed on osteoclast precursors; (2) RANK ligand (RANKL), which is expressed on osteoblasts and marrow stromal cells; and (3) osteoprotegerin (OPG), a secreted decoy receptor made by osteoblasts and several other types of cells that binds RANKL and thus prevents its interaction with RANK. When stimulated by RANKL, RANK signaling activates NF- κ B, which is essential for the generation and survival of osteoclasts. A second important pathway involves macrophage colony-stimulating factor (M-CSF), a factor produced by osteoblasts that is also crucial for the generation of osteoclasts. Finally, Wnt proteins produced by osteoprogenitor cells bind to LRP5 and LRP6 on osteoblasts to activate β -catenin signaling and OPG synthesis (Fig. 26.5). Conversely, osteocytes produce sclerostin, which inhibits Wnt/ β -catenin signaling, thereby reducing bone formation. The importance of these pathways is emphasized by rare germline mutations in the genes encoding OPG, RANK, RANKL, LRP5, and sclerostin, all of which severely disrupt bone metabolism and produce congenital bone disorders (described later).

The balance between bone formation and resorption is modulated by RANK and Wnt signaling. For example, because RANKL and OPG oppose one another, bone resorption or formation can be favored by increasing or decreasing the RANK-to-OPG ratio. Systemic factors affecting this balance include hormones (parathyroid hormone, estrogen,

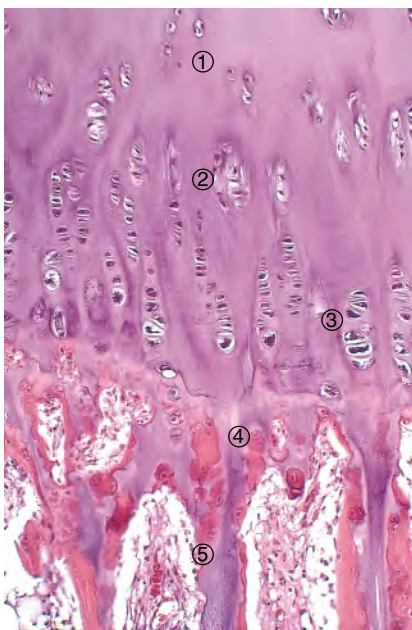


Fig. 26.3 Active growth plate with ongoing endochondral ossification. 1, Reserve zone. 2, Zone of proliferation. 3, Zone of hypertrophy. 4, Zone of mineralization. 5, Primary spongiosa.

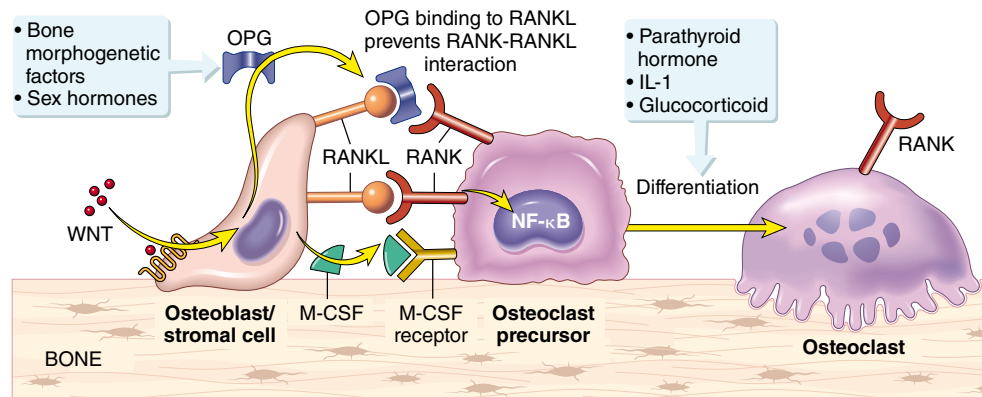


Fig. 26.4 Paracrine mechanisms that regulate osteoclast formation and function. Osteoclasts are derived from monocytes. Osteoblast/stromal cell membrane-associated RANKL binds to its receptor RANK located on the cell surface of osteoclast precursors. Signals transduced by RANK and macrophage colony-stimulating factor (*M-CSF*) receptor cause the precursor cells to differentiate into functional osteoclasts. In contrast, Wnt protein binding triggers osteoblasts/stromal cells to secrete osteoprotegerin (*OPG*), a “decoy” receptor that prevents RANKL from binding to RANK. Consequently, *OPG* prevents bone resorption by inhibiting osteoclast differentiation. *IL-1*, Interleukin-1; *NF-κB*, nuclear factor kappa-B; *RANK*, receptor activator of *NF-κB*; *RANKL*, receptor activator of *NF-κB* ligand.

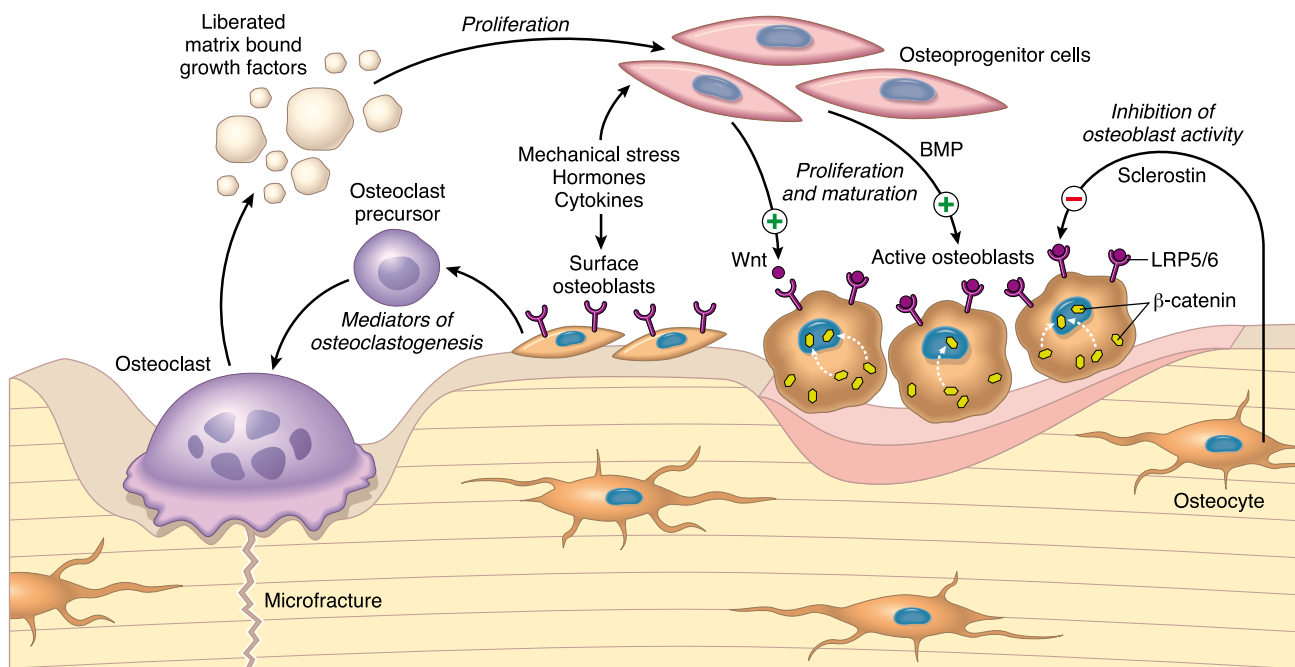


Fig. 26.5 Bone cells and their interrelated activities. Hormones, cytokines, growth factors, and signal-transducing molecules are instrumental in bone formation and maturation and allow communication between osteoblasts and osteoclasts. Bone resorption and formation during remodeling are coupled processes controlled by systemic factors and local cytokines, some of which are deposited in the bone matrix. *BMP*, Bone morphogenetic protein; *LRP5/6*, LDL receptor-related proteins 5 and 6.

testosterone, and glucocorticoids), vitamin D, inflammatory cytokines (e.g., interleukin [IL]-1), and growth factors (e.g., BMPs). Each of these acts by altering RANK/*NF-κB* and Wnt/*β-catenin* signaling. Parathyroid hormone, IL-1, and glucocorticoids promote osteoclast differentiation and bone turnover. Alternatively, some BMPs and sex hormones block osteoclast differentiation or activity by promoting *OPG* expression, alterations that favor bone deposition.

Another level of control involves paracrine signaling between osteoblasts and osteoclasts. Matrix breakdown by osteoclasts liberates and activates

growth factors, cytokines, and enzymes (e.g., collagenase), some of which stimulate osteoblasts. Thus, substances that initiate bone deposition are released into the microenvironment during bone resorption (see Fig. 26.5).

Peak bone mass is achieved in early adulthood after the cessation of growth of the skeleton. The peak bone mass set point is determined by several factors including polymorphisms in the vitamin D receptor gene, nutrition, and physical activity. Beginning in the fourth decade, resorption exceeds formation, resulting in a steady decline in skeletal mass.

DEVELOPMENTAL DISORDERS OF BONE AND CARTILAGE

Developmental abnormalities of the skeleton result from germline mutations and become apparent during the earliest stages of bone formation. By contrast, acquired diseases usually appear in adulthood. The spectrum of disorders of bone development is broad, and there is no standard approach to their classification; we will categorize the major diseases according to their pathogenesis.

Developmental anomalies can result from localized disordered migration and condensation of mesenchyme (dysostosis) or global disorganization of bone and/or cartilage (dysplasia). Dysostoses may occur in isolation or as part of more complex syndromes and are caused by defects in mesenchymal condensation and differentiation into cartilage anlage. The most common forms include complete absence of a bone or entire digit (aplasia), extra bones or digits (supernumerary digit), and abnormal fusion of bones (syndactyly, craniosynostosis). Genetic alterations that affect genes encoding transcription factors (especially homeobox genes), cytokines, and cytokine receptors are especially common among dysostoses. In contrast, dysplasias arise from mutations in genes that control development or remodeling of the entire skeleton. In bone biology, the term dysplasia is used to imply abnormal growth and does not denote a premalignant lesion (Chapter 7).

More than 350 skeletal dysostoses and dysplasias are recognized, most of which are extremely rare. The classification has evolved from clinical and radiographic descriptions to one centered on the causative genetic defects. Table 26.1 lists some of the better characterized developmental abnormalities and their associated defective genes. The relationships between specific mutations and phenotypes are complex; different point mutations in a single gene (e.g., *COL2A1*) can produce distinct phenotypes, while mutations in disparate genes (e.g., *LRP5*, *RANKL*) can lead to similar phenotypes.

Defects in Nuclear Proteins and Transcription Factors

Defects in transcription factors, especially homeobox proteins, result in disorganized mesenchymal condensation and dysregulated osteoblast and chondrocyte differentiation, leading to abnormal bone development.

- *Brachydactyly types D and E* are caused by mutations in the homeobox *HOXD13* gene and are characterized by shortening of the terminal phalanges of the thumb and big toe, respectively.
- Loss-of-function mutations in *RUNX2* result in *cleidocranial dysplasia*, an autosomal dominant disorder characterized by patent fontanelles, delayed closure of cranial sutures, Wormian bones (extra bones that occur within a cranial suture), delayed eruption of secondary teeth, primitive clavicles, and short stature.

TABLE 26.1 Diseases of the Skeleton With Identified Genetic Defects

Disorder	Gene Symbol	Affected Molecule	Clinical Phenotype
Defects in Transcription Factors Producing Abnormalities in Mesenchymal Condensation and Related Cell Differentiation			
Brachydactyly types D and E	<i>HOXD13</i>	Transcription factor	Short, broad terminal phalanges of first digits
Campomelic dysplasia	<i>SOX9</i>	Transcription factor	Sex reversal, abnormal skeletal development
Cleidocranial dysplasia	<i>RUNX2</i>	Transcription factor	Abnormal clavicles, wormian bones, supernumerary teeth
Holt-Oram syndrome	<i>TBX5</i>	Transcription factor	Congenital abnormalities, forelimb anomalies
Nail-patella syndrome	<i>LMX1B</i>	Transcription factor	Hypoplastic nails, hypoplastic or aplastic patellas, dislocated radial head, progressive nephropathy
Waardenburg syndrome types I and III	<i>PAX3</i>	Transcription factor	Hearing loss, abnormal pigmentation, craniofacial abnormalities
Defects in Hormones and Signal Transduction Proteins Producing Abnormal Proliferation or Maturation of Osteoblasts, Osteoclasts, or Chondrocytes			
Achondroplasia	<i>FGFR3</i>	Receptor	Short stature, rhizomelic shortening of limbs, frontal bossing, midface deficiency
Hypochondroplasia	<i>FGFR3</i>	Receptor	Disproportionately short stature, micromelia, relative macrocephaly
Osteopetrosis, autosomal dominant	<i>LRP5</i>	Receptor	Increased bone density, hearing loss, skeletal fragility
Osteopetrosis, infantile form	<i>RANKL</i>	Receptor ligand	Increased bone density
Osteoporosis-pseudoglioma syndrome	<i>LRP5</i>	Receptor	Congenital or infant-onset loss of vision, skeletal fragility
Thanatophoric dysplasia	<i>FGFR3</i>	Receptor	Severe limb shortening and bowing, frontal bossing, depressed nasal bridge
Defects in Extracellular Structural Proteins			
Achondrogenesis type 2	<i>COL2A1</i>	Type II collagen	Short trunk
Metaphyseal dysplasia, Schmid type	<i>COL10A1</i>	Type X collagen	Mildly short stature
Osteogenesis imperfecta types I to IV	<i>COL1A1</i> , <i>COL1A2</i>	Type I collagen	Bone fragility
Defects in Metabolic Enzymes and Transporters			
Osteopetrosis with renal tubular acidosis	<i>CA2</i>	Carbonic anhydrase	Increased bone density, fragility, renal tubular acidosis
Osteopetrosis, late onset type 2	<i>CLCN7</i>	Chloride channel	Increased bone density, fragility

Modified from Mundlos S, Olsen BR. Heritable diseases of the skeleton. Part I: Molecular insights into skeletal development—transcription factors and signaling pathways. *FASEB J* 1997;11(2):125-132; Mundlos S, Olsen BR. Heritable diseases of the skeleton. Part II: Molecular insights into skeletal development—matrix components and their homeostasis. *FASEB J* 1997;11(4):227-233; Superti-Furga A, Bonafé L, Rimoin DL. Molecular-pathogenetic classification of genetic disorders of the skeleton. *Am J Med Genet*. 2001;106(4):262-293; Krakow D, Rimoin DL. The skeletal dysplasias. *Genet Med*. 2010;12(6):327-341.

Defects in Hormones and Signal Transduction Proteins

Achondroplasia is the most common skeletal dysplasia and a major cause of disproportionate short stature. It is an autosomal dominant disorder caused by gain-of-function mutations in the fibroblast growth factor receptor 3 (*FGFR3*) gene, ~90% of which stem from new mutations in the paternal allele. Upon binding to its cognate ligand, FGF, *FGFR3* activation normally inhibits endochondral growth. This effect is exaggerated by *FGFR3* gain-of-function mutations. The delayed cartilage growth results in shortened proximal extremities, an enlarged head with bulging forehead (frontal bossing), and depression of the root of the nose. These skeletal abnormalities are not usually associated with changes in longevity, intelligence, or fertility.

Thanatophoric dysplasia, the most common lethal form of dwarfism, is characterized by diminished proliferation of chondrocytes and disorganization of the zone of proliferation. Like achondroplasia, thanatophoric dysplasia is caused by gain-of-function mutations in *FGFR3*, but the mutations in thanatophoric dysplasia result in stronger *FGFR3* signaling activity and thus a more severe phenotype. It occurs in approximately 1 in 20,000 live births. Affected individuals have disproportionately short (micromelic) limbs, frontal bossing, relative macrocephaly, a small chest cavity, and a bell-shaped abdomen. The underdeveloped thoracic cavity leads to respiratory insufficiency, and these individuals frequently die at birth or soon after.

Defects in Extracellular Structural Proteins

Mutations in the major bone and cartilage collagens (types I, II, IX, X, and XI) give rise to highly variable disorders ranging from lethal disease to premature osteoarthritis (OA).

Type I Collagen Diseases (Osteogenesis Imperfecta)

Osteogenesis imperfecta (OI), or brittle bone disease, is the most common inherited disorder of connective tissue. Approximately 80% to 90% of OI cases are caused by autosomal dominant mutations in the genes encoding the $\alpha 1$ or $\alpha 2$ chains of type I collagen that result in impaired type I collagen synthesis. OI principally affects

bone, but other tissues rich in type I collagen (joints, eyes, ears, skin, and teeth) are also impacted. Normal collagen synthesis and extracellular transport require triple helix formation. Many of the over 800 causative collagen mutations identified in patients with OI result in the replacement of a glycine residue within the triple-helical domain with another amino acid, leading to misfolding of collagen polypeptides and defective assembly of higher-order collagen chains. Mutant collagens also interfere with assembly of wild-type collagen chains, exerting a dominant negative effect that explains the autosomal dominant inheritance pattern. A minority (10% to 20%) of OI is caused by mutations in other genes that variously control collagen synthesis, osteoblast signaling, or mineralization.

The fundamental abnormality in OI is too little bone, resulting in extreme skeletal fragility. Other findings include so-called *blue sclerae*, caused by decreased collagen content that makes the sclera translucent, allowing partial visualization of the underlying choroid; hearing loss related to a sensorineural deficit and impeded conduction due to abnormalities in the bones of the middle and inner ear; and dental imperfections (small, misshapen, and blue-yellow teeth), secondary to dentin deficiency.

For treatment and prognosis, most cases of OI are clinically categorized into four major subtypes of varying severity (Table 26.2). Mutations that result in decreased synthesis of qualitatively normal collagen are associated with mild skeletal abnormalities. Individuals with type I OI have a normal life span but experience childhood fractures that decrease in frequency following puberty. More severe or lethal phenotypes are associated with mutant collagens that interfere with triple helix formation. Type II OI, which is uniformly fatal in utero or during the perinatal period, is characterized by extraordinary bone fragility and multiple intrauterine fractures (Fig. 26.6).

Defects in Metabolic Pathways (Enzymes, Ion Channels, and Transporters)

Osteopetrosis

Osteopetrosis, also known as marble bone disease, refers to a group of rare genetic diseases caused by deficient osteoclast development

TABLE 26.2 Subtypes of Osteogenesis Imperfecta

Subtype	Collagen Defect	Inheritance	Major Clinical Features	Prognosis
I	Decreased synthesis of pro- $\alpha 1(1)$ chain Abnormal pro- $\alpha 1(1)$ or pro- $\alpha 2(1)$ chains	Autosomal dominant	Postnatal fractures Normal stature Skeletal fragility Dentinogenesis imperfecta Hearing impairment Joint laxity Blue sclerae	Compatible with survival
II	Abnormally short pro- $\alpha 1(1)$ chain Unstable triple helix Abnormal or insufficient pro- $\alpha 2(1)$	Most autosomal recessive Some autosomal dominant New mutations	Death in utero or within days of birth Skeletal deformity with excessive fragility and multiple fractures Blue sclerae	Perinatal lethal
III	Altered structure of pro-peptides of pro- $\alpha 2(1)$ Impaired formation of triple helix	Autosomal dominant (75%) Autosomal recessive (25%)	Compatible with survival Growth retardation Multiple fractures Progressive kyphoscoliosis Blue sclerae at birth that become white Hearing impairment Dentinogenesis imperfecta	Progressive, deforming
IV	Short pro- $\alpha 2(1)$ chain Unstable triple helix	Autosomal dominant	Postnatal fractures, normal sclerae Moderate skeletal fragility Short stature Sometimes dentinogenesis imperfecta	Compatible with survival



Fig. 26.6 Skeletal radiograph of a fetus with lethal type II osteogenesis imperfecta. Note the numerous fractures of virtually all bones, resulting in accordion-like shortening of the limbs.

and function, leading to reduced bone resorption and diffuse, symmetric skeletal sclerosis. Although the term *osteopetrosis* refers to the stonelike quality of the bones, the bones are abnormally brittle and fracture easily. Osteopetrosis is classified into subtypes based on the mode of inheritance and the severity of clinical findings.

Pathogenesis

Most of the mutations underlying osteopetrosis interfere with acidification of the osteoclast resorption pit, which is required for the resorption of calcium hydroxyapatite within the matrix by osteoclasts. For example, a mild autosomal dominant form of osteopetrosis is caused by mutation of *CLCN7*, which encodes a proton-chloride exchanger on the osteoclast cell membrane that is required for resorption pit acidification. Similarly, most cases of autosomal recessive osteopetrosis are caused by a mutation of *TCIRG1*, which encodes a subunit of the osteoclast vacuolar H^+ -ATPase, an enzyme that is also necessary for acidification of the resorption pit. Other causes of autosomal recessive osteopetrosis include defects in carbonic anhydrase 2 (CA2), which like other carbonic anhydrase isozymes generates protons and bicarbonate from carbon dioxide and water. CA2 facilitates both resorption pit acidification by osteoclasts and urinary acidification by renal tubular epithelial cells. Osteopetrosis due to CA2 mutations is, therefore, accompanied by renal tubular acidosis. Alternatively, mutations in the X-linked gene *IKBKG*, which encodes NEMO (NF- κ B essential modulator), the regulatory subunit of the inhibitor of kappa B kinase (IKK) complex, causes osteopetrosis by impairing NF- κ B function, osteoclastogenesis, and osteoclast survival (see Fig. 26.4).

MORPHOLOGY

Due to deficient osteoclast activity, bones involved by osteopetrosis lack a medullary canal and the ends of long bones are bulbous (**Erlenmeyer flask deformity**) and misshapen (Fig. 26.7A). The neural foramina are small and can compress exiting nerves. The primary spongiosa, which is normally removed during growth, persists and fills the medullary cavity, leaving no room for the hematopoietic marrow and preventing the formation of mature trabeculae (Fig. 26.7B). Deposited bone is not remodeled and tends to be woven rather than lamellar. Depending on the underlying genetic defect, the number of osteoclasts may be normal, increased, or decreased.

Clinical Features

Severe infantile osteopetrosis is an autosomal recessive disorder that usually becomes evident in utero or soon after birth. Fracture, anemia,

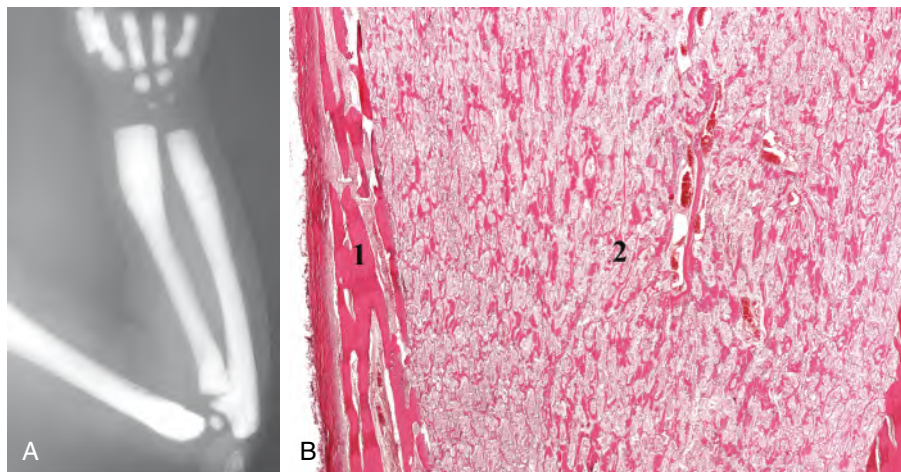


Fig. 26.7 Osteopetrosis. (A) Radiograph of the upper extremity in an individual with osteopetrosis. The bones are diffusely sclerotic, and the distal metaphyses of the ulna and radius are poorly formed. (B) Section of proximal tibial diaphysis from a fetus with osteopetrosis. The cortex (1) is present, but the medullary cavity (2) is filled with primary spongiosa, which replaces the hematopoietic elements.

and hydrocephaly are often seen, resulting in postpartum mortality. Affected individuals who survive into their infancy have cranial nerve defects (optic atrophy, deafness, and facial paralysis) and repeated—often fatal—infections because of leukopenia due to reduced marrow space and decreased hematopoiesis. Compensatory extramedullary hematopoiesis may lead to hepatosplenomegaly. Autosomal dominant forms are milder and may not be detected until adolescence or adulthood when discovered on x-ray studies, either incidentally or because of repeated fractures.

Osteopetrosis was the first genetic bone disease treated with hematopoietic stem cell transplantation, which is effective because osteoclasts are derived from hematopoietic precursors. The normal osteoclasts produced from donor stem cells reverse many of the skeletal abnormalities.

Diseases Associated With Defects in Degradation of Macromolecules

Mucopolysaccharidoses

The mucopolysaccharidoses, discussed in Chapter 5, are lysosomal storage diseases. They are caused by deficiencies in enzymes, primarily acid hydrolases that degrade dermatan sulfate, heparan sulfate, and keratan sulfate. Mesenchymal cells, such as chondrocytes, degrade extracellular matrix mucopolysaccharides. In these diseases, mucopolysaccharides accumulate within chondrocytes and induce apoptosis. Extracellular mucopolysaccharide accumulation leads to structural defects in articular cartilage. Consequently, many of the skeletal manifestations of the mucopolysaccharidoses result from abnormalities in the cartilage anlage, growth plates, costal cartilages, and articular surfaces. Affected individuals are frequently of short stature and have chest wall abnormalities and malformed bones.

KEY CONCEPTS

Developmental Disorders of Bone and Cartilage

Abnormalities in a single bone or a localized group of bones are called dysostoses and arise from defects in mesenchyme migration and condensation. They manifest as absent, supernumerary, or abnormally fused bones. Global disorganization of bone and/or cartilage is termed dysplasia. Developmental abnormalities can be categorized by the associated genetic defect.

- **Transcription factors:** Homeobox genes, such as *HOXD13*, are frequently mutated in brachydactyly syndromes.
- **Signal transduction molecules:** *FGFR3* mutations are responsible for achondroplasia and thanatophoric dysplasia, both of which manifest as dwarfism.
- **Structural proteins:** Mutations in the genes for type I collagen underlie most types of osteogenesis imperfecta, characterized by defective bone formation and skeletal fragility (brittle bone disease).
- **Metabolic enzymes and transporters:** Mutations that interfere with osteoclast acidification of the resorption pit or osteoclastogenesis cause osteopetrosis, in which bones are hard but brittle (marble bone disease). Depending on the gene mutated, extraskeletal manifestations may also be present.

METABOLIC DISEASES OF BONE

Osteopenia and Osteoporosis

Osteopenia refers to decreased bone mass; osteoporosis is defined as osteopenia that is severe enough to significantly increase the risk of fracture. Radiographically, osteoporosis is considered bone mass at

least 2.5 standard deviations below mean peak bone mass in young adults, whereas osteopenia is 1 to 2.5 standard deviations below the mean. The disorder may be localized to a certain bone or region, as in disuse osteoporosis of a limb, or may involve the entire skeleton. Most generalized osteoporosis is primary but a minority of cases result from endocrine, gastrointestinal, or other underlying conditions or from drugs (Table 26.3). Each year, nearly 2 million Americans experience an osteoporosis-related fracture, which are associated with with an estimated cost of \$57 billion.

Pathogenesis

Although numerous factors affect bone mass, the most common forms of osteoporosis are *senile* and *postmenopausal*. Peak bone mass is achieved during young adulthood. Its magnitude is influenced by hereditary factors, especially polymorphisms in the genes that affect bone metabolism (Fig. 26.8). Physical activity, muscle strength, diet, and hormonal state also contribute to peak bone mass. Once maximal skeletal mass is attained, bone resorption slightly exceeds formation, resulting in age-related bone loss that averages 0.7% per year. Both sexes are affected equally. Important factors include:

- **Age-related changes.** Compared to younger individuals, osteoblasts in older individuals show a reduced proliferative and biosynthetic capacity and an attenuated response to growth factors, resulting in a diminished ability to make bone. This form of osteoporosis, known as *senile osteoporosis*, is categorized as a “low-turnover” form of osteoporosis.
- **Reduced physical activity.** The decreased physical activity sometimes associated with aging results in reduced bone mass. Osteocytes

TABLE 26.3 Categories of Generalized Osteoporosis

Primary
Idiopathic Postmenopausal Senile
Secondary
Endocrine Disorders
Addison disease Diabetes, type 1 Hyperparathyroidism Hyperthyroidism Hypothyroidism Pituitary tumors Neoplasia Carcinomatosis Multiple myeloma
Gastrointestinal
Hepatic insufficiency Malabsorption Malnutrition Vitamin C and D deficiencies
Drugs
Alcohol Anticoagulants Anticonvulsants Chemotherapy Corticosteroids
Miscellaneous
Anemia Homocystinuria Immobilization Osteogenesis imperfecta Pulmonary disease

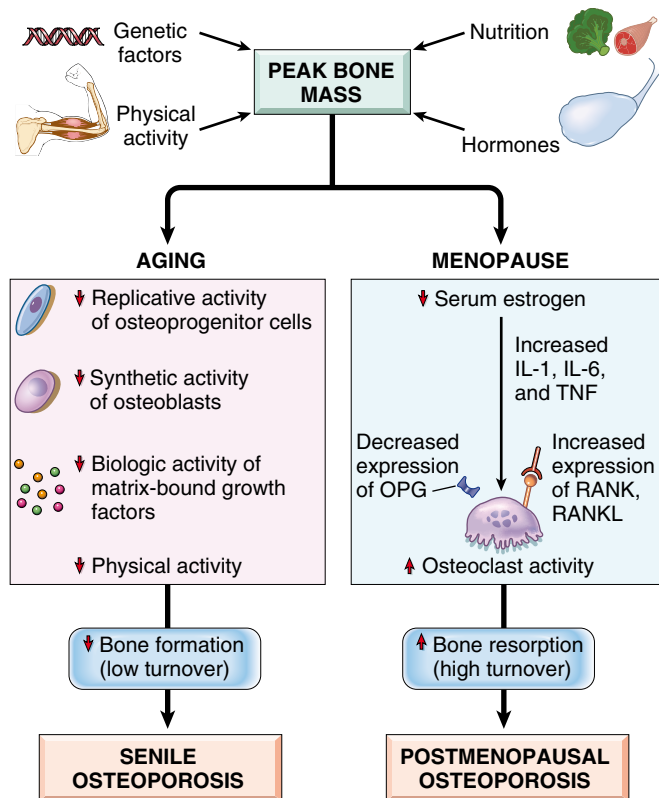


Fig. 26.8 Pathophysiology of postmenopausal and senile osteoporosis (see text). *IL*, Interleukin; *OPG*, osteoprotegerin; *RANK*, receptor activator of NF- κ B; *RANKL*, receptor activator of NF- κ B ligand; *TNF*, tumor necrosis factor.

respond to mechanical load by stimulating or inhibiting osteoblasts and osteoclasts to remodel normal bone. Alterations in the mechanical load sensed by osteocytes underlies localized bone loss in an immobilized or paralyzed extremity, generalized bone loss in the microgravity of outer space, and increased bone density in athletes. The type of exercise is important, as load magnitude influences bone density more than the number of load cycles. Thus, resistance exercises such as weight training are more effective stimuli for increasing bone mass than repetitive endurance activities such as bicycling.

- **Genetic factors.** Single-gene defects account for only a small fraction of osteoporosis cases. However, polymorphisms in certain genes have been linked to osteoporosis by genome-wide association studies. The most strongly linked genes include *RANK*, *RANKL*, and *OPG*, all of which encode key osteoclast regulators; the human leukocyte antigen (HLA) locus (for unknown reasons); and the estrogen receptor gene (discussed later).
- **Calcium nutritional state.** Adolescents often have insufficient dietary calcium intake, which can lead to reduced peak bone mass and increased risk of osteoporosis. Relative calcium deficiency, reduced vitamin D, and elevated PTH levels may also contribute to development of senile osteoporosis.
- **Hormonal influences.** Estrogen promotes increased bone density through a variety of mechanisms, including inhibition of osteoblast apoptosis, stimulation of osteoclast apoptosis, and suppression of RANKL production (see Fig. 26.8). In the decade after menopause, up to 2% of cortical bone and 9% of cancellous bone may be lost each year. *Estrogen deficiency* plays the major role in this phenomenon, and close to 40% of postmenopausal women are affected by postmenopausal osteoporosis. Although decreased estrogen increases both bone formation and resorption, the latter dominates, resulting in *high-turnover* osteoporosis. Menopause leads to increased secretion of inflammatory cytokines, such as IL-6, tumor necrosis factor (TNF), and IL-1 by innate immune cells and IL-17 by Th17 cells in the blood and marrow through unknown mechanisms. The resulting increases in RANKL and decreases in OPG stimulate osteoclast recruitment and activity.

MORPHOLOGY

The hallmark of osteoporosis is histologically normal bone that is decreased in quantity. The entire skeleton is affected in senile and postmenopausal osteoporosis, but certain bones tend to be more severely impacted. Postmenopausal osteoporosis affects mainly bones or portions of bones that have increased surface area, such as the cancellous compartment of vertebral bodies. The trabecular plates become perforated and thinned and lose their interconnections, leading to microfractures and eventually to vertebral collapse (Fig. 26.9A). In senile osteoporosis, the cortex is thinned by subperiosteal and endosteal resorption, and the Haversian systems are widened (Fig. 26.9B).

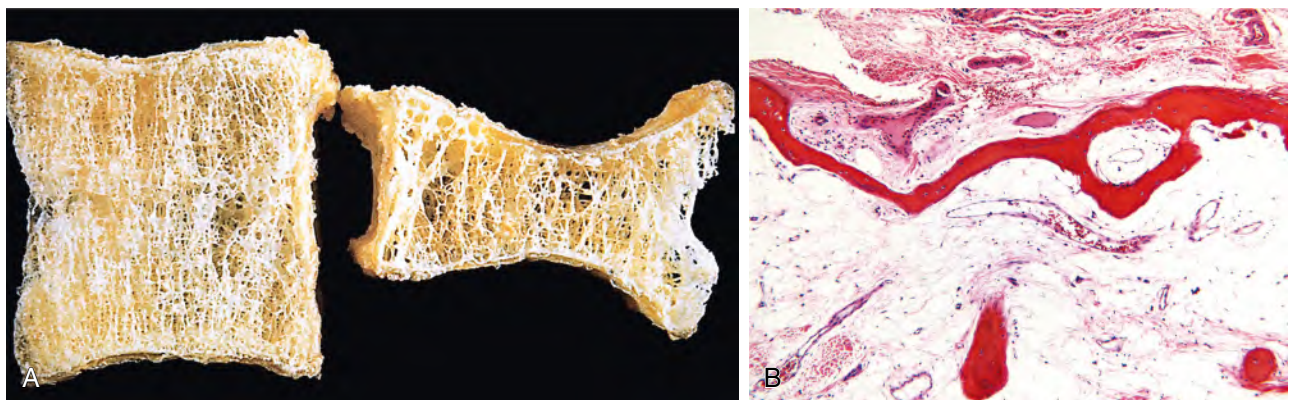


Fig. 26.9 Osteoporosis. (A) Osteoporotic vertebral body (right) shortened by compression fractures compared with a normal vertebral body (left). Note that the osteoporotic vertebra has a characteristic loss of horizontal trabeculae and thickened vertical trabeculae. (B) In advanced osteoporosis, both the trabecular bone of the medulla (bottom) and the cortical bone (top) are markedly thinned.

Clinical Features

The clinical manifestations of osteoporosis depend on which bones are involved. Vertebral fractures in the thoracic and lumbar regions are painful and when multiple can cause significant loss of height and deformities such as lumbar *lordosis* and *kyphoscoliosis*. Fractures of the femoral neck, pelvis, or spine lead to immobilization and complications such as pulmonary embolism and pneumonia, resulting in over 50,000 deaths per year.

Osteoporosis cannot be reliably detected in plain radiographs until 30% to 40% of the bone mass is lost and blood levels of calcium, phosphorus and alkaline phosphatase are usually normal. Thus, screening asymptomatic people for osteoporosis is difficult. The best estimates of bone loss are made using specialized radiographic imaging techniques, such as dual-energy x-ray absorptiometry and quantitative computed tomography, both of which measure bone density.

Prevention and treatment of senile and postmenopausal osteoporosis include exercise, appropriate calcium and vitamin D intake, and pharmacologic agents that reduce bone resorption (particularly bisphosphonates). Denosumab, an anti-RANKL antibody, has shown promise in treating some forms of postmenopausal osteoporosis. Menopausal hormone therapy with estrogen agonists has been used to slow bone loss and reduce associated fractures, but complications (particularly deep vein thrombosis and stroke) have limited the use of estrogen in the treatment of osteoporosis.

Osteomalacia and Rickets

Osteomalacia and rickets are manifestations of impaired mineralization of bone matrix. Most examples of undermineralized matrix result from abnormal vitamin D metabolism or vitamin D deficiency (detailed in Chapter 9). *Rickets* refers to the disorder in children, in whom it interferes with the deposition of bone in the growth plates. *Osteomalacia* is the adult counterpart, in which bone formed during remodeling is undermineralized, causing a predisposition to fractures. Osteomalacia contrasts with osteoporosis, in which mineralization of bone is normal but bone mass is decreased.

Hyperparathyroidism

In hyperparathyroidism, excess production and activity of parathyroid hormone (PTH) causes increased osteoclast activity, bone resorption, and osteopenia.

Pathogenesis

As discussed in Chapter 24, parathyroid hormone plays a central role in calcium homeostasis through the following effects.

- **Osteoclast activation**, which increases bone resorption and calcium mobilization. PTH mediates this effect indirectly by increasing RANKL expression on osteoblasts.
- **Increased calcium resorption** by the renal tubules
- **Increased urinary phosphate excretion**
- **Increased synthesis of active vitamin D**, $1,25(\text{OH})_2\text{-D}$ by renal α -1 hydrolase, which in turn increases intestinal calcium absorption and mobilizes bone calcium by inducing RANKL expression on osteoblasts

The net result of the above alterations is elevated serum calcium, which under normal circumstances provides negative feedback to the parathyroid glands to decrease PTH production. Excessive or inappropriate PTH release may stem from autonomous parathyroid secretion (*primary hyperparathyroidism*) or may be the result of renal disease (*secondary hyperparathyroidism*) (see Chapter 24). In both instances, hyperparathyroidism produces changes throughout the skeleton due to unrestrained osteoclast activity. While elevated PTH is responsible for

bone changes in primary hyperparathyroidism, in chronic renal failure the combination of elevated PTH, inadequate $1,25(\text{OH})_2$ -vitamin D synthesis, hyperphosphatemia, and metabolic acidosis lead to bone changes as part of a syndrome known as *renal osteodystrophy* (discussed later). With time, loss of bone mass increases susceptibility to fractures, bone deformation, and joint problems.

MORPHOLOGY

Symptomatic, untreated primary hyperparathyroidism manifests with three inter-related skeletal abnormalities: osteoporosis, brown tumors, and osteitis fibrosa cystica. **Osteoporosis** is generalized but is most severe in the phalanges, vertebrae, and proximal femur. Hyperparathyroidism most prominently enhances osteoclast activity within cortical bone (subperiosteal and endosteal surfaces), but cancellous bone can also be affected. At these sites, osteoclasts may tunnel into and dissect centrally along the length of the trabeculae, leaving adjacent marrow spaces to be replaced by fibrovascular tissue. This creates a railroad track-like appearance that is referred to as **dissecting osteitis** (Fig. 26.10A), a misnomer as this is not an inflammatory lesion. Radiographs show decreased bone density.

Bone loss in hyperparathyroidism may lead to microfractures, secondary hemorrhage, macrophage recruitment, and ingrowth of reparative fibrous tissue to create a mass lesion called **brown tumor** (Fig. 26.10B). The brown color reflects vascularity, hemorrhage, and hemosiderin. Brown tumors often undergo cystic degeneration. The combination of increased bone cell activity, peritrabecular fibrosis, and cystic brown tumors is the hallmark of severe hyperparathyroidism and is known as **generalized osteitis fibrosa cystica** (**von Recklinghausen disease of bone**).

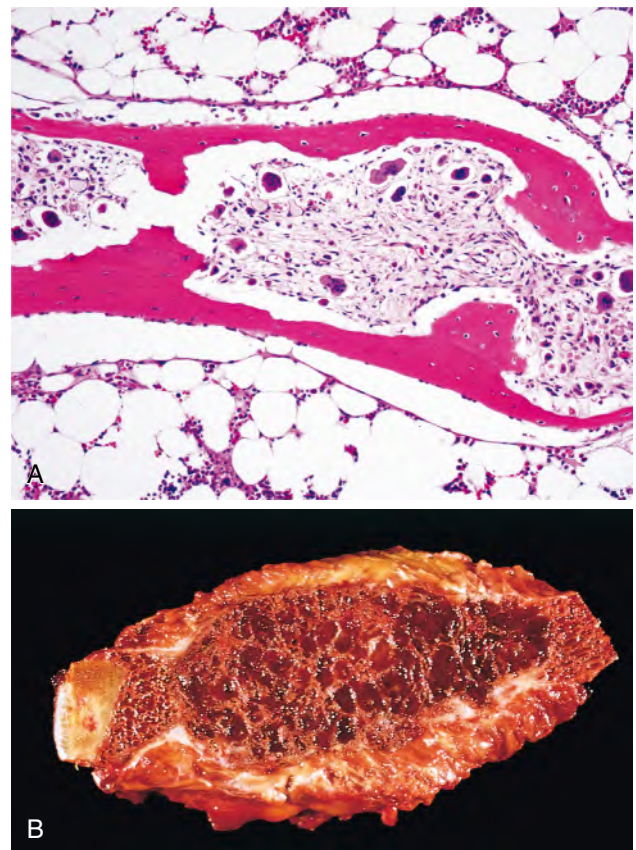


Fig. 26.10 Hyperparathyroidism. (A) Osteoclasts boring into the center of the trabecula producing a “railroad track” appearance (dissecting osteitis). (B) Resected rib, harboring an expansile brown tumor adjacent to the costal cartilage.

Clinical Features

As bone mass decreases, affected patients are increasingly susceptible to fractures, bone deformation, and joint problems. Osteitis fibrosa cystica is now uncommon because hyperparathyroidism is usually diagnosed on routine blood tests and treated at an early stage. In symptomatic patients, surgical removal of autonomously functioning parathyroid glands (typically due to involvement by parathyroid adenoma; Chapter 24) can completely reverse the bone changes. Secondary hyperparathyroidism can cause similar changes, but the process is usually less severe and less prolonged, so skeletal abnormalities tend to be milder. Asymptomatic cases are followed by monitoring serum calcium and creatinine and bone mineral density.

Renal Osteodystrophy

The term **renal osteodystrophy** describes the collective skeletal changes that occur in chronic renal disease, including those associated with dialysis. The manifestations include many of the entities described earlier, including (1) osteopenia/osteoporosis, (2) osteomalacia, (3) secondary hyperparathyroidism, and (4) growth retardation in skeletally immature patients. As medical advances have prolonged the lives of individuals with renal disease, the impact of abnormal renal function on skeletal homeostasis has assumed greater clinical importance. Histologic bone changes in individuals with end-stage renal failure can be divided into three major types of disorders:

- *High-turnover osteodystrophy* is characterized by increased bone resorption and bone formation, with the former predominating.
- *Low-turnover or aplastic disease* is manifested by adynamic bone (little osteoclastic and osteoblastic activity) and, less commonly, osteomalacia.
- *Mixed pattern disease* is heterogeneous, with areas of high turnover and low turnover.

Pathogenesis

Kidney disease causes skeletal abnormalities through several interrelated mechanisms (Fig. 26.11):

- *Tubular dysfunction*, leading to tubular and systemic acidosis. Acidosis in turn leads to increased expression of RANKL on osteoblasts and increased osteoclast activity, resulting in matrix resorption and osteopenia.

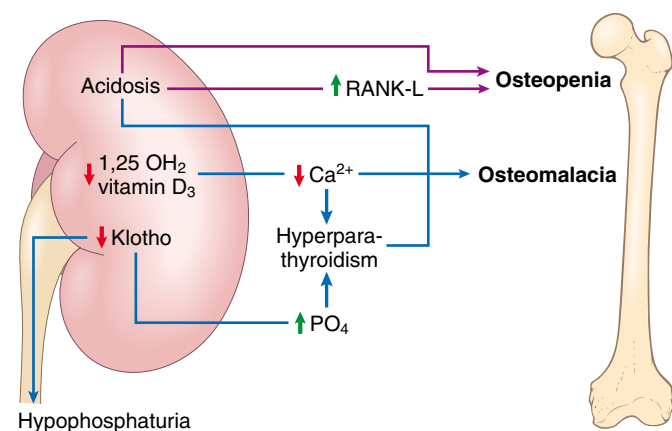


Fig. 26.11 Mechanisms of renal osteodystrophy involve metabolic abnormalities and altered endocrine signaling between bone and kidney. See text for details. *RANKL*, receptor activator of NF- κ B ligand.

- *Secondary hyperparathyroidism*, due to hypophosphaturia, hyperphosphatemia, and hypocalcemia. These abnormalities stem from decreased expression of renal Klotho, a receptor for the hormone FGF-23, which is made by osteocytes and promotes phosphate excretion in the urine; and from decreased renal production of 1,25(OH)₂ vitamin D₃, which decreases calcium absorption in the gut. The net effect of these alterations is diminished calcification of bone matrix, leading to osteomalacia.

KEY CONCEPTS

Metabolic Diseases of Bone

- Osteopenia and osteoporosis are conditions in which bone is normally mineralized but decreased in quantity. Osteoporosis is defined as bone loss sufficient to increase fracture risk and is associated with significant morbidity and mortality from fractures. Multiple factors including peak bone mass, age, physical activity, genetics, nutrition, and hormonal influences contribute to its pathogenesis.
- Osteomalacia is defined by the presence of bone that is insufficiently mineralized. In the developing skeleton, this results in rickets.
- Hyperparathyroidism arises from either autonomous or compensatory hypersecretion of PTH and can lead to osteoporosis, brown tumors, and osteitis fibrosa cystica. In high-income countries where early diagnosis is the norm, these manifestations are rare.
- Renal osteodystrophy is marked by a constellation of bone abnormalities (osteopenia, osteomalacia, hyperparathyroidism, and growth retardation) that occur because of chronic renal failure. The bone changes stem from decreased tubular, glomerular, and hormonal renal functions.

Paget Disease of Bone (Osteitis Deformans)

Paget disease of bone is a disorder associated with increased but disordered and structurally abnormal bone formation. It develops in three sequential phases: (1) an initial osteolytic stage; (2) a mixed osteoclastic-osteoblastic stage, which ends with a predominance of osteoclastic activity; and (3) a final burned-out, quiescent, osteosclerotic stage (Fig. 26.12).

Paget disease usually presents in late adulthood and becomes progressively more common with increasing age. An estimated 1% of the US population older than age 40 years is affected. The exact incidence is unknown because many affected individuals are asymptomatic. Paget disease is relatively common in the United Kingdom, central Europe, and Greece as well as areas colonized by European immigrants (e.g., the United States and Australia). By contrast, Paget disease is rare in native populations of Scandinavia, China, Japan, and Africa. A decrease in new cases has been observed in some countries over the past 30 years.

Pathogenesis

The cause of Paget disease remains uncertain but current evidence suggests both genetic and environmental contributions. Approximately 60% of cases of familial Paget disease and 10% of sporadic cases are associated with germline mutations in the *SQSTM1* gene, which encodes a protein known as sequestosome-1. Mutations in *SQSTM1* appear to lead to increased NF- κ B activity and thus increase osteoclast activity. Activating mutations in the *RANK* gene and inactivating mutations in the *OPG* gene account for some cases of juvenile Paget disease. The geographic variation in the incidence of the disease suggests involvement of an environmental factor that remains unknown. In vitro studies have raised the possibility that chronic infection of osteoclast precursors by measles or other RNA viruses may play a role, but firm links to the disease have yet to be established.

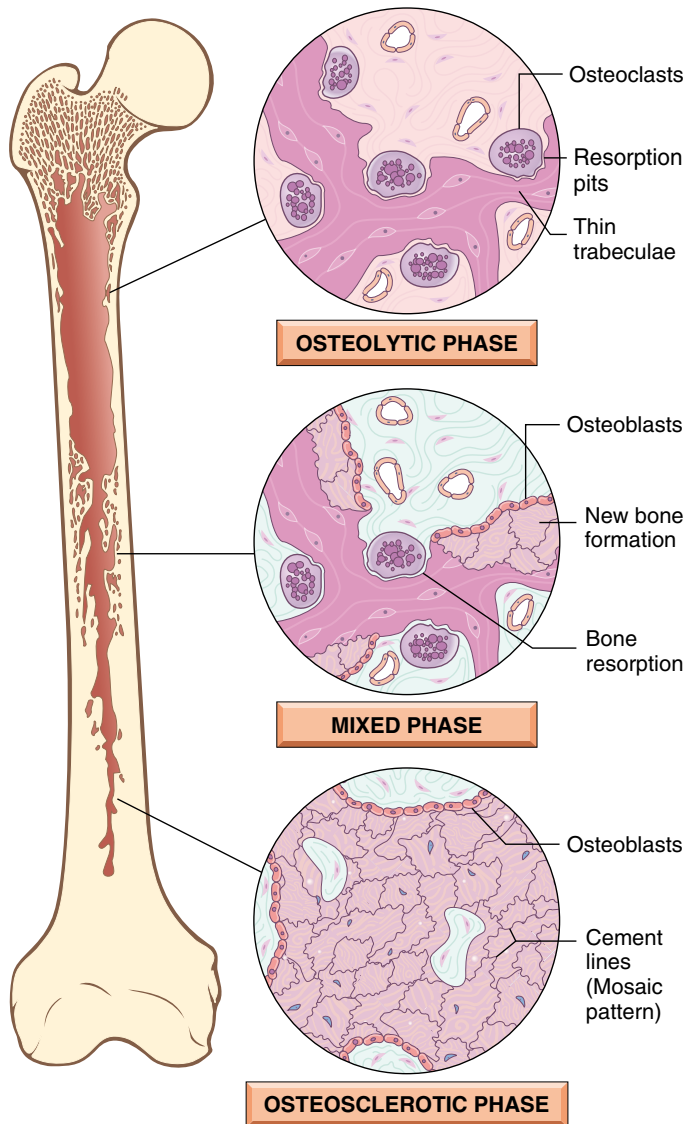


Fig. 26.12 Diagrammatic representation of Paget disease of bone demonstrating the three phases of disease evolution.

MORPHOLOGY

Paget disease of bone shows remarkable histologic variation over time between affected bones. The initial lytic phase is characterized by numerous abnormally large osteoclasts and resorption pits. The osteoclasts may have 100 or more nuclei (more than four times the normal number). Osteoclasts persist in the mixed phase but, in addition, increased osteoblastic activity and poorly mineralized woven bone are observed. The marrow adjacent to the bone-forming surface is replaced by loose connective tissue that contains osteoprogenitor cells and numerous blood vessels. The hallmark of Paget disease is a mosaic pattern of lamellar bone (Fig. 26.13A) that develops in the sclerotic phase. This jigsaw puzzle–like appearance is produced by unusually **prominent cement lines**, which join haphazardly oriented units of lamellar bone. As the mosaic pattern takes hold and cell activity decreases, periosteal fibrovascular tissue recedes and is replaced by normal marrow. In the end, the coarsely thickened trabeculae and soft and porous cortices lack structural stability and make the bone vulnerable to fractures.

Clinical Features

Paget disease is *monostotic* in about 15% of cases and *polyostotic* in the remaining 85%. The axial skeleton or proximal femur is involved in up to 80% of cases. Most cases are asymptomatic and are discovered as an incidental radiographic finding (e.g., during staging studies for a new diagnosis of carcinoma). Symptoms, if present, include pain in the affected bone due to microfractures or bone overgrowth that compresses spinal and cranial nerve roots. Enlargement of the craniofacial skeleton may produce *leontiasis ossea* (lion face) and a cranium so heavy that it is difficult for the person to hold their head erect. The weakened pagetic bone may lead to invagination of the skull base (*platybasia*) and compression of the posterior fossa. Weight bearing causes anterior bowing of the femurs and tibiae (Fig. 26.13B) and distorts the femoral heads, resulting in the development of severe *secondary osteoarthritis*. *Chalk stick–type fractures* are common and usually involve the long bones of the lower extremities. Compression fractures of the spine can result in kyphosis and spinal cord injury. Rarely, the hypervascularity of pagetic bone warms the overlying skin, and in severe polyostotic Paget disease the increased blood flow can act as an arteriovenous shunt, occasionally leading to high-output heart failure or exacerbation of underlying cardiac disease. Many affected individuals have elevated serum alkaline phosphatase levels, but serum calcium and phosphate levels are normal.

A variety of neoplasms develop in Paget disease, albeit rarely. Secondary osteosarcoma occurs in less than 1% of individuals with Paget disease overall but in 5% to 10% of those with severe polyostotic disease, in whom it is rapidly fatal. In the absence of malignant transformation, Paget disease is usually not a serious or life-threatening disease. Most affected individuals have mild symptoms that are readily managed by treatment with antiresorptive agents such as calcitonin and bisphosphonates, which act by reducing osteoclast activity.

FRACTURES

A fracture is a loss of bone integrity resulting from mechanical injury and/or diminished bone strength. Fractures are the most

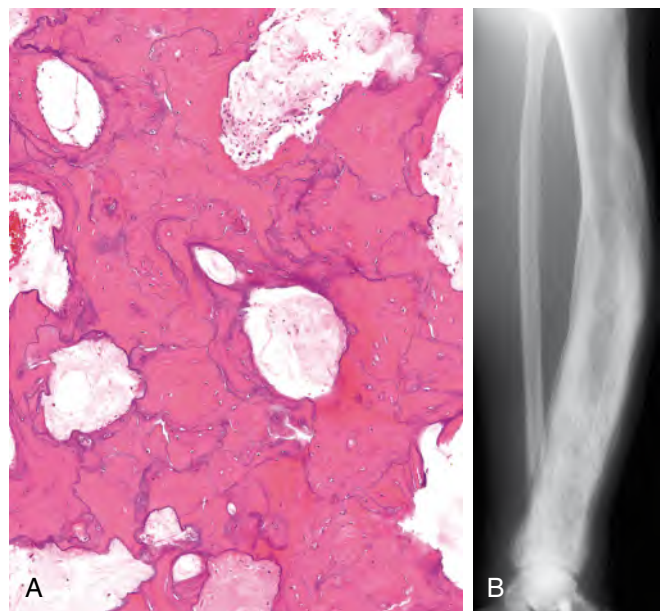


Fig. 26.13 Paget disease of bone. (A) Mosaic pattern of lamellar bone in Paget disease. (B) Plain radiograph of the leg in severe Paget disease. The tibia is bowed. The affected portion is enlarged and sclerotic, and it exhibits irregular thickening of both cortical and cancellous bone.

common pathologic conditions affecting bone. The following qualifiers describe fracture types and affect treatment:

- **Simple:** the overlying skin is intact
- **Compound:** the bone communicates with the skin surface
- **Comminuted:** the bone is fragmented
- **Displaced:** the ends of the bone at the fracture site are not aligned
- **Stress:** a slowly developing fracture that follows a period of increased physical activity in which the bone is subjected to repetitive loads
- **Greenstick:** extending only partially through the bone; these fractures are common in infants and young children whose bones are soft
- **Pathologic:** involving bone weakened by an underlying disease process, such as a tumor

Healing of Fractures

Bone has a remarkable capacity for repair. At the time of fracture, rupture of blood vessels results in a hematoma that fills and surrounds the area of injury (Fig. 26.14). The clotted blood seals off the fracture site and provides a fibrin scaffold that guides the influx of inflammatory cells and ingrowth of fibroblasts and new capillaries. Simultaneously, degranulated platelets and migrating inflammatory cells release PDGF, TGF- β , FGF, and other growth factors that activate osteoprogenitor cells in the periosteum, medullary cavity, and surrounding soft tissues and stimulate osteoclastic and osteoblastic activity. By the end of the first week, a mass of predominantly uncalcified tissue, called *soft callus* or *procallus*, bridges the ends of the fractured bones providing some anchorage but not structural rigidity for weight bearing. By approximately 2 weeks after injury, the procallus is converted into a *bony callus* by the deposition of woven bone. The activated osteoprogenitor cells deposit subperiosteal trabeculae of woven bone oriented perpendicular to the cortical axis within the medullary cavity. In some cases, activated soft tissue mesenchymal cells may also differentiate into chondrocytes that produce fibrocartilage and hyaline cartilage. The cartilage then undergoes endochondral ossification, forming new bone. As a result, fractured bone ends are bridged and with progressive mineralization, the stiffness and strength of the callus increases to allow weight bearing.

In the early stages of callus formation, excess fibrous tissue, cartilage, and woven bone is produced. As the callus matures and is subjected to weight-bearing forces, portions that are not physically stressed are resorbed. This remodeling reduces the size of the callus until the shape and outline of the fractured bone are reestablished as *lamellar bone*. The healing process is completed by restoration of the medullary cavity. In simple fractures with adequate immobilization and absent comorbidities, near-perfect union is the norm.

The healing of a fracture may be impeded by several factors.

- **Displaced and comminuted fractures** frequently result in some deformity.
- **Bone neoplasms** (primary or metastatic) that produce a pathologic fracture often inhibit the normal healing process.
- **Inadequate immobilization** permits movement of the callus and interferes with normal maturation, which may result in *delayed union* or *nonunion*. If nonunion persists, the malformed callus undergoes cystic degeneration and the luminal surface can become lined by synovial-like cells, creating a false joint or *pseudoarthrosis*.
- **Infection of the fracture site** is a serious obstacle to healing and is especially common in open fractures.
- **Malnutrition, diabetes, and skeletal dysplasia** (e.g., osteoporosis) also hinder fracture healing.

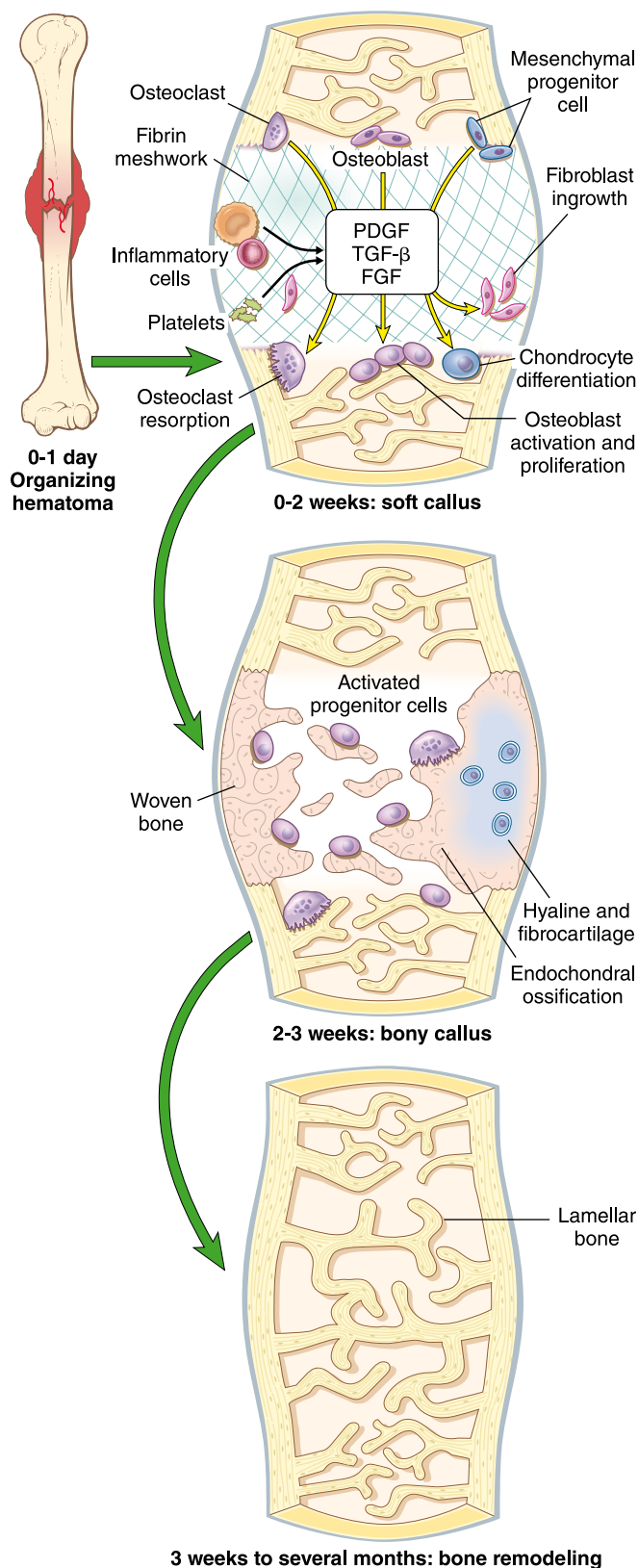


Fig. 26.14 Healing of a fracture. Healing begins with an organizing hematoma. Within 2 weeks, the two ends of the bone are bridged by a fibrin meshwork in which osteoclasts, osteoblasts, and chondrocytes differentiate from precursors. These cells produce cartilage and bone matrix, which, with adequate immobilization, remodels into normal lamellar bone. FGF, Fibroblast growth factor; PDGF, platelet-derived growth factor; TGF β , transforming growth factor β .

OSTEONECROSIS (AVASCULAR NECROSIS)

Osteonecrosis refers to infarction (ischemic necrosis) of bone and marrow. A diverse set of conditions predispose to bone ischemia, including vascular injury (trauma, vasculitis), drugs (e.g., corticosteroids, bisphosphonates), systemic disease (e.g., sickle cell crisis), and radiation. Osteonecrosis of the jaw is a rare complication of long term treatment of osteoporosis with bisphosphonates. In about 25% of cases the cause is unknown (idiopathic). Mechanisms of avascular necrosis include mechanical disruption of vessels (e.g., fracture), thrombotic occlusion, and extravascular compression. Osteonecrosis peaks in middle adulthood and accounts for about 10% of hip arthroplasties in the United States.

MORPHOLOGY

Bone infarcts may be medullary or subchondral. Regardless of etiology, medullary infarcts are geographic in shape and involve both trabecular bone and marrow but spare the cortex, which is protected by collateral blood flow from the periosteum. In subchondral infarcts, a triangular segment with subchondral bone plate at its base undergoes necrosis; the overlying articular cartilage remains viable due to nutrients within synovial fluid. Dead bone is recognized histologically by empty lacunae lacking osteocytes. The marrow fat contains necrotic adipocytes. The released fatty acids bind calcium and form insoluble calcium soaps. The remaining trabeculae act as scaffolding for deposition of new bone, while osteoclasts resorb necrotic trabeculae. The pace of repair in subchondral infarcts is slow, resulting in collapse and fracturing of the necrotic bone and sloughing of the articular cartilage (Fig. 26.15).

Clinical Features

Symptoms depend on location and extent of infarct. Typically, subchondral infarcts cause pain that is initially associated with activity but becomes constant as secondary changes supervene. As mentioned above, subchondral infarcts often collapse, resulting in severe secondary osteoarthritis. Treatment ranges from conservative measures (limiting weight-bearing activities, immobilization) to surgery. Medullary infarcts are usually small and clinically silent.

OSTEOMYELITIS

Osteomyelitis is bone and marrow inflammation, virtually always secondary to infection. Osteomyelitis may be a complication of any systemic infection but frequently manifests as a solitary primary focus.

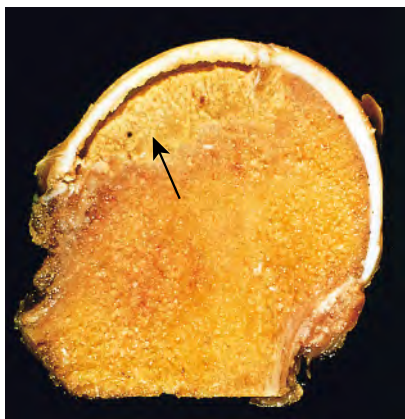


Fig. 26.15 Osteonecrosis. Femoral head with a subchondral, triangle-shaped pale area of necrotic bone (arrow). The space between the overlying articular cartilage and bone is caused by trabecular compression fractures without repair.

All types of organisms, including viruses, parasites, fungi, and bacteria, can produce osteomyelitis, but certain pyogenic bacteria and mycobacteria are the most common causes.

Pyogenic Osteomyelitis

Pyogenic osteomyelitis is almost always caused by bacterial infection.

Organisms may reach the bone by (1) hematogenous spread, (2) extension from a contiguous site, or (3) direct implantation after compound fracture or orthopedic procedures. In otherwise healthy children, most osteomyelitis is hematogenous in origin and is more common in long bones. The initiating bacteremia may stem from seemingly trivial mucosal injuries or from minor skin infections. In adults, osteomyelitis is most often a complication of open fractures, surgical procedures, or diabetes, which predisposes to infections of the feet. Osteomyelitis is particularly worrisome in infants, in whom epiphyseal infection can spread to a joint, resulting in septic or suppurative arthritis, articular cartilage destruction, and permanent disability.

Causative pathogens vary depending on the anatomic location and clinical setting. *Staphylococcus aureus* is responsible for 80% to 90% of culture-positive pyogenic osteomyelitis. Staphylococcal cell wall proteins bind to bone matrix components such as collagen, facilitating bacterial adherence to bone. In the neonatal period, group B streptococci and *E. coli* are likely pathogens, whereas in older children gram-positive organisms such as *S. aureus* are the most likely cause. Mixed bacterial infections are common in the setting of direct spread, surgery, or open fractures. In patients with sickle cell anemia, areas of osteonecrosis serve as a nidus of bacterial seeding and loss of splenic function impairs the immune response; in this setting, *Salmonella* and other gram-negative organisms are the most frequent pathogens. Microbiological cultures are obtained whenever possible to identify the causative organism. Specific organisms are only identified in approximately 50% of patients; it is hoped that introduction of sensitive PCR assays that identify microbe-specific DNA sequences will increase this number.

The location of bone infections is influenced by the osseous vascular circulation, which varies with age. In neonates, metaphyseal vessels penetrate the growth plate, resulting in frequent infection of the metaphysis, epiphysis, or both. In older children, involvement of the metaphysis is typical. The epiphyses and subchondral regions are more commonly involved in adults after growth plate closure, which creates anastomoses between metaphyseal and epiphyseal vessels that provide a route for bacterial spread.

MORPHOLOGY

Changes associated with osteomyelitis depend on the stage (acute, subacute, and chronic) and location of the infection. In the acute phase, bacteria proliferate and induce a neutrophilic reaction. Necrosis of bone cells and marrow ensues within 48 hours. Bacteria and the associated inflammatory response then spread through the Haversian canals to reach the periosteum. Because the periosteum is loosely attached to the cortex in children, subperiosteal abscesses may dissect along the bone surface. Lifting of the periosteum further impairs blood supply and contributes to the necrosis. Soft tissue abscesses may also form after periosteal rupture and may channel to the skin as draining sinuses.

As the inflammatory process evolves during the first week, chronic inflammatory cells are recruited that release cytokines that stimulate bone resorption, fibrous tissue ingrowth, and deposition of reactive bone at the periphery of the infection. The dead bone is known as **sequestrum**. The new reactive bone can form a shell of living tissue known as **involucrum** around the devitalized, infected bone (Fig. 26.16). The histologic findings of chronic osteomyelitis are more variable but typically involve marrow fibrosis, sequestrum, and a chronic inflammatory infiltrate of lymphocytes and plasma cells.



Fig. 26.16 Resected femur in a person with draining osteomyelitis. The drainage tract in the subperiosteal shell of viable new bone (involucrum) reveals the inner native necrotic cortex (sequestrum).

Clinical Features

Hematogenous osteomyelitis may present acutely as a systemic illness with malaise, fever, chills, leukocytosis, and intense throbbing pain over the affected region. In other instances, the presentation is subtle, with only unexplained fever (infants) or localized pain (adults). Plain radiographs characteristically show a lytic focus of bone destruction and associated reactive bone; MRI is more specific and sensitive for identifying the changes of osteomyelitis. Biopsy and microbial culture are required to identify the pathogen in most instances.

The combination of antibiotics and surgical drainage is usually curative, but in 5% to 25% of cases acute osteomyelitis fails to resolve. This unfavorable outcome is particularly likely in the setting of delayed diagnosis, extensive bone necrosis, inadequate antibiotic treatment or surgical debridement, or weakened host immunity. The course of such chronic infections may be punctuated by spontaneous flares that may occur after years of dormancy. Complications of chronic osteomyelitis include pathologic fracture, secondary (reactive) amyloidosis, endocarditis, sepsis, and the development of squamous cell carcinoma in draining sinus tracts or sarcoma in the infected bone.

Mycobacterial Osteomyelitis

Mycobacterial osteomyelitis tends to be more destructive and resistant to control than pyogenic osteomyelitis. Mycobacterial osteomyelitis was previously limited to lower resource countries, but the incidence has risen worldwide due to immigration and increased numbers of immunocompromised individuals. Overall, approximately 1% to 3% of individuals with pulmonary or extrapulmonary tuberculosis develop osseous infection, which may be present for years prior to diagnosis. The typical patient presents with localized pain, low-grade fever, chills, or weight loss. Infection is usually solitary in immunocompetent patients but can disseminate in the immunocompromised.

Mycobacteria that infect bone are usually bloodborne and originate from a focus of active visceral disease elsewhere. Direct extension (e.g., from a pulmonary focus into a rib or from tracheobronchial nodes into adjacent vertebrae) may also occur. The principal histologic feature, the presence of caseating granulomas typical of other involved sites, is also seen in tuberculous osteomyelitis. Mycobacterial osteomyelitis

tends to be more destructive and resistant to control than pyogenic osteomyelitis.

The spine is involved in 40% of mycobacterial osteomyelitis cases (*Pott disease*). Infection often tracks through intervertebral discs to involve multiple vertebrae and surrounding soft tissues (Fig. 26.17). Discs and vertebral destruction result in permanent compression fractures that produce scoliosis, kyphosis, and neurologic deficits secondary to spinal cord and nerve compression. Tuberculous osteomyelitis may also cause tuberculous arthritis, sinus tract formation, abscesses, and amyloidosis.

Skeletal Syphilis

Syphilis (*Treponema pallidum*) and yaws (*Treponema pertenue*) can involve bone. Although the incidence of syphilis is increasing, bone involvement remains rare because diagnosis and treatment typically occur before bone disease can develop.

In congenital syphilis, bone lesions appear in the fifth month of gestation and are fully developed at birth. The spirochetes concentrate at sites of active endochondral ossification and within the periosteum to cause osteochondritis and periostitis, respectively. The characteristic *saber shin* is produced by reactive periosteal bone deposition on the medial and anterior surfaces of the tibia.

In acquired syphilis, bone disease begins early in the tertiary stage, usually 2 to 5 years after the initial infection. Bones of the nose, palate, skull, and extremities, especially the long tubular bones such as the tibia, are involved most frequently.

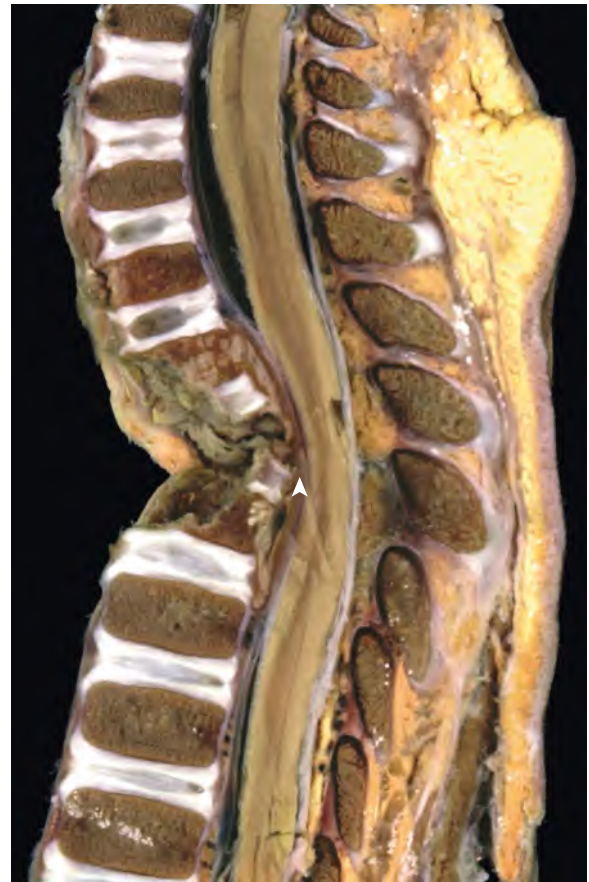


Fig. 26.17 Mycobacterial osteomyelitis of the spine (Pott disease). Infection has destroyed vertebral bodies and intervertebral discs (arrowhead), leading to compression fractures and kyphosis.

MORPHOLOGY

Spirochetes can be detected with silver stains or by immunohistochemistry. Edematous granulation tissue with numerous plasma cells and necrotic bone characterizes syphilitic bone disease. **Obliterative endarteritis** typically accompanies caseous necrosis as part of gummas, which may be present in congenital or acquired syphilis (Chapter 8).

BONE TUMORS

Primary bone tumors are rare and are vastly outnumbered by metastases and hematopoietic tumors in the marrow. Nevertheless, mortality is high for many of these neoplasms and disfiguring surgery is often required for treatment, making management of bone malignancies challenging. Therapy for the ~2400 new bone sarcomas diagnosed each year in the United States aims to optimize survival and maintain function of affected body parts.

The predilection of specific types of tumors to affect certain age groups and particular anatomic sites provides a diagnostic clue. For example, osteosarcoma incidence peaks during adolescence and most frequently involves the distal femur and proximal tibia. By contrast, chondrosarcoma affects older adults and arises in the pelvic bones and proximal extremities.

Bone tumors may present in a variety of ways. Benign lesions are often asymptomatic incidental findings while malignant tumors cause pain, a growing mass, or pathologic fractures. Radiologic imaging defines tumor location and extent and can identify features that narrow the differential diagnosis, but biopsy is necessary for definitive diagnosis in almost all cases.

Bone tumors are classified according to the normal cell types they recapitulate or matrix they produce. Lesions that do not have normal tissue counterparts are grouped according to clinicopathologic features (Table 26.4). Benign tumors greatly outnumber malignant tumors in incidence and occur with greatest frequency during the first 3 decades of life. In older adults, a bone tumor is more likely to be malignant.

Bone-Forming Tumors

Tumors in this category produce unmineralized osteoid or mineralized woven bone.

Osteoid Osteoma and Osteoblastoma

Osteoid osteoma and osteoblastoma are benign bone-producing tumors with similar histologic appearances but different clinical and radiographical features. By definition, osteoid osteomas are less than 2 cm in diameter. They are most common in young males. About 50% of cases involve the cortex of the femur or tibia. A thick rim of reactive cortical bone may be the only radiographic clue. Despite their small size, osteoid osteomas present with severe nocturnal pain that is probably caused by prostaglandin E₂ (PGE₂) produced by the proliferating osteoblasts and is relieved by aspirin and other nonsteroidal antiinflammatory drugs (NSAIDs). Osteoblastomas are larger than 2 cm, typically involve the posterior components of the vertebrae (laminae and pedicles), cause pain that is unresponsive to aspirin, and usually do not induce reactive cortical bone. Osteoid osteoma can be treated by radiofrequency ablation, but osteoblastoma usually requires curettage or en bloc excision. Malignant transformation is rare.

TABLE 26.4 Classification of Major Nonhematopoietic Primary Bone Tumors

Category	Behavior	Tumor Type	Common Locations	Age (Years)	Morphology
Cartilage forming	Benign	Osteochondroma	Metaphysis of long bones	10–30	Bony excrescence with cartilage cap
		Chondroma	Small bones of hands and feet	30–50	Circumscribed hyaline cartilage nodule in medulla
		Chondroblastoma	Epiphysis of long bones	10–20	Circumscribed, pericellular calcification
		Chondromyxoid fibroma	Tibia, pelvis	20–30	Collagenous to myxoid matrix, stellate cells
	Malignant	Chondrosarcoma (conventional)	Pelvis, shoulder	40–60	Extends from medulla through cortex into soft tissue, chondrocytes with increased cellularity and atypia
Bone forming	Benign	Osteoid osteoma	Metaphysis of long bones	10–20	Cortical, interlacing microtrabeculae of woven bone
		Osteoblastoma	Vertebral column	10–20	Posterior elements of vertebra, histology similar to osteoid osteoma
	Malignant	Osteosarcoma	Metaphysis of distal femur, proximal tibia	10–20	Extends from medulla to lift periosteum, malignant cells producing woven bone
Unknown origin	Benign	Giant cell tumor	Epiphysis of long bones	20–40	Destroys medulla and cortex, sheets of mononuclear stromal cells with scattered multinucleate giant cells
		Aneurysmal bone cyst	Proximal tibia, distal femur, vertebra	10–20	Vertebral body, hemorrhagic spaces separated by cellular, fibrous septae
	Malignant	Ewing sarcoma	Diaphysis of long bones	10–20	Sheets of primitive small round cells
		Adamantinoma	Tibia	30–40	Cortical, fibrous bone matrix with epithelial islands
Notochordal	Malignant	Chordoma	Clivus, sacrum	30–60	Destroys medulla and cortex, foamy cells in myxoid matrix

Modified from Unni KK, Inwards CY. *Dahlin's Bone Tumors*. 6th ed. Lippincott-Williams & Wilkins; 2010:5.

MORPHOLOGY

Osteoid osteoma and osteoblastoma are round-to-oval masses of hemorrhagic, gritty tan tissue. They are well circumscribed and composed of delicate inter-connected trabeculae of woven bone rimmed by a single layer of prominent osteoblasts (Fig. 26.18A) and surrounded by loose connective tissue with many dilated and congested capillaries. Reactive bone encircles the actual neoplasm, or nidus, which forms a small radiolucent core that may be centrally mineralized (Fig. 26.18B). The relatively small size, well-defined margins, and benign cytomorphology of the neoplastic osteoblasts help distinguish these tumors from osteosarcoma.

Osteosarcoma

Osteosarcoma is a malignant tumor that produces osteoid matrix or mineralized bone. Excluding hematopoietic tumors, osteosarcoma is the most common primary malignant tumor of bone. The age distribution of osteosarcoma is bimodal, with 75% occurring before 20 years of age. A second smaller peak occurs in older adults, in whom it is frequently associated with predisposing conditions (*secondary osteosarcoma*) such as Paget disease of bone, bone infarcts, and chronic osteomyelitis. Males are affected slightly more often than females (1.6:1). Although any bone can be involved, tumors usually arise in the metaphyseal region of the long bones; almost 50% are near the knee in the distal femur or proximal tibia.

Osteosarcoma often presents as a painful, progressively enlarging mass, sometimes with a pathologic fracture. Radiographically, the enlarging tumor forms a destructive, mixed lytic and sclerotic mass with infiltrative margins (Fig. 26.19A). The tumor frequently breaks through the cortex and lifts the periosteum, inducing reactive periosteal bone formation. The triangular shadow between the cortex and raised periosteal ends, known radiographically as *Codman' triangle*, indicates an aggressive tumor.

Pathogenesis

The peak incidence of osteosarcoma is during the adolescent growth spurt. The tumor occurs most frequently in the growth plate of rapidly growing bones, where increased proliferation may predispose to mutations that drive oncogenesis. Osteosarcomas typically have complex karyotypes with numerous chromosomal aberrations and mutations in

well-known tumor suppressor genes and oncogenes (Chapter 7), including:

- *RB* mutations are present in up to 70% of sporadic osteosarcomas and germline *RB* mutations confer a 1000-fold increased risk of osteosarcoma.
- *TP53* is mutated in the germline of individuals with Li-Fraumeni syndrome, who have a greatly increased incidence of osteosarcoma. Inactivating mutations in *TP53* are common in sporadic osteosarcomas.
- *CDKN2A*, which encodes two tumor suppressors, p16 and p14, is inactivated in many osteosarcomas.
- *MDM2* and *CDK4*, which inhibit p53 and RB function, respectively, are overexpressed in many low-grade osteosarcomas, often through amplification of these genes on chromosome 12q14.
- *MYC* amplification is seen in up to half of osteosarcomas and may be associated with a particularly poor prognosis.

MORPHOLOGY

Osteosarcomas are large, gritty, gray-white tumors that often contain hemorrhage and cystic degeneration (Fig. 26.19B). The tumor frequently destroys the surrounding cortices to produce a soft tissue mass, spreads extensively in the medullary canal, and replaces hematopoietic marrow. Tumors infrequently penetrate the epiphyseal plate or enter the joint, where they may grow along tendoligamentous structures or through the attachment site of the joint capsule.

Osteosarcomas demonstrate pleomorphism, large hyperchromatic nuclei, bizarre tumor giant cells, and abundant mitoses, including abnormal (e.g., tripolar) forms. Extensive necrosis and intravascular invasion are also common.

Diagnosis requires the presence of malignant tumor cells producing unmineralized osteoid or mineralized bone (Fig. 26.20), which is typically fine and lacelike, but can also form broad sheets or primitive trabeculae. In some cases, malignant cells also produce cartilage; if abundant, such tumors are classified as **chondroblastic osteosarcoma**.

Clinical Features

Osteosarcomas are generally assumed to have occult metastases at the time of diagnosis. As a result, treatment typically consists of neoadjuvant chemotherapy, surgery, and postoperative adjuvant chemotherapy. Chemotherapy has greatly improved osteosarcoma prognosis,

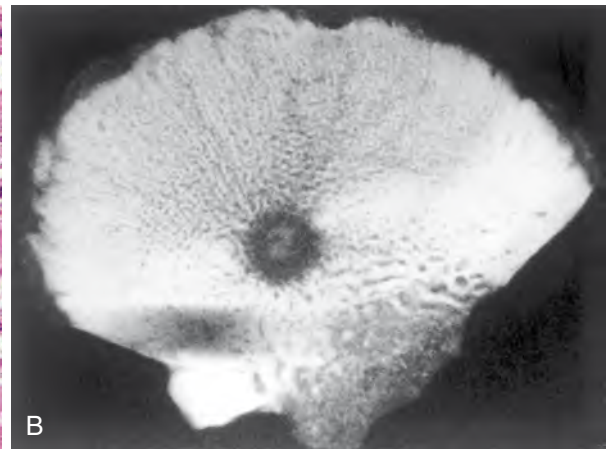
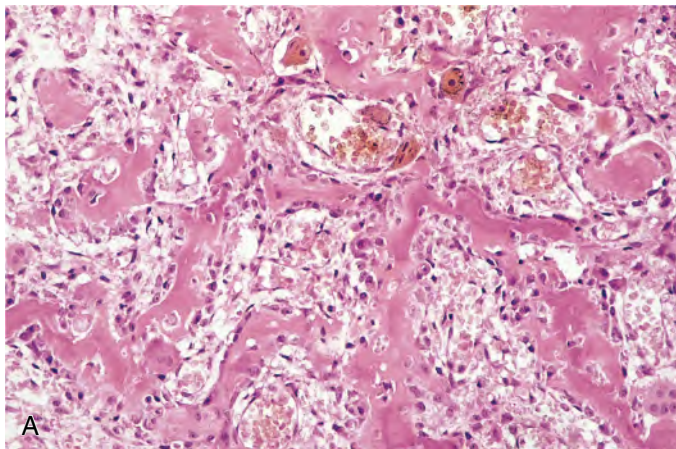


Fig. 26.18 Osteoid osteoma. (A) Microscopically, the tumor is composed of haphazardly interconnecting trabeculae of woven bone that are rimmed by prominent osteoblasts. The intertrabecular spaces are filled by vascularized loose connective tissue. (B) Specimen radiograph of intracortical osteoid osteoma. The round radiolucency with central mineralization represents the lesion, which is surrounded by abundant reactive bone that has massively thickened the cortex.

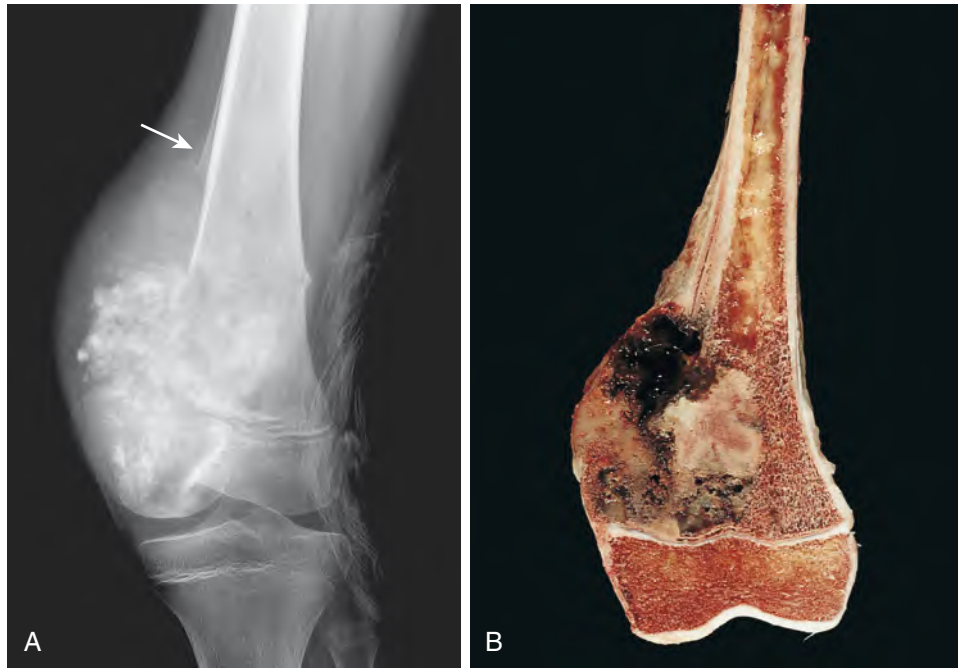


Fig. 26.19 Osteosarcoma. (A) Distal femoral osteosarcoma with prominent bone formation extending into the soft tissues. The periosteum, which has been lifted, has laid down a proximal triangular shell of reactive bone known as a Codman triangle (*arrow*). (B) Osteosarcoma of distal femur. There is extensive cortical disruption and subperiosteal expansion. Tumor is confined to the metaphyseal side of the cartilaginous growth plate. The hemorrhagic area represents the previous biopsy site. (A, From Czerniak B. *Dorfman and Czerniak's Bone Tumors*. 2nd ed. Elsevier; 2016.)

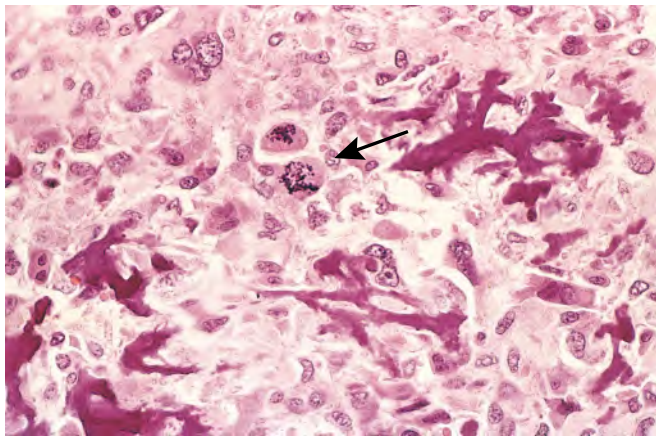


Fig. 26.20 Fine, lacelike pattern of neoplastic bone produced by anaplastic malignant tumor cells in an osteosarcoma. Note the abnormal mitotic figures (*arrow*).

with 5-year survival reaching 70% in individuals without overt metastases at initial diagnosis. Osteosarcoma metastasizes hematogenously to the lungs, bones, brain, and other sites. The outcome for those with clinically evident metastases, recurrent disease, or secondary osteosarcoma remains guarded, with a 5-year survival rate less than 20%.

Cartilage-Forming Tumors

These tumors are characterized by the formation of hyaline cartilage by neoplastic cells. Benign cartilage tumors are much more common than malignant lesions. Together, cartilage-forming tumors account for most primary nonhematopoietic bone tumors.

Osteochondroma

Osteochondroma, or exostosis, is a benign cartilage-capped tumor that arises from the bone surface. It may be sessile or pedunculated with a bony stalk. About 85% of osteochondromas are solitary and sporadic. The remainder occur as part of autosomal dominant *multiple hereditary exostosis syndrome* (described later). Solitary osteochondromas are usually diagnosed in late adolescence and early adulthood, whereas multiple hereditary exostosis syndrome becomes apparent during childhood. In those with sporadic disease, males are affected three times more often than females. Osteochondromas develop only in bones of endochondral origin. The most common site is the metaphysis near the growth plate of long tubular bones, especially near the knee, followed by the bones of the pelvis, scapula, and ribs, where they tend to have short stalks.

Osteochondromas present as slow-growing masses, which can be painful if they impinge on a nerve or if the stalk is fractured. In many cases, they are detected incidentally. In multiple hereditary exostosis syndrome, the underlying bones may be bowed and shortened, reflecting an associated disturbance in epiphyseal growth.

Pathogenesis

Hereditary exostoses are associated with germline mutations in the *EXT1* or *EXT2* genes and subsequent loss of the remaining wild-type allele in chondrocytes of the growth plate. (Fig. 26.21A). Most mutations result in a truncated or nonfunctional protein. Reduced expression of *EXT1* or *EXT2* has also been observed in sporadic osteochondromas. These genes encode glycosyltransferases involved in heparan sulfate biosynthesis. Loss of heparan sulfate may allow mutated chondrocyte progenitors in the growth plate to proliferate and lose polarity, leading to secondary perturbation of signaling through the Hedgehog, BMP, FGFR3, and Wnt signaling pathways.

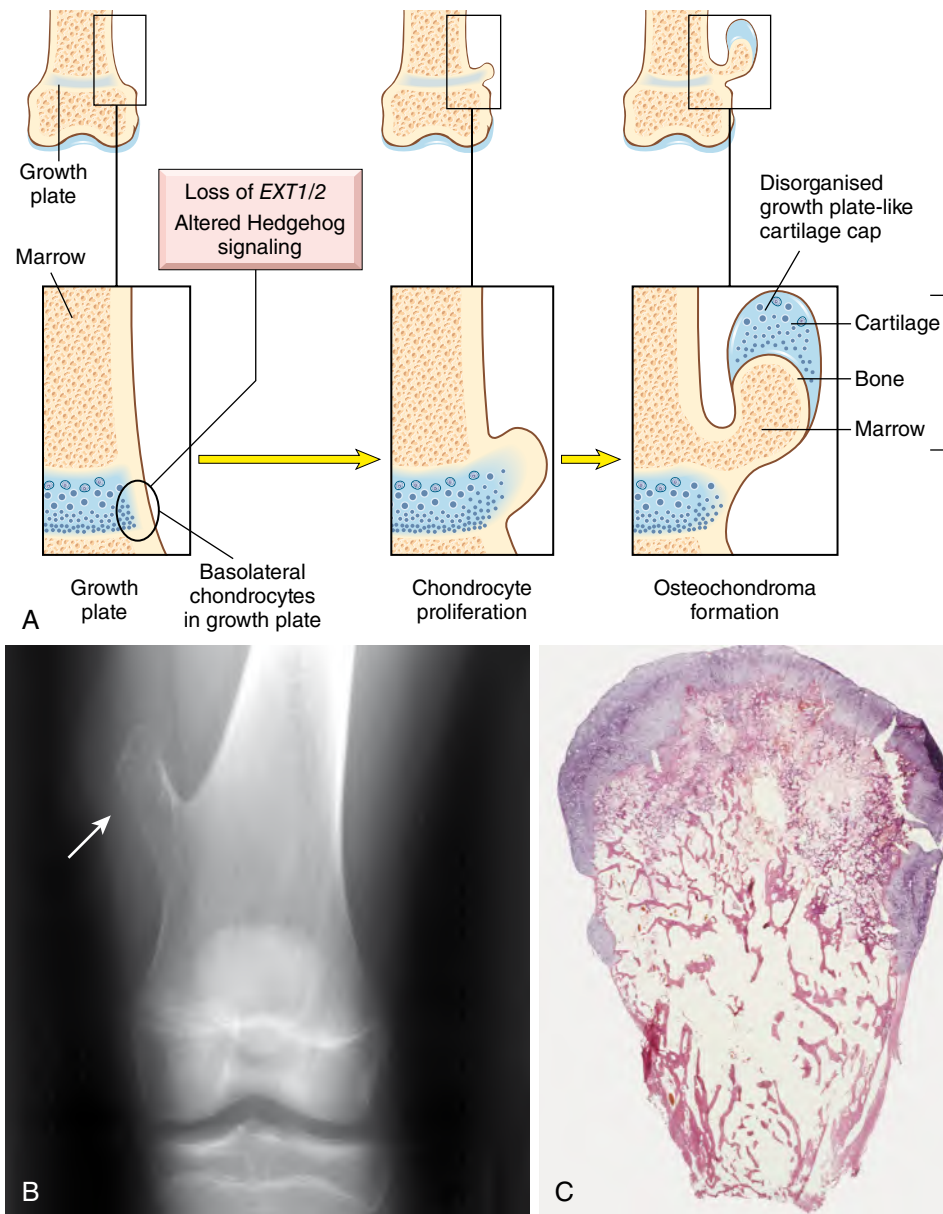


Fig. 26.21 Osteochondroma. (A) The development of an osteochondroma beginning with an outgrowth from the epiphyseal cartilage (B) Radiograph of an osteochondroma arising from the distal femur (arrow). (C) Whole mount section of an osteochondroma with its surface cap of disorganized cartilage. (C, From Czerniak B. *Dorfman and Czerniak's Bone Tumors*. 2nd ed. Elsevier: 2016.)

MORPHOLOGY

Osteochondromas are sessile or pedunculated and range in size from 1 to 20 cm. The cortex of the stalk merges with the cortex of the host bone, so that the medullary cavity of the osteochondroma is in continuity with that of the bone from which it arises (Fig. 26.21B). On plain radiographs the cap is oriented away from the nearest physis. The cap is composed of benign hyaline cartilage (Fig. 26.21C) and is covered by perichondrium. The cartilage has the histologic appearance of a disorganized growth plate and undergoes endochondral ossification, with newly made bone forming the inner portion of the head and stalk. Osteochondromas in multiple hereditary exostoses syndrome are sometimes more cellular and display more atypia than the sporadic variety.

Clinical Features

Osteochondromas usually stop expanding at the time of growth plate closure and, when symptomatic, are cured by simple excision. Secondary chondrosarcoma occurs rarely, usually in tumors associated with multiple hereditary exostosis.

Chondroma

Chondroma is a benign tumor of hyaline cartilage that occurs in bones of endochondral origin. Tumors can arise within the medullary cavity (*enchondroma*) or on the bone surface (*juxtacortical chondroma*). Enchondroma is the most common cartilage-producing tumor. It typically presents as a solitary metaphyseal lesion of tubular bones of the hands and feet. Radiographs show a

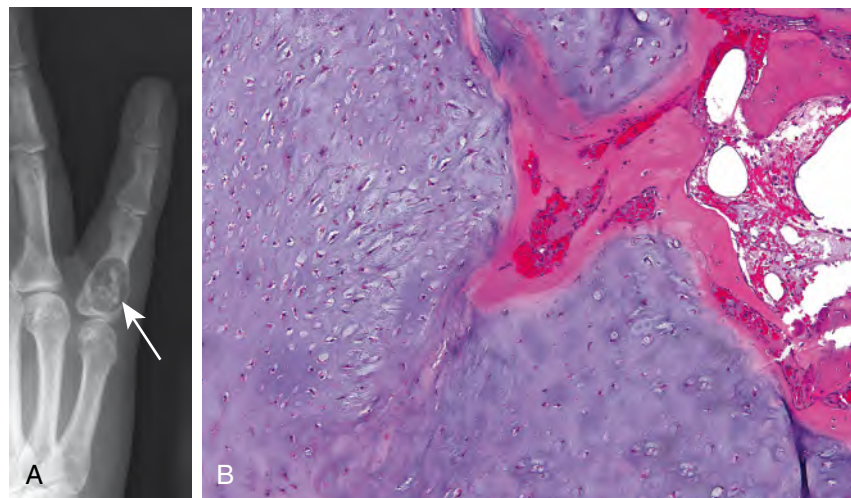


Fig. 26.22 Enchondroma of the proximal phalanx. (A) The radiolucent nodule of cartilage with central calcification (arrow) thins but does not penetrate the cortex. (B) Enchondroma is composed of a nodule of hyaline cartilage encased by a thin layer of reactive bone.

circumscribed lucent lesion with central irregular calcifications, a sclerotic rim, and an intact cortex (Fig. 26.22A). *Ollier disease* and *Maffucci syndrome* are nonhereditary disorders characterized by multiple enchondromas (*enchondromatosis*). Maffucci syndrome is distinguished from Ollier disease by also being associated with spindle cell hemangiomas and other noncartilage neoplasms.

Enchondromas are most often diagnosed between 20 and 50 years of age. When they involve large bones, they are usually asymptomatic and detected incidentally. Occasionally, they may be painful or cause pathologic fractures. In enchondromatosis, the tumors may be numerous and large, producing severe deformities.

Pathogenesis

Heterozygous mutations in the *IDH1* and *IDH2* genes, which encode two isoforms of the isocitrate dehydrogenase enzyme, are present in most enchondromas. Individuals with enchondromatosis syndromes are genetically mosaic, harboring *IDH* mutations in a subset of otherwise normal cells throughout their bodies. Similarly, *IDH* mutations are present in only a subset of tumor cells in syndromic and sporadic enchondromas. The *IDH* mutations cause the encoded proteins to acquire a new enzymatic activity that leads to the synthesis of 2-hydroxyglutarate. As discussed in Chapter 7, this “oncometabolite” interferes with regulation of DNA methylation, an effect that likely alters the expression of several genes that regulate differentiation, cell growth, and survival. Of note, excessive 2-hydroxyglutarate promotes differentiation of mesenchymal stem cells toward the chondrocyte lineage.

MORPHOLOGY

Enchondromas are well-circumscribed nodules that are usually smaller than 3 cm, gray-blue, and translucent. Histologically, enchondromas are composed of hyaline cartilage containing cytologically benign chondrocytes (Fig. 26.22B). Peripheral endochondral ossification may occur, and the center can calcify and infarct. The enchondromas in Ollier disease and Maffucci syndrome are sometimes more cellular and display more atypia than sporadic enchondromas.

Clinical Features

The growth potential of enchondromas is limited. Treatment depends on the clinical situation and if necessary, usually consists of curettage. Solitary chondromas rarely undergo sarcomatous transformation, but those associated with enchondromatosis do so more frequently. Individuals with Maffucci syndrome develop multiple enchondromas and spindle cell hemangiomas and are also at risk of developing other malignancies, including brain gliomas, another type of cancer associated with *IDH* gene mutations (Chapter 28).

Chondrosarcoma

Chondrosarcomas are malignant tumors that produce cartilage.

After osteosarcoma, chondrosarcoma is the second most common malignant matrix-producing tumor of bone. These tumors are subclassified into hyaline cartilage-producing (*conventional*), *clear cell*, *dedifferentiated*, and *mesenchymal* subtypes.

Conventional chondrosarcomas represent approximately 90% of all chondrosarcomas and arise in individuals in their fourth decade or older. Men are affected twice as frequently as women. Conventional chondrosarcomas commonly arise in the axial skeleton, especially the pelvis, shoulder, and ribs, but unlike benign enchondroma, distal extremities are rarely involved. About 15% of conventional chondrosarcomas are secondary and arise from preexisting enchondromas or osteochondromas. On imaging, the calcified matrix of chondrosarcomas appears as foci of flocculent densities. Slow-growing low-grade tumors cause reactive cortical thickening, whereas aggressive high-grade neoplasms destroy the cortex and form soft tissue masses (Fig. 26.23A).

The clear cell and mesenchymal variants occur in younger people in their teens or 20s. The clear cell variant is unique in that it originates in the epiphyses of long tubular bones.

Pathogenesis

Although chondrosarcomas are genetically heterogeneous, a few recurrent driver mutations are recognized. Chondrosarcomas arising in multiple osteochondroma syndrome exhibit mutations in the *EXT1* or *EXT2* genes that regulate the synthesis of cartilaginous matrix proteins. Both chondromatosis-related and sporadic chondrosarcomas may have *IDH1* or *IDH2* mutations. Silencing of the *CDKN2A* tumor suppressor locus by DNA methylation is common in sporadic tumors.

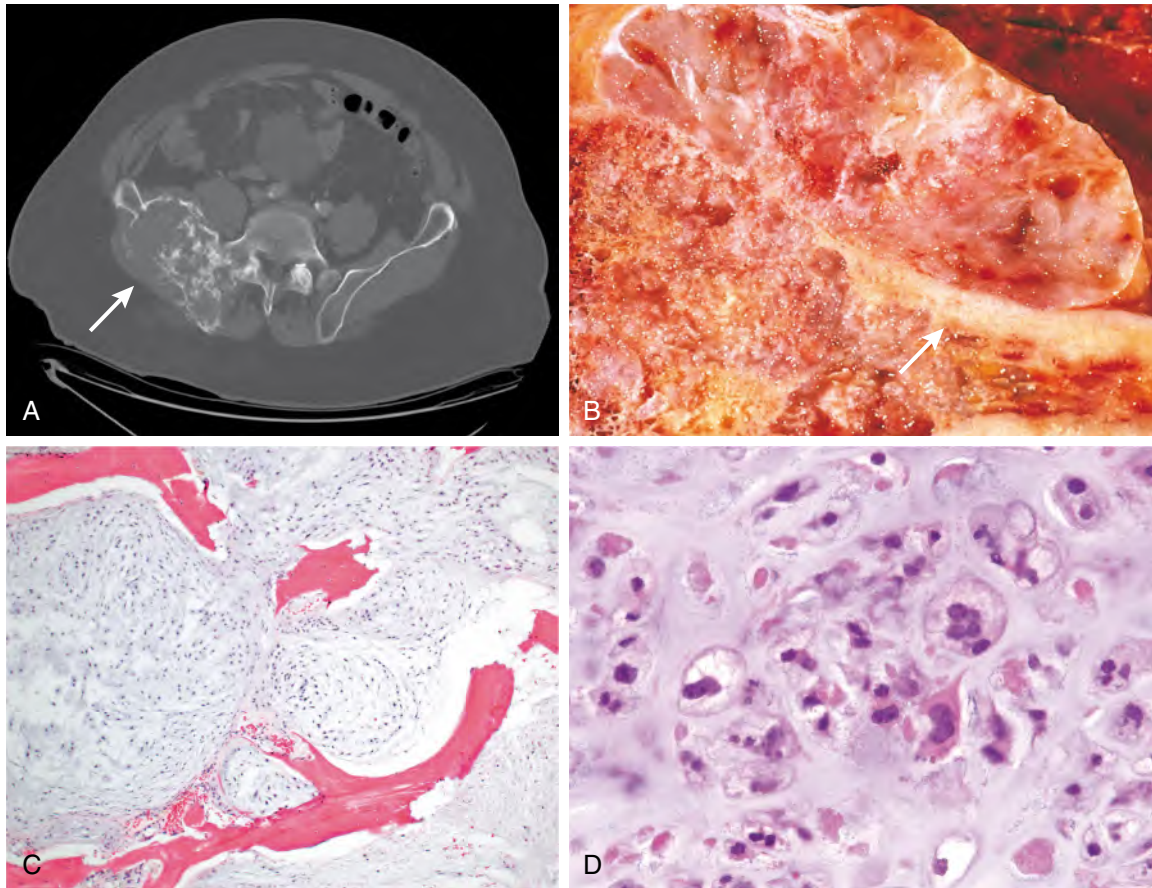


Fig. 26.23 Chondrosarcoma. (A) CT image of the pelvis shows a large chondrosarcoma of the right ilium disrupting the cortex and extending into soft tissue (arrow). (B) Nodules of hyaline and myxoid cartilage permeate the medullary cavity, grow through the cortex, and form a relatively well-circumscribed soft tissue mass (arrow). (C) Conventional chondrosarcoma appearing as a hypercellular cartilaginous mass with entrapped normal bone trabeculae. (D) Anaplastic chondrocytes amid hyaline cartilage matrix in a grade 3 chondrosarcoma.

MORPHOLOGY

Conventional chondrosarcomas are bulky tumors made up of nodules of glistening gray-white, translucent cartilage, but the matrix is often gelatinous or myxoid and can ooze from the cut surface (Fig. 26.23B). Focal calcifications may be present and central necrosis may create cystic spaces. Extracortical extension into surrounding muscle and fat is common. Histologically, the cartilage infiltrates the marrow space and surrounds preexisting bony trabeculae (Fig. 26.23C). Chondrosarcomas vary in cellularity, degree of cytologic atypia, and mitotic activity and on this basis are assigned a grade from 1 to 3 that correlates well with prognosis. Grade 1 tumors have low cellularity and contain neoplastic chondrocytes with plump vesicular nuclei and small nucleoli. By contrast, grade 3 chondrosarcomas show high cellularity, extreme pleomorphism and frequent mitotic figures (Fig. 26.23D).

Dedifferentiated chondrosarcoma consists of a low-grade chondrosarcoma with a second, high-grade component that does not produce cartilage.

Clear-cell chondrosarcoma contains sheets of large, atypical chondrocytes that have abundant clear cytoplasm, numerous osteoclast-type giant cells, and intralesional bone formation. **Mesenchymal chondrosarcoma** is composed of islands of well-differentiated hyaline cartilage surrounded by sheets of primitive appearing small round cells.

Clinical Features

Chondrosarcomas typically present as painful, progressively enlarging masses. The histologic grade and anatomic site correlate with biologic behavior. Grade 1 tumors rarely metastasize, locally recur in approximately 10% of cases, and have 5-year survival rates of close to 100%. Mortality derives from local recurrence, which is most problematic in the limb girdles and axial skeleton. Accordingly, grade 1 tumors in the long bones of the extremities have a good prognosis. Consequently, the diagnosis *atypical cartilaginous tumor* has been proposed for grade 1 chondrosarcomas of the extremities. By contrast, most grade 3 tumors spread hematogenously, especially to the lungs, and 5-year survival is less than 50%. Treatment ranges from curettage in select cases of grade 1 chondrosarcoma to wide surgical excision of higher grade or nonappendicular skeleton tumors.

Other Bone Tumors of Mesenchymal Origin

Ewing Sarcoma

Ewing sarcoma is a malignant bone tumor that is strongly associated with oncogenic fusion genes composed of portions of the *EWSR1* gene and genes encoding ETS family transcription factors. Ewing sarcoma accounts for approximately 10% of primary malignant bone tumors. While it is the second most common bone sarcoma in

children after osteosarcoma. It is a rare tumor; about 200 cases are diagnosed in the United States each year. Approximately 80% of patients are younger than 20 years. Boys are affected slightly more often than girls.

Ewing sarcoma usually arises in the diaphysis of long tubular bones, especially the femur and the flat bones of the pelvis. However, approximately 20% of cases occur in extraskelatal locations. It typically presents as a painful, enlarging mass. The affected site is frequently tender, warm, and swollen, and there may be systemic findings that mimic infection, including fever, elevated red cell sedimentation rate, anemia, and leukocytosis. Radiographs show a destructive lytic tumor with permeative or moth-eaten margins that extend into surrounding soft tissues. As the tumor advances radially into soft tissue, layers of reactive bone deposit, producing a characteristic *onion-skin periosteal reaction*.

Pathogenesis

Nearly all (>90%) of Ewing sarcomas contain a balanced translocation involving the *EWSR1* gene on chromosome 22 and the *FLI1* gene on chromosome 11. *EWSR1* and *FLI1* are members of the FET and ETS family of transcription factors, respectively. The fusion gene encodes a chimeric *EWSR1/FLI1* protein that binds to chromatin and dysregulates transcription. A minority of Ewing sarcomas (<10%) have fusion genes involving other members of the FET or ETS gene families that are believed to act in an analogous fashion. The cell of origin is not certain; mesenchymal stem cells and primitive neuroectodermal cells are possible sources.

MORPHOLOGY

Ewing sarcoma usually arises in the medullary cavity and invades the cortex, periosteum, and soft tissue. Grossly, the tumor is soft, tan-white, and frequently contains areas of hemorrhage and necrosis. Microscopically, it is composed of sheets of uniform small, round cells that are more cohesive than lymphocytes (Fig. 26.24). They have scant cytoplasm that may be clear because of the presence of glycogen. **Homer-Wright rosettes**, circular groups of cells with a central fibrillary core, may be present. Fibrous septa may be seen, but the tumor cells do not produce bone or cartilage.

Clinical Features

Although Ewing sarcoma is aggressive, neoadjuvant chemotherapy followed by surgical excision and adjuvant chemotherapy, with or without radiation, has achieved 75% 5-year survival and 50% long-term cure

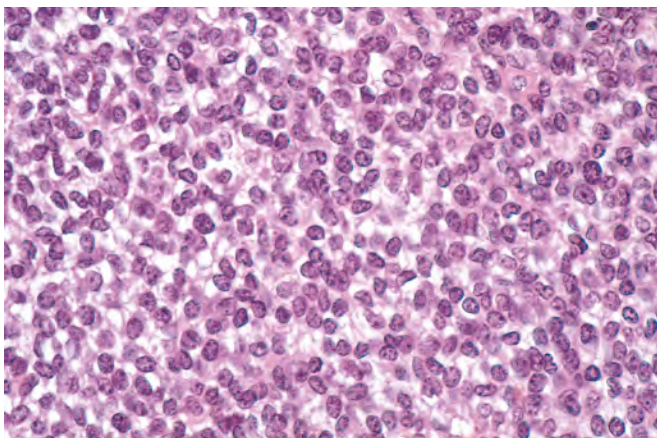


Fig. 26.24 Ewing sarcoma composed of sheets of uniform, small round cells with minimal clear cytoplasm.

rates. Neoadjuvant chemotherapy-induced necrosis is a positive prognostic indicator.

Giant Cell Tumor

Giant cell tumor is a neoplasm composed of monotonous mononuclear stromal cells, macrophages, and osteoclast-type giant cells. This uncommon tumor usually arises in the third through fifth decades of life, typically in the epiphyses of long bones. While benign, they can be locally aggressive.

Pathogenesis

Most of the cells within giant cell tumors are nonneoplastic osteoclasts and their precursors. The neoplastic cells are primitive osteoblast precursors that express high levels of RANKL, which promotes the proliferation of osteoclast precursors and their differentiation into mature osteoclasts. The absence of normal feedback between osteoblasts and osteoclasts results in localized, but highly destructive, bone resorption. The neoplastic cells have acquired mutations in the *H3-3A* gene encoding histone 3.3, a chromatin packaging protein; precisely how this leads to tumor formation is unknown.

Giant cell tumors develop within the epiphysis and may extend into the metaphysis. The majority occur near the knee in the distal femur or proximal tibia, but virtually any bone can be involved. Because they typically arise near joints, giant cell tumors may cause arthritis-like symptoms. They may also present with pathologic fractures.

MORPHOLOGY

Giant cell tumors often destroy the overlying cortex, resulting in a bulging soft tissue mass delineated by a thin shell of reactive bone (Fig. 26.25A). Grossly, the tumors are large, red-brown masses that frequently undergo cystic degeneration. Histologically, the tumors consist of uniform oval mononuclear tumor cells and abundant osteoclast-type giant cells with 100 or more nuclei (Fig. 26.25B). Necrosis and mitotic activity may be prominent.

Clinical Features

Giant cell tumors are treated by curettage, but 40% to 60% recur locally. Although up to 4% metastasize to the lungs, these regress spontaneously or can be cured by excision. The RANKL inhibitor denosumab has shown promise as an alternative to surgery in cases in which resection would lead to significant morbidity.

Aneurysmal Bone Cyst

Aneurysmal bone cyst (ABC) is a benign neoplasm characterized by multiloculated blood-filled spaces. All age groups are affected, but most cases present in adolescence. ABC develops most frequently in the femur, tibia, and vertebral body posterior elements (lamina and pedicles).

Pathogenesis

The spindle-shaped tumor cells of ABC are of uncertain origin and demonstrate rearrangements of chromosome 17p13 that result in fusion of the coding region of the *USP6* gene with the regulatory regions of numerous other genes, most commonly *CDH11*, leading to overexpression of USP6 protein. USP6 is a deubiquitinating enzyme that enhances the activity of the transcription factor NF- κ B. This in turn up-regulates expression of proteins such as matrix metalloproteases, leading to cystic bone resorption.

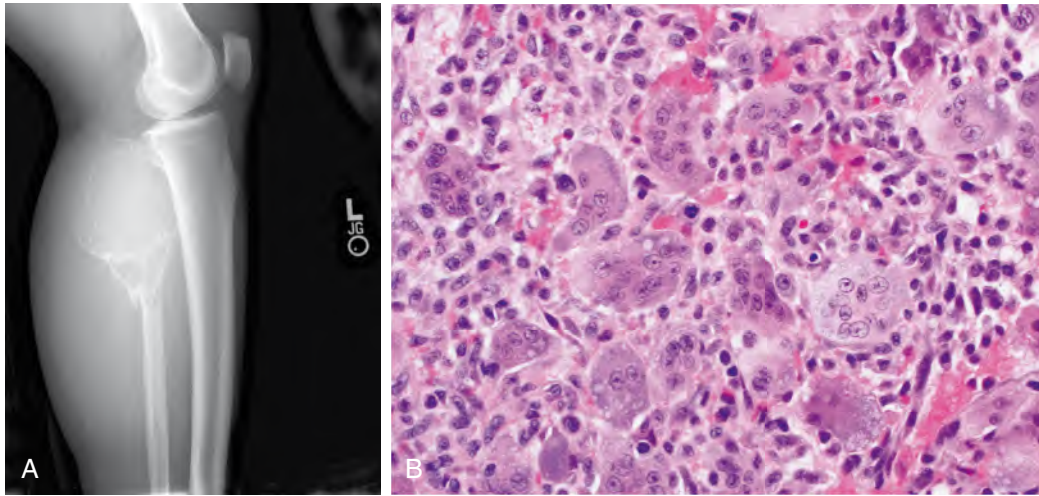


Fig. 26.25 Giant cell tumor of bone. (A) Giant cell tumor of the proximal fibula is predominantly lytic and expansile with destruction of the cortex. A pathologic fracture is also present. (B) Giant cell tumor illustrating an abundance of osteoclast-type multinucleated giant cells with background mononuclear stromal cells.

MORPHOLOGY

Radiographically, ABC is usually an eccentric, expansile, well-circumscribed lytic lesion with well-defined margins. Most show central lysis and a thin sclerotic “eggshell” of reactive bone at the periphery. Computed tomography and magnetic resonance imaging may demonstrate internal septa and characteristic fluid-fluid levels (Fig. 26.26A). Grossly, ABCs consist of multiple blood-filled cystic spaces separated by thin, tan-white septa (Fig. 26.26B). Microscopically, the septa are composed of plump spindle cells, multinucleated osteoclast-like giant cells, and reactive woven bone lined by osteoblasts (Fig. 26.27). The septa lack an endothelial lining and so are not true aneurysms. Bone deposition typically follows the contours of fibrous septa. Approximately one-third of cases contain an unusual densely calcified, basophilic metaplastic matrix referred to as **blue bone**. Necrosis is uncommon unless a pathologic fracture is present.

Clinical Features

ABC presents with localized pain and swelling. Vertebral lesions may produce nerve compression. Despite being benign, ABC is locally aggressive. Treatment is by curettage or excision. Recurrence occurs in 10% to 50% of cases.

Nonossifying Fibroma

Nonossifying fibroma is a benign spindle cell tumor with osteoclast-type giant cells. These lesions are present in up to 50% of children older than 2 years of age and typically present as an incidental finding in adolescents. The vast majority arise eccentrically in the metaphysis of the distal femur and proximal tibia. Multifocal nonossifying fibromas may occur in the setting of *neurofibromatosis type 1* (NF1) and *Jaffe-Campanacci syndrome*, which may represent a subtype of NF1. Most are less than 0.5 cm in diameter, but they occasionally may grow to 5 cm or larger in size.

Pathogenesis

Activating mutations in *KRAS* and *FGFR1* genes have been identified in sporadic tumors, whereas mutations in *NF1* are present in NF1 and Jaffe-Campanacci patients with nonossifying fibromas. These findings indicate that nonossifying fibromas are true neoplasms, albeit with limited growth potential, that are driven by activation of the RAS pathway.

MORPHOLOGY

Nonossifying fibromas are sharply demarcated radiolucent masses surrounded by a thin rim of sclerosis (Fig. 26.28A). Grossly, they are grey to yellow-brown and lack mineralization. Microscopically, they consist of bland fibroblasts that are frequently arranged in a **storiform (pinwheel) pattern** and macrophages that can take the form of clustered cells with foamy cytoplasm or multinucleated giant cells (Fig. 26.28B). Hemosiderin is commonly present.

Clinical Features

Nonossifying fibromas are asymptomatic and are detected incidentally on radiographic studies. The radiographic findings are sufficiently specific that biopsy is rarely necessary. Most small lesions have limited growth potential and undergo spontaneous resolution over several years as they are replaced by normal cortical bone. The few that progressively enlarge may present with pathologic fracture and require biopsy to exclude other types of tumors. They are treated with curettage and may require bone grafting.

Fibrous Dysplasia

Fibrous dysplasia is a benign neoplasm of bone characterized by disorganized and inadequately mineralized bone and intervening fibrous tissue. The lesions arise during skeletal development and may be sporadic or syndromic. They present in several ways, as follows:

- **Monostotic:** involvement of a single bone
- **Polyostotic:** involvement of multiple bones
- **Mazabraud syndrome:** fibrous dysplasia and soft tissue myxomas
- **McCune-Albright syndrome:** polyostotic fibrous dysplasia, café-au-lait skin pigmentations, and endocrine abnormalities, especially precocious puberty

Pathogenesis

All forms of fibrous dysplasia result from somatic gain-of-function mutations in the gene *GNAS1* during fetal development. *GNAS1* encodes the stimulatory α subunit of the GTP-binding protein G_s , and is also mutated in pituitary neuroendocrine tumors (Chapter 24). The mutations produce constitutive activation of G_s , which leads to increased intracellular levels of the signaling molecule cyclic adenosine monophosphate (cAMP). This in turn alters the activity of

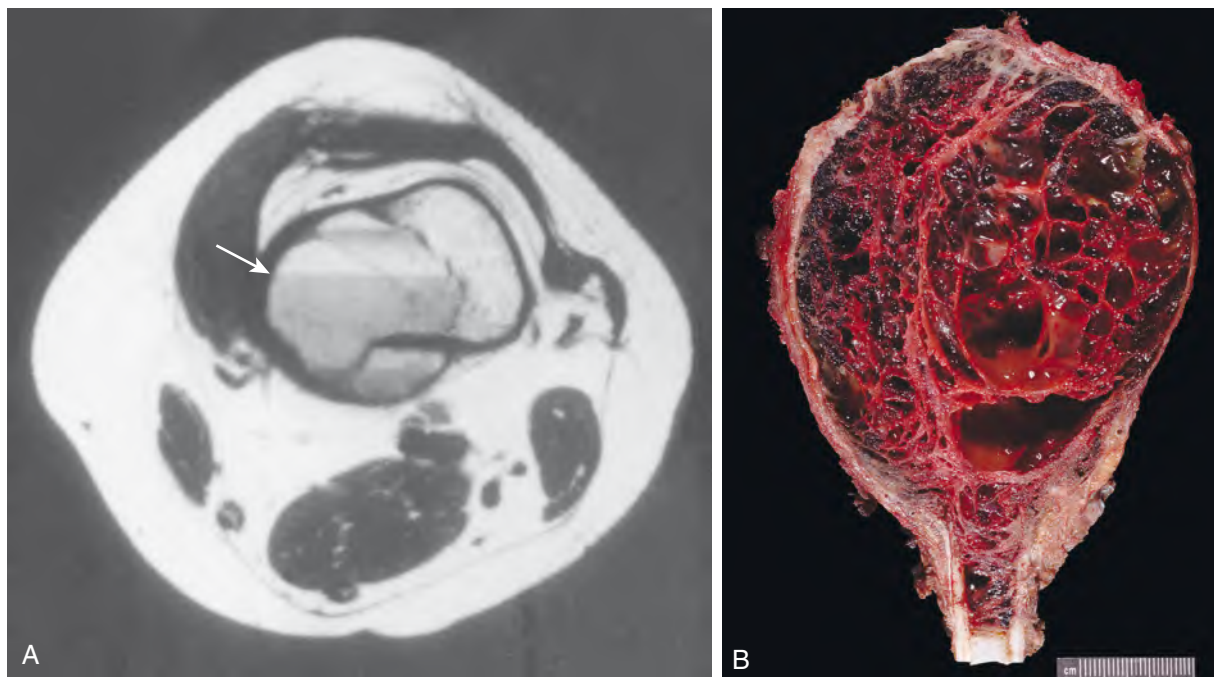


Fig. 26.26 Aneurysmal bone cyst. (A) Axial magnetic resonance image demonstrating characteristic fluid-fluid levels (arrow). (B) Grossly, the lesion appears hemorrhagic and spongelioid in this bisected portion of proximal fibula.

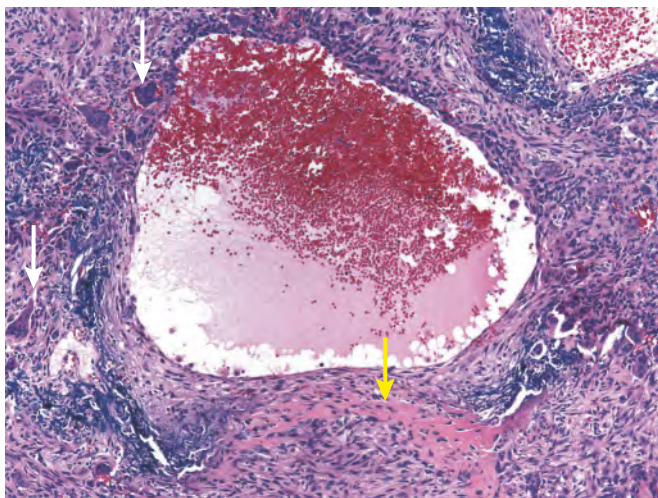


Fig. 26.27 Aneurysmal bone cyst with blood-filled cystic space surrounded by a fibrous wall containing proliferating fibroblasts, reactive woven bone (yellow arrow), and osteoclast-type giant cells (white arrows).

several signaling pathways including the Wnt pathway, thereby promoting cellular proliferation and inhibiting osteoblast differentiation. The phenotype depends on (1) the stage of embryogenesis when the mutation is acquired and (2) the fate of the cell harboring the mutation. At one extreme, mutation during very early embryogenesis produces McCune-Albright syndrome. By contrast, mutation in an osteoblast precursor during or after formation of the skeleton results in monostotic fibrous dysplasia.

MORPHOLOGY

Fibrous dysplasia is an intramedullary lytic lesion that may expand, causing bowing and cortical thinning. Periosteal reaction usually is absent. Grossly, lesional tissue is tan-white and gritty. Microscopically, the tumor is composed of curvilinear trabeculae of woven bone without prominent osteoblastic rimming surrounded by a moderately cellular fibroblastic proliferation and loose collagen (Fig. 26.29). Cystic degeneration, hemorrhage, and foamy macrophages are also commonly present.

Clinical Features

Monostotic fibrous dysplasia appears in early adolescence and often stops enlarging at the time of growth plate closure. The femur, tibia, ribs, jawbones, and calvarium are the most commonly affected bones. The lesion is frequently asymptomatic and usually discovered incidentally but may cause pain and pathologic fracture. Symptomatic lesions are treated by curettage, but recurrence is common.

Polyostotic fibrous dysplasia manifests at a slightly earlier age than the monostotic type and may continue to cause problems into adulthood. The femur, tibia, and skull are most frequently affected, and craniofacial involvement is common. Involvement of the shoulder and pelvic girdles may result in progressive disease that produces deformities and fractures that require surgical repair. Bisphosphonates can be used to reduce the severity of the bone pain. A rare complication, usually in the setting of polyostotic fibrous dysplasia, is malignant transformation of a lesion into a sarcoma.

Mazabraud syndrome presents with skeletal features of polyostotic fibrous dysplasia in childhood followed by the appearance of intramuscular myxomas in adulthood, often in the same anatomic region as existing fibrous dysplasia. Although benign, myxomas can cause local compression symptoms or limb deformity; they are cured by surgical excision.

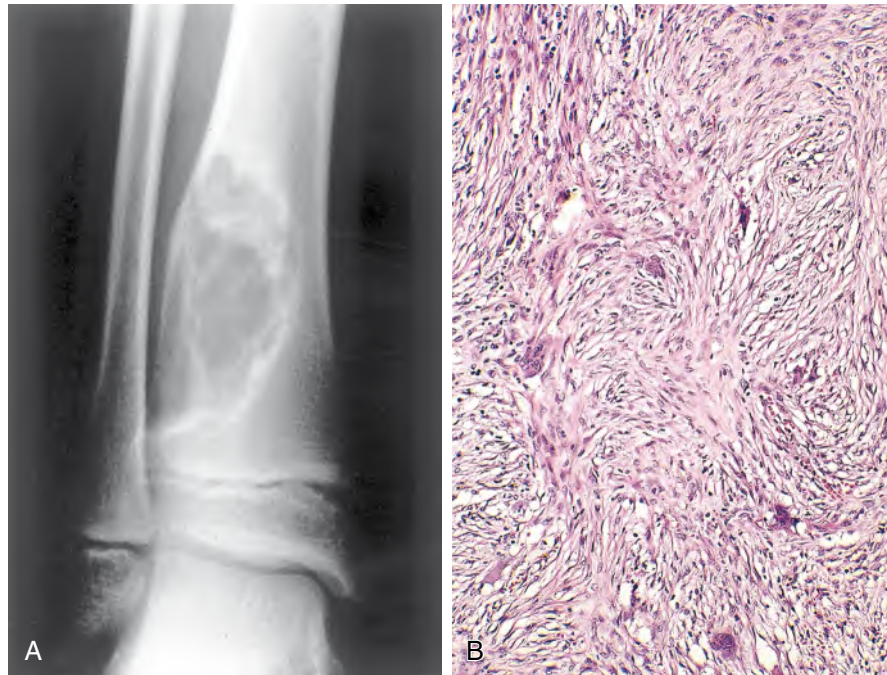


Fig. 26.28 Non-ossifying fibroma. (A) Plain radiograph of the distal leg and ankle demonstrates a metaphyseal eccentric, lobulated radiolucency surrounded by a sclerotic margin. (B) Storiform pattern created by benign spindle cells with scattered osteoclast-type giant cells.

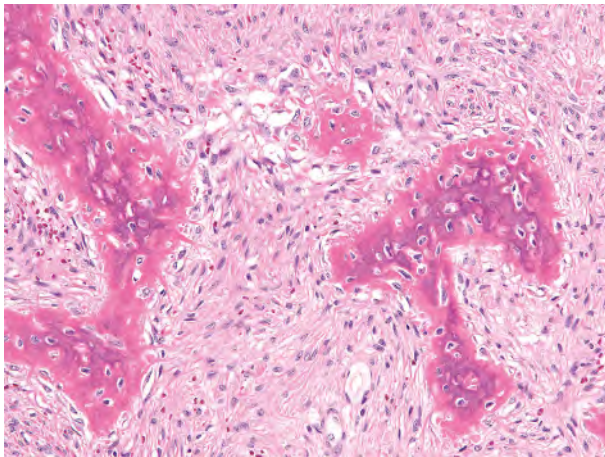


Fig. 26.29 Fibrous dysplasia composed of curvilinear trabeculae of woven bone that lack conspicuous osteoblastic rimming and arise in a background of fibrous tissue.

The most common clinical presentation of McCune-Albright syndrome is precocious sexual development, primarily in girls. The syndrome can include other endocrinopathies such as hyperthyroidism, pituitary neuroendocrine tumors that secrete growth hormone, and primary adrenal hyperplasia (see Chapter 24). Bone lesions are often unilateral and skin lesions, when present, are limited to the same side of the body. Skeletal manifestations are managed as for polyostotic fibrous dysplasia, while the endocrinopathies are treated medically (e.g., with aromatase inhibitors for precocious puberty).

Metastatic Tumors

Cancers of extraosseous origin are the most common form of malignancy to involve the skeleton, greatly outnumbering primary

bone cancers. The pathways of spread to bone include (1) direct extension, (2) lymphatic or hematogenous dissemination, and (3) intraspinal seeding (via the Batson plexus of veins). Any cancer can spread to bone, but in adults more than 75% of skeletal metastases originate from carcinomas of the prostate, breast, kidney, and lung. In children, neuroblastoma, Wilms tumor, osteosarcoma, Ewing sarcoma, and rhabdomyosarcoma most commonly produce metastases to bone. Skeletal metastases are typically multifocal and tend to involve the axial skeleton, especially the vertebral column. However, carcinomas of the kidney and thyroid may present as solitary skeletal metastases.

The radiographic appearance of metastases may be purely *lytic* (bone destroying), purely *blastic* (bone forming), or *mixed* (lytic and blastic). Some carcinomas, for example prostatic adenocarcinoma, produce predominantly blastic metastases, whereas kidney and breast carcinomas produce lytic lesions. Bidirectional interactions between metastatic cancer cells and native bone cells account for these varied features. Tumor cells do not directly resorb bone in lytic lesions, but they secrete substances such as prostaglandins, cytokines, and PTH-related peptide that stimulate osteoclast activity. Conversely, as bone is resorbed, matrix-bound growth factors (e.g., TGF- β , IGF-1, and FGF) are released that support the growth of metastatic tumor cells. Blastic metastases may result from tumor cells secreting Wnt proteins that stimulate osteoblastic bone formation.

The presence of bone metastases generally portends a poor prognosis because it indicates wide disease dissemination, but metastatic breast and prostate cancers can sometimes be managed for many years with hormone therapy. For other cancers, treatment options include systemic chemotherapy or immunotherapy, localized radiation, and bisphosphonates. Surgery may be necessary to stabilize pathologic fractures.

KEY CONCEPTS

Bone Tumors

Most bone tumors are classified according to the normal cell they resemble or the matrix they produce; the remainder are grouped by clinicopathologic features. Most primary bone tumors are benign. Metastases, especially adenocarcinomas, are more common than primary bone neoplasms.

Major categories of primary bone tumors include:

- Bone forming: Osteoblastoma and osteoid osteoma consist of benign osteoblasts that synthesize osteoid. Osteosarcoma is a tumor of malignant osteoblasts with an aggressive clinical course that predominantly involves adolescents.
- Cartilage forming: Osteochondroma is an exostosis with a cartilage cap. Syndromic forms are most often associated with germline mutations in *EXT* genes. Chondromas are benign intramedullary tumors that produce hyaline cartilage and usually arise in the digits. Chondrosarcomas are malignant tumors of cartilage involving the axial skeleton in adults.
- Ewing sarcoma is an aggressive malignant small round cell tumor most often associated with fusion of the *EWSR1* and *FLI1* genes.
- Nonossifying fibroma and fibrous dysplasia are benign osseous neoplasms driven by activation of the RAS pathway and gain-of-function mutations in the *GNAS1* gene, respectively.

JOINTS

Joints allow movement while providing mechanical stability. They are classified as solid (nonsynovial) and caviated (synovial). The solid joints, or *synarthroses*, provide structural integrity, lack a joint space, and allow only minimal movement. They are grouped according to the type of connective tissue (fibrous tissue or cartilage) that bridges the ends of the bones. Fibrous synarthroses include the cranial sutures and the bonds between roots of teeth and the jawbones. Cartilage synarthroses include the sternocostal joints and pubic symphysis. Synovial joints, by contrast, have a joint space that allows a wide range of motion. Synovial membranes enclose the joints and are firmly anchored to an overlying fibrous capsule reinforced by ligaments and muscle. Synovial membranes are lined by two types of cells arranged in layers. *Type A synoviocytes* are specialized macrophages with phagocytic activity. *Type B synoviocytes* are similar to fibroblasts and synthesize hyaluronic acid and various proteins. The synovial lining lacks a basement membrane, which allows for efficient exchange of nutrients, waste, and gases between blood and synovial fluid, a hyaluronic acid–rich plasma filtrate that acts as a viscous lubricant and provides nutrition for the articular cartilage.

Hyaline cartilage is a unique connective tissue ideally suited to serve as an elastic shock absorber and wear-resistant surface. It is composed of water (70%), type II collagen (10%), proteoglycans (8%), and chondrocytes. The collagen fibers enable resistance to tensile stresses and transmit vertical loads; water and proteoglycans limit compression and friction. Hyaline cartilage lacks blood vessels, lymphatics, and nerves. Chondrocytes synthesize the matrix and secrete enzymes that remodel it. The activity of these enzymes is held in check by inhibitors that are also secreted by chondrocytes. Diseases that destroy articular cartilage alter this balance by increasing the activity of degradative enzymes and decreasing the production of their inhibitors. Among the factors implicated in triggering the degradative process are cytokines such as IL-1 and TNF, which may be released by inflammatory cells and chondrocytes, synoviocytes, fibroblasts, and inflammatory cells.

ARTHRITIS

Arthritis is inflammation of joints. The most important forms of arthritis are osteoarthritis and rheumatoid arthritis, which differ in pathogenesis and clinical and pathologic manifestations (discussed later). Other types of arthritis are caused by immune reactions, infections, and crystal deposition.

Osteoarthritis

Osteoarthritis (OA), also called degenerative joint disease, is characterized by cartilage degeneration that results in structural and functional failure of synovial joints. It is the most common type of joint disease. Although the term osteoarthritis implies an inflammatory disease, it is primarily a degenerative disease of articular cartilage, with inflammation acting as a secondary contributor.

In most instances, OA appears insidiously, without apparent initiating cause, as an aging phenomenon (*idiopathic* or *primary osteoarthritis*). In these cases, the disease usually affects few joints (oligoarticular) but may be generalized. In about 5% of cases, OA appears in younger individuals with predisposing conditions, such as a joint deformity, a previous joint injury, or an underlying systemic disease (diabetes, obesity, ochronosis) that places joints at risk. In these settings, the disease is called *secondary osteoarthritis*. The association with aging is strong; the prevalence of OA increases exponentially beyond 50 years of age, and about 40% of people older than 70 years are affected.

Pathogenesis

The lesions of OA stem from degeneration of articular cartilage and disordered repair. Biomechanical stress is the principal pathogenic mechanism, but genetic factors, including polymorphisms in genes encoding components of the matrix and signaling molecules, may predispose to chondrocyte injury that causes matrix alteration (Fig. 26.30). In the early stages of OA, chondrocytes proliferate (likely in response to matrix loss) and continue to synthesize proteoglycans, but degradation exceeds synthesis. Concurrently, the water content of the matrix increases. Chondrocytes also secrete MMPs that degrade the type II collagen network. The normal horizontally arranged collagen fibers are cleaved, producing fissures and clefts at the articular surface. Cytokines and diffusible factors from chondrocytes and synovial cells, particularly TGF- β (which induces the production of MMPs), IL-1, IL-6, prostaglandins, and nitric oxide, are also implicated in OA, and chronic low-level inflammation contributes to disease progression. Bone morphogenetic proteins and TGF- β appear to play a central role in the development of osteophytes, bony outgrowths at the periphery of the articular surface.

MORPHOLOGY

In early OA, chondrocytes proliferate and form clusters. As type II collagen and proteoglycans are degraded, the resulting fissures and clefts create a granular and soft articular surface (Fig. 26.31A). In advanced OA, sloughed full-thickness portions of cartilage tumble into the joint, forming **loose bodies (joint mice)**. The exposed subchondral bone becomes the new articular surface, which is burnished by friction with the opposing surface, giving it the appearance of polished ivory (**bone eburnation**) (Fig. 26.31B). Small fractures develop in the subarticular bone, creating gaps that allow synovial fluid to be forced into the subchondral regions in a ball valve-like mechanism, leading to fibrous-walled cysts. Mushroom-shaped bony outgrowths (**osteophytes**) develop at the margins of the articular surface and are capped by fibrocartilage and hyaline cartilage that gradually ossify. Advanced OA manifests radiographically as loss of joint space, subchondral sclerotic bone, and osteophytes (Fig. 26.32), findings that are sufficiently diagnostic that biopsy is not required. The synovium is usually only mildly congested and fibrotic and contains few inflammatory cells.

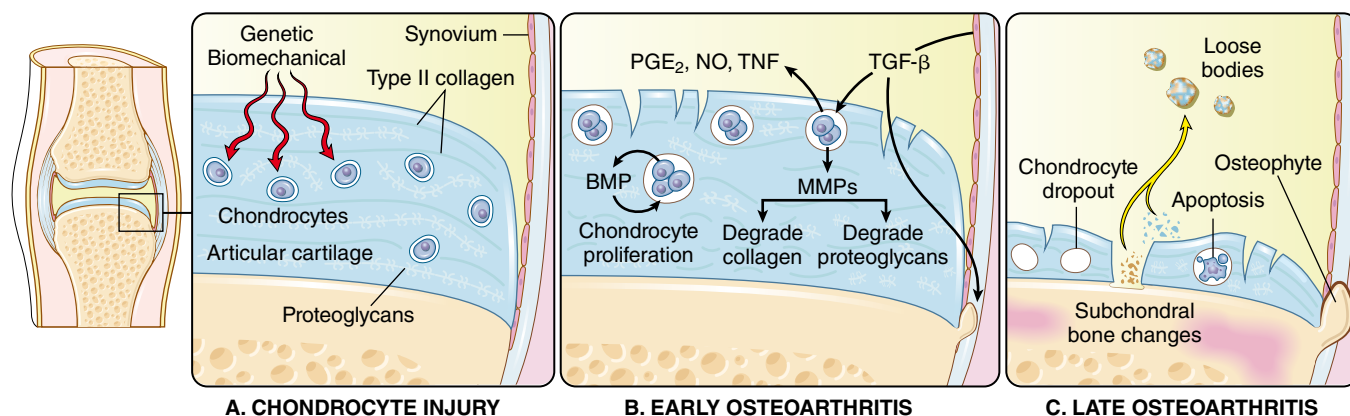


Fig. 26.30 Schematic view of osteoarthritis (OA). (A) OA is thought to be initiated by chondrocyte injury in a genetically predisposed patient leading to changes in the extracellular matrix. (B) Although chondrocytes may proliferate and attempt to repair damaged matrix, continued degradation exceeds repair in early OA. (C) Late OA is evidenced by loss of both matrix and chondrocytes with subchondral bone damage. *BMP*, Bone morphogenetic protein; *MMPs*, matrix metalloproteinases; *NO*, nitric oxide; *PGE₂*, prostaglandin E₂; *TGF-β*, transforming growth factor β; *TNF*, tumor necrosis factor.



Fig. 26.31 Severe osteoarthritis of the hip. The joint space is narrowed, and there is subchondral sclerosis with scattered oval radiolucent cysts and peripheral osteophyte lipping (arrows).

Clinical Features

Primary OA typically presents in individuals age 50 or older. Manifestations of OA in a young patient warrant a search for an underlying cause. Characteristic symptoms include deep joint pain that worsens with use, morning stiffness that improves with activity, crepitus, and limitation of range of movement. Impingement on spinal foramina by osteophytes results in cervical and lumbar nerve root compression and radicular pain, muscle spasms, muscle atrophy, and neurologic deficits. Typically, only one or a few joints are involved, except in the uncommon generalized variant. The joints commonly involved include the

hips, knees, lower lumbar and cervical vertebrae, proximal and distal interphalangeal joints of the fingers, first carpometacarpal joints, and first tarsometatarsal joints. *Heberden nodes*, prominent osteophytes at the distal interphalangeal joints, are common in females. Joint deformities may develop but unlike rheumatoid arthritis (discussed next), joint fusion does not occur (Fig. 26.33). The level of disease severity detected radiographically does not correlate well with pain and disability. Treatment includes pain management, antiinflammatory drugs, intraarticular corticosteroid injection, activity modification, and (in severe cases) arthroplasty.

Rheumatoid Arthritis

Rheumatoid arthritis (RA) is a chronic autoimmune disorder that principally attacks the joints, producing a nonsuppurative, proliferative, and inflammatory synovitis. The prevalence of RA in the United States is approximately 0.25% to 1% and its incidence peaks in the third to fourth decades. It is three times more common in females than in males. RA often progresses to articular cartilage destruction and, in some cases, joint fusion (*ankylosis*). Extraarticular lesions may also occur in the skin, heart, blood vessels, and lungs, leading to clinical manifestations that overlap with other autoimmune disorders such as systemic lupus erythematosus and scleroderma.

Pathogenesis

The autoimmune response in RA is initiated by CD4+ helper T cells reacting against a joint antigen. Numerous joint antigens have been proposed as initiators and targets of the pathologic immune response, including collagen and chemically modified peptides. The pathologic changes in RA are the result of inflammation mediated by CD4+ T cells and perhaps antibodies (Fig. 26.34). Implicated inflammatory mediators include:

- *TNF*, *IL-1*, and *IL-6* from macrophages recruit activated leukocytes and stimulate synovial cells to secrete proteases that destroy hyaline cartilage.
- *IL-17* from *Th17* cells recruits neutrophils and monocytes.
- *RANKL* expressed on activated T cells stimulates osteoclasts, resulting in bone resorption.
- *IFN-γ* from *Th1* cells activates macrophages and resident synovial cells.

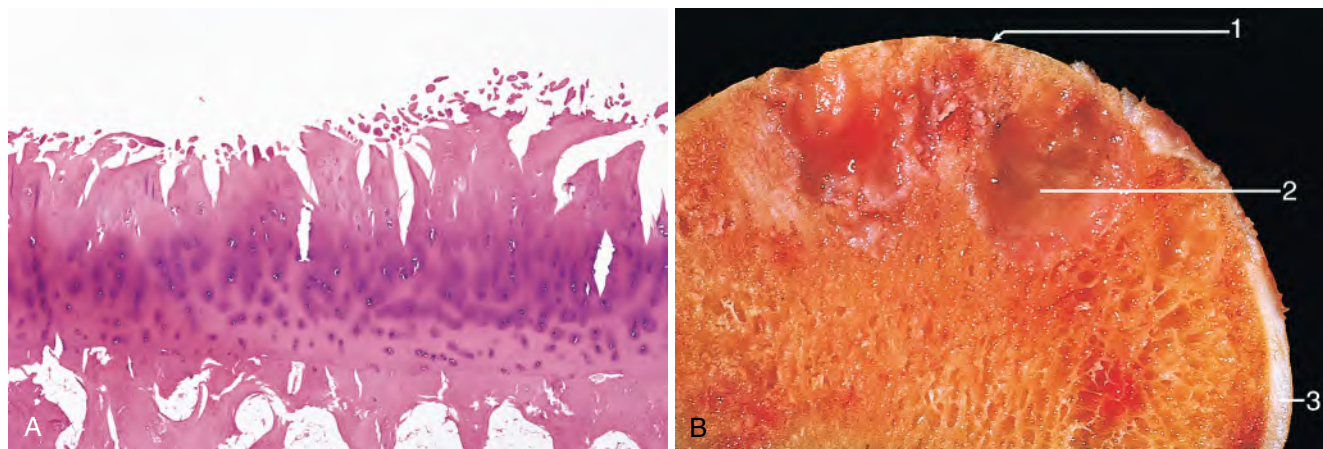


Fig. 26.32 Osteoarthritis. (A) Histologic demonstration of the characteristic fibrillation of articular cartilage. (B) Eburnated articular surface exposing subchondral bone (1), subchondral cysts (2), and residual articular cartilage (3).

Based on the therapeutic effect of TNF antagonists, TNF is believed to be a key mediator of inflammation and joint damage in RA. Antibodies may also contribute to joint damage. The synovium in RA often contains germinal centers with secondary follicles and abundant plasma cells. Some of these plasma cells secrete antibodies that recognize self-antigens. Many of these autoantibodies, which are produced in lymphoid organs as well as synovium, are specific for *citrullinated peptides* (CCPs) in which arginine residues are posttranslationally converted to citrulline. The modified epitopes are present in several proteins found in joints, including fibrinogen, type II collagen, α -enolase, and vimentin. *Anti-citrullinated peptide antibody* (ACPA) is a diagnostic marker that can be detected in the serum of up to 70% of RA patients. Some data suggest that, in combination with T cells reactive to citrullinated proteins, ACPAs contribute to disease severity and persistence. IgM and IgA autoantibodies that bind to the Fc region of IgG are present in 80% of individuals with RA. These autoantibodies, collectively referred to as *rheumatoid factor*, may be deposited in joints as immune complexes, but are not present in all patients and can be detected in some individuals without RA.

As in other autoimmune diseases, genetic predisposition contributes to the development, progression, and persistence of disease. It is estimated that 50% of the risk of developing RA is related to inherited genetic susceptibility. The HLA-DR4 allele is associated with ACPA-positive RA. Evidence suggests that an epitope on citrullinated proteins mimics epitopes on some microbes and can be presented by the class II HLA-DR4 molecule (*molecular mimicry*).

Environmental factors that promote autoimmunity may contribute to the pathogenesis of RA but have not been clearly defined. Perhaps infection (including periodontitis) and smoking promote citrullination of self proteins, creating new epitopes that trigger autoimmune reactions.

MORPHOLOGY

RA typically manifests as a symmetric arthritis of the small joints of the hands and feet. The synovium becomes edematous, thickened, and hyperplastic, transforming its smooth contour to one covered by delicate and bulbous villi (Fig. 26.34). The characteristic histologic features include (1) **synovial cell hyperplasia** and proliferation; (2) **dense inflammatory infiltrates** of CD4+ helper T cells, B cells (frequently forming lymphoid follicles), plasma cells, dendritic cells, and macrophages; (3) increased vascularity due to angiogenesis; (4) fibrin-rich exudate containing neutrophils on the synovial and joint surfaces; and

(5) osteoclastic activity in subchondral bone, allowing the inflamed synovium to penetrate the bone and cause periarticular erosions and subchondral cysts. Together, these changes produce the **pannus**, a mass of edematous synovium, inflammatory cells, granulation tissue, and fibroblasts that grows over and causes erosion of articular cartilage. In time, after the cartilage has been destroyed, the pannus bridges the apposing bones to form a **fibrous ankylosis**, which eventually ossifies and results in bone fusion, or bony ankyloses.

Rheumatoid nodules are an infrequent manifestation of RA and typically occur in the subcutaneous tissue of the forearm, elbows, occiput, and lumbosacral area. These are small, firm, nontender, oval masses. Microscopically, they have a central zone of fibrinoid necrosis surrounded by a prominent rim of activated macrophages and numerous lymphocytes and plasma cells (Fig. 26.35). Severe disease may be associated with **leukocytoclastic vasculitis** (Chapter 11), an acute necrotizing vasculitis of small and large arteries that may involve pleura, pericardium, or lung and evolve into a chronic fibrosing process. Ocular changes such as uveitis and keratoconjunctivitis (similar to Sjögren syndrome, Chapter 6) may also occur.

Clinical Features

RA may be distinguished from other forms of polyarticular inflammatory arthritis serologically by ACPA detection and radiographically by characteristic changes. It begins with malaise, fatigue, and generalized musculoskeletal pain in about half of patients; joint involvement develops after weeks to months. The pattern of joint involvement varies but is generally symmetrical and affects small joints before larger ones. Symptoms usually develop in the hands and feet, followed in decreasing frequency by the wrists, ankles, elbows, and knees. In the hands, the metacarpophalangeal and proximal interphalangeal joints are involved, in contrast to OA (see earlier). While laboratory tests for ACPA and rheumatoid factor aid in diagnosis, about 20% of patients with RA are seronegative; in these cases, diagnosis is based on the clinical and radiographic findings. Tissue biopsy is rarely necessary.

Involved joints are swollen, warm, and painful. Unlike OA, pain is usually worse in the morning or following inactivity. Typically, RA pursues a chronic, waxing and waning course that leads to progressive joint enlargement and decreasing range of motion. In a minority of individuals, especially those lacking ACPA and rheumatoid factor, the disease may stabilize or even regress. A small number of patients

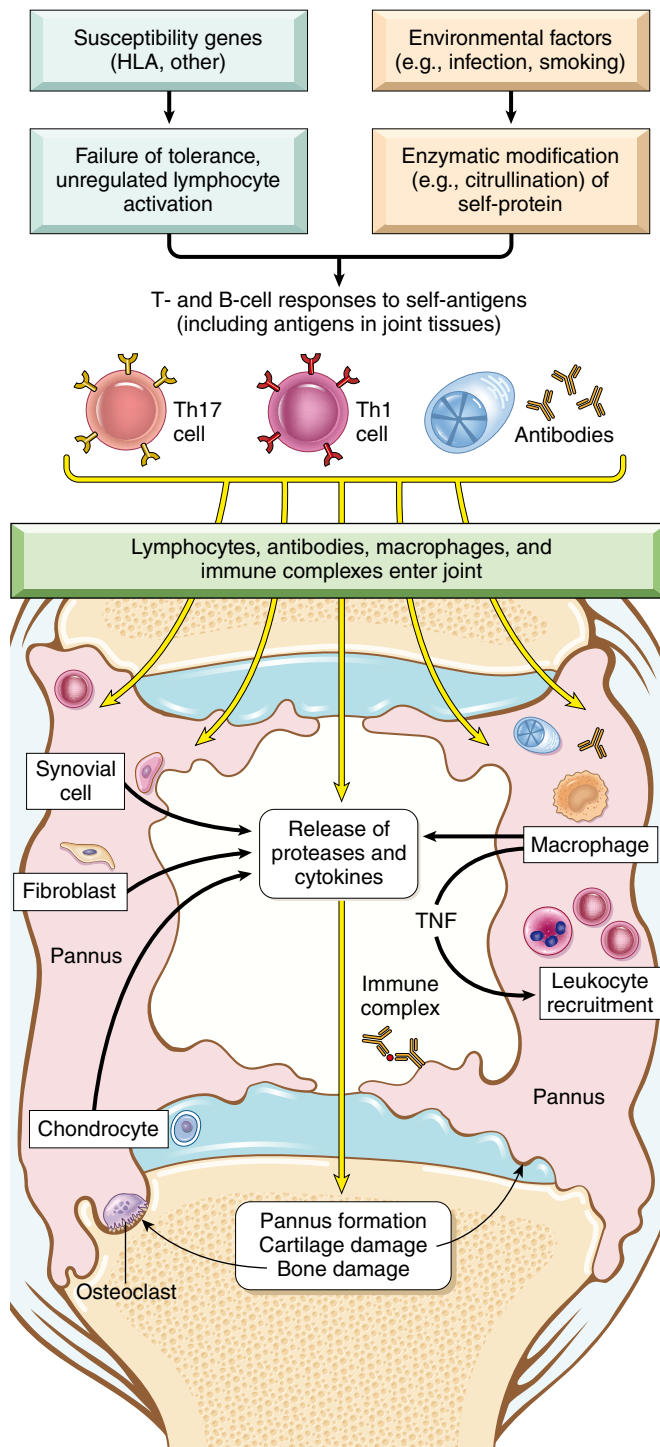


Fig. 26.33 Major processes involved in the pathogenesis of rheumatoid arthritis. *HLA*, Human leukocyte antigen; *TNF*, tumor necrosis factor.

(about 10%) develop acute-onset RA, with severe symptoms and polyarticular involvement that develops over several days.

Inflammation in tendons, ligaments, and occasionally adjacent skeletal muscle frequently accompanies the arthritis and produces characteristic ulnar deviation of the fingers (*swan-neck deformity*) and flexion-hyperextension of the fingers (*boutonnière deformity*) (Fig. 26.36A). Radiographic hallmarks include joint effusions and

juxtaarticular osteopenia, erosions and narrowing of the joint space, and loss of articular cartilage (Fig. 26.36B).

RA treatment is aimed at relieving pain and inflammation and slowing or arresting joint destruction. Therapies include corticosteroids, other immunosuppressants such as methotrexate, and, most notably, TNF antagonists. TNF antagonists are not curative so that patients must be maintained on anti-TNF therapy to avoid disease flares. This long-term treatment with anti-TNF agents predisposes individuals to infection with opportunistic organisms such as *M. tuberculosis*. Other biologic agents that interfere with T- and B-lymphocyte responses have also been approved for therapeutic use.

Comparative Features of Osteoarthritis and Rheumatoid Arthritis

Before turning to less common types of arthritis, it is worth pausing to summarize the salient features that distinguish OA from RA, which differ in terms of their pathogenesis and distribution of disease, as well as their characteristic joint pathology. These distinctions are summarized in Table 26.5 and Fig. 26.37, respectively.

Juvenile Idiopathic Arthritis

Juvenile idiopathic arthritis (JIA) is a heterogeneous group of disorders of unknown cause that present with arthritis before 16 years of age and persist for at least 6 weeks. In the United States, 30,000 to 50,000 individuals are affected. Relative to RA, JIA is more commonly associated with oligoarthritis, systemic disease, and involvement of large joints. Antinuclear antibody (ANA) seropositivity is typical, but rheumatoid nodules are usually absent. JIA pathogenesis is undefined, but it shares features with adult RA, including associations with certain HLA and *PTPN22* gene variants and studies that link its pathogenesis to involvement of Th1 and Th17 cells and inflammatory mediators such as IL-1, IL-17, TNF, and IFN- γ .

Current subclassification of JIA is based on clinical manifestations (e.g., oligoarticular, systemic) and laboratory tests (ANA, rheumatoid factor titers). Treatment of all subgroups is similar to adult RA, and there has been some success using a systemic IL-6 receptor antibody. Long-term prognosis is variable. Many patients have chronic disease, but only about 10% develop serious functional disability.

Seronegative Spondyloarthropathies

Spondyloarthropathies are a heterogeneous group of chronic inflammatory autoimmune disorders that share the following features:

- Absence of rheumatoid factor
- Pathologic changes in the ligamentous attachments (entheses) rather than synovium
- Sacroiliac and vertebral joint involvement
- Association with the HLA-B27 MHC class I allele
- Bony proliferation leading to ankylosis

The disease is likely caused by T-cell responses, presumably directed against as-yet undefined microbial antigens that cross-react with antigens expressed on musculoskeletal system components.

Ankylosing Spondylitis

Ankylosing spondylitis causes destruction of articular cartilage and bony ankyloses, especially of the sacroiliac and vertebral apophyseal joints between tuberosities and processes. The disease presents as lower back pain and spinal immobility, usually in the second and third decades of life. Peripheral joints, such as the hips, knees, and shoulders, are involved in at least one-third of cases. Approximately 90% of patients are HLA-B27 positive. The role of HLA-B27 is unknown; it is presumably related to the ability of this MHC molecule to

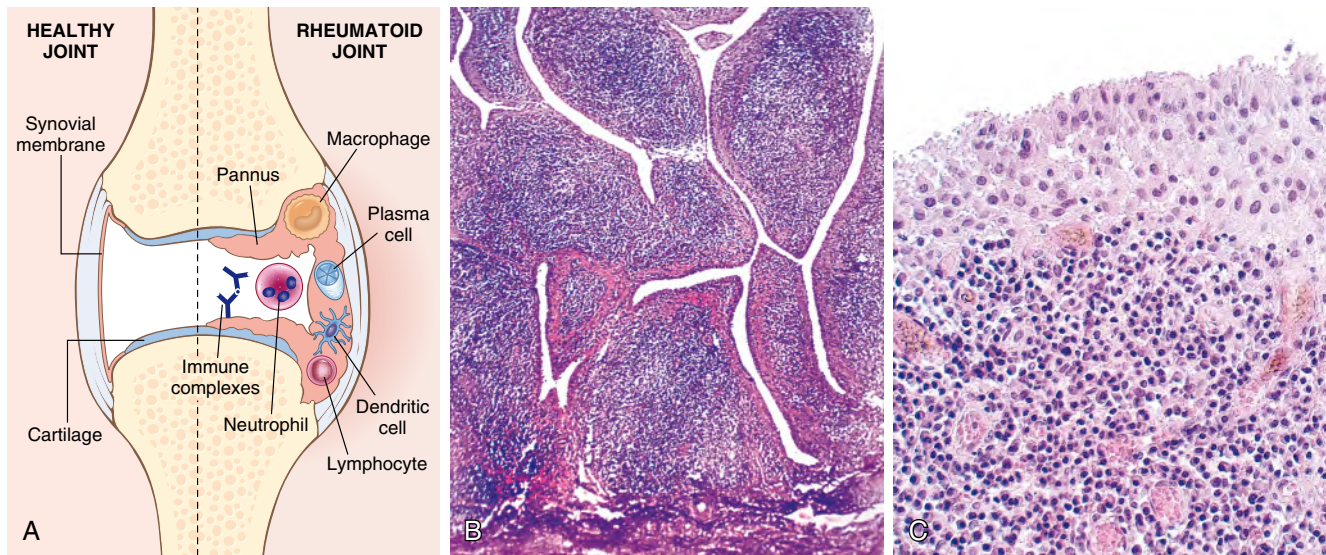


Fig. 26.34 Rheumatoid arthritis. (A) Schematic view of the joint lesion. (B) Low magnification reveals marked synovial hypertrophy with formation of villi. (C) At higher magnification, subsynovial tissue containing a dense lymphoid aggregate. (A, Modified from Feldmann M. Development of anti-TNF therapy for rheumatoid arthritis. *Nat Rev Immunol.* 2002;2(5):364-371.)

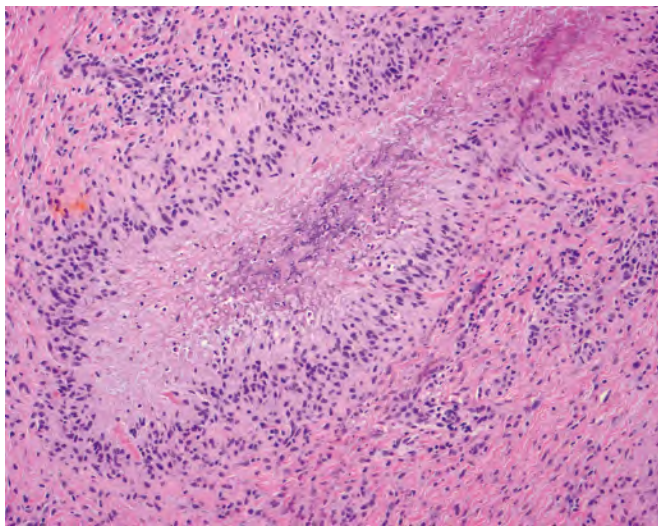


Fig. 26.35 Rheumatoid nodule composed of central necrosis rimmed by palisaded macrophages.

present one or more antigens that trigger the disease, but neither the antigen nor the pathogenic immune cell is known.

Reactive Arthritis

Reactive arthritis is defined by the triad of arthritis, nongonococcal urethritis or cervicitis, and conjunctivitis following an infection. More recently, the definition has been broadened to include mono- or oligoarticular arthritis that occurs days to weeks after genitourinary (*Chlamydia*) or gastrointestinal (*Shigella*, *Salmonella*, *Yersinia*, *Campylobacter*, and *Clostridioides difficile*) infections. HIV-positive patients may also be affected; the reactive arthritis in these individuals is thought to be caused by pathogens other than HIV. HLA-B27 is common in patients with reactive arthritis. The disease most often presents in young adults, with a global prevalence of approximately 30 per

100,000 adults. Arthritic episodes usually wax and wane over a period of months, but nearly 50% of patients have recurrent arthritis, tendonitis, and lumbosacral pain. Spondyloarthritis can occur in these patients, and spine involvement can be indistinguishable from ankylosing spondylitis.

Patients initially experience an acute-onset asymmetric oligoarthritis, often affecting the lower extremities, especially the knee. Inflammatory low back pain is a common accompanying symptom. *Enthesitis* occurs frequently and often involves the insertions of the Achilles tendon and plantar fascia on the calcaneus, resulting in swelling of the heel. *Dactylitis* (inflammation of the digits) is present in a minority of patients in whom synovitis of a digital tendon sheath produces sausage-like swelling of digits. This may progress to ossification at tendoligamentous insertion sites and development of calcaneal spurs and bony outgrowths. Extraarticular involvement includes conjunctivitis and uveitis, urethritis, balanitis, cervicitis, mucosal ulcers, cutaneous manifestations (e.g., psoriasis-like nail changes) and cardiac valvular disease (e.g., aortic insufficiency).

Infectious Arthritis

Microorganisms of all types can seed joints during hematogenous dissemination. Articular structures may also become infected by direct inoculation or contiguous spread from a soft tissue abscess or from osteomyelitis. Because cartilage, unlike bone, has a limited regenerative capacity, the rapid joint destruction that may ensue can result in permanent deformities.

Suppurative Arthritis

Bacterial infections that cause acute suppurative arthritis usually enter the joints from distant sites by hematogenous spread. As with osteomyelitis, the etiologic agent depends on the anatomic location and clinical setting. In neonates, contiguous spread from underlying epiphyseal osteomyelitis and group B *Streptococcus* may be responsible. With the advent of effective vaccination, the incidence of *H. influenzae* arthritis has fallen, and *S. aureus* is now the most common cause of bacterial arthritis in all age groups in the United States. Individuals

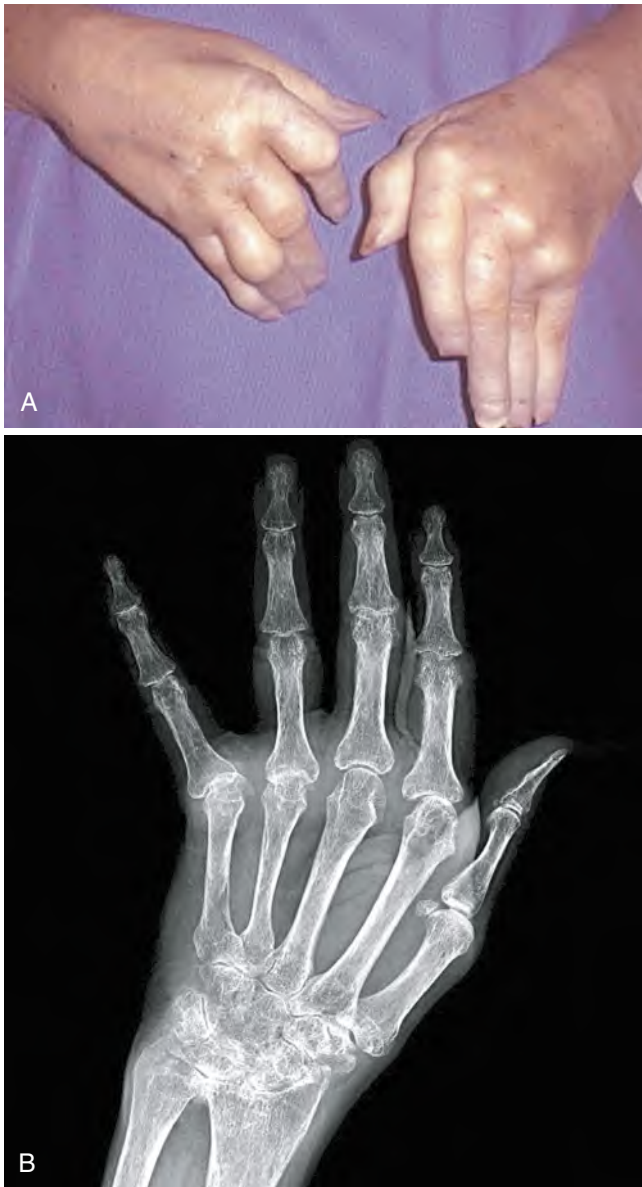


Fig. 26.36 Rheumatoid arthritis of the hand. (A) Prominent ulnar deviation and flexion-hyperextension of the fingers. (B) Radiographic features include diffuse osteopenia, periarticular bony erosions, and marked loss of the joint spaces of the carpal, metacarpal, phalangeal, and interphalangeal joints. (A, From Klatt EC. *Robbins and Cotran Atlas of Pathology*. 4th ed. Elsevier; 2021.)

with sickle cell disease are prone to *Salmonella* infection at any age. Individuals with deficiencies of components of the complement membrane attack complex (C5–C9) are especially susceptible to disseminated gonococcal infections and arthritis. Other predisposing conditions include immunodeficiencies (congenital and acquired), debilitating illness, joint trauma, chronic arthritis of any cause, and intravenous substance use.

The classic presentation is sudden development of an acutely painful, swollen joint with a restricted range of motion. Fever, leukocytosis, and elevated C-reactive protein are common. The infection usually involves only a single joint, most commonly the knee, hip, shoulder, elbow, wrist, or sternoclavicular joint. Axial joint involvement is most frequent in injected substance users. Joint aspiration is diagnostic if it yields purulent fluid in which the infectious organism can be identified. Prompt recognition and effective antimicrobial therapy can prevent joint destruction.

Mycobacterial Arthritis

Mycobacterial arthritis is a chronic progressive joint disorder caused by *M. tuberculosis* infection. It occurs primarily in adults as a complication of adjacent osteomyelitis or after hematogenous dissemination from a visceral (usually pulmonary) site of infection. Onset is insidious and associated with gradually increasing pain. Mycobacterial seeding of the joint induces formation of confluent caseating granulomas. The synovium may grow as a pannus over the articular cartilage and erode the bone along the joint margins. Chronic disease results in fibrous ankylosis and obliteration of the joint space. Weight-bearing joints, especially the hips, knees, and ankles, in descending order of frequency, are most commonly affected.

Viral Arthritis

Arthritis can occur with a variety of viral infections. The most common are alphavirus, parvovirus B19, rubella, Epstein-Barr virus, and hepatitis B and C viruses. The manifestations range from acute to subacute arthritis. Joint symptoms may be caused by direct infection of the joint by the virus, as in rubella and some alphavirus infections, or by an autoimmune reaction triggered by the infection.

Lyme Arthritis

Lyme arthritis is caused by the spirochete *Borrelia burgdorferi*, which is transmitted by deer ticks of the *Ixodes ricinus* complex. It is the leading arthropod-borne disease in the United States. It most often occurs in New England, the mid-Atlantic states, and the upper Midwest, but its geographic distribution is expanding. In the classic form, Lyme disease involves multiple organ systems during several successive clinical phases (Chapter 8). The initial skin infection is

TABLE 26.5 Comparative Features of Osteoarthritis and Rheumatoid Arthritis

	Osteoarthritis	Rheumatoid Arthritis
Primary pathogenic abnormality	Mechanical injury to articular cartilage	Autoimmunity
Role of inflammation	May be secondary; inflammatory mediators exacerbate cartilage damage	Primary: cartilage destruction is caused by T cells and antibodies reactive with joint antigens
Joints involved	Primarily weight bearing (knees, hips)	Often begins with small joints of fingers; progression leads to involvement of multiple joints
Pathology	Cartilage degeneration and fragmentation, bone spurs, subchondral cysts; minimal inflammation	Inflammatory pannus invading and destroying cartilage; severe chronic inflammation; joint fusion (ankylosis)
Serum antibodies	None	Various, including ACPA, rheumatoid factor
Involvement of other organs	No	Yes (lungs, heart, other organs)

ACPA, Anti-citrullinated peptide antibody.

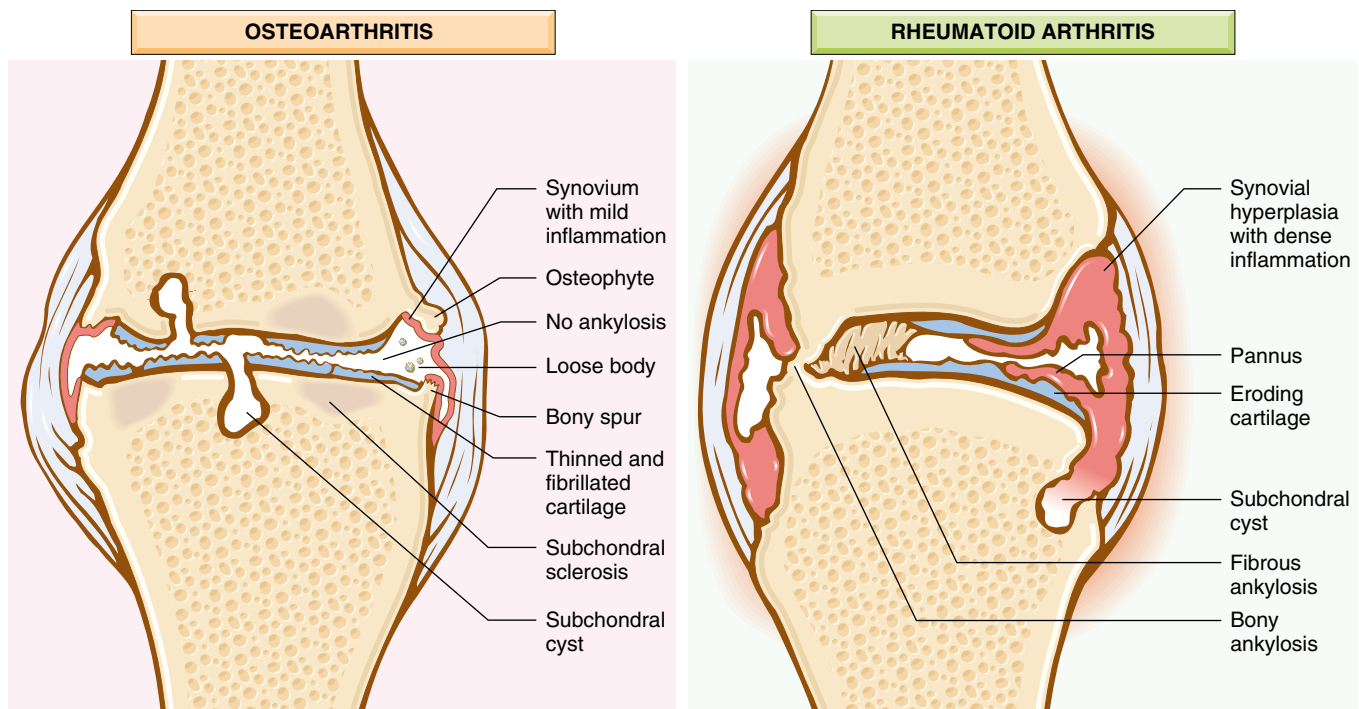


Fig. 26.37 Comparison of the morphologic features of rheumatoid arthritis and osteoarthritis.

followed within days or weeks by dissemination of the organism to other cutaneous sites, cranial nerves, heart, and meninges. Currently, arthritis occurs in less than 10% of cases of *Borrelia* infection because most individuals are treated and cured at an earlier stage. However, without timely treatment up to 80% of individuals develop migratory arthritis (*Lyme arthritis*) that persists for weeks to months. Large joints, especially the knees, shoulders, elbows, and ankles, in descending order of frequency, are usually involved. Initial attacks last for a few weeks to months and are usually limited to one or two joints at a time. Arthritis may subsequently develop at new sites.

Borrelia DNA can be detected by PCR performed on synovial fluid in approximately 70% of patients with Lyme arthritis, but diagnosis is usually based on serologic detection of anti-*Borrelia* antibodies. Lyme disease treatment with antibiotics active against *Borrelia* is curative in approximately 90% of patients.

Uncommonly, arthritis may persist after treatment with antibiotics, a condition referred to as postinfectious or antibiotic-refractory Lyme arthritis. In many such cases, *Borrelia* cannot be detected in the joint fluid, even by PCR. It has been proposed that antibiotic-refractory arthritis is an autoimmune disease triggered by cross-reactive immune responses to *Borrelia* outer surface protein A and some unknown self-antigen(s).

MORPHOLOGY

Microscopically, infected synovium exhibits chronic synovitis marked by synovocyte hyperplasia, fibrin deposition, mononuclear cell infiltrates (especially CD4+ T cells), and onion-skin–like thickening of arteriolar walls (**obliterative endarteritis**). In severe cases, the histopathology mimics rheumatoid arthritis.

Crystal-Induced Arthritis

Articular crystal deposits are associated with a variety of acute and chronic joint disorders. Endogenous crystals include monosodium urate (*gout*), calcium pyrophosphate dihydrate (*pseudogout*), and basic

calcium phosphate. Exogenous crystals, such as the biomaterials used in prosthetic joints, can also induce arthritis as debris accumulates with wear. Endogenous and exogenous crystals produce disease by triggering inflammatory reactions that destroy cartilage.

Gout

Gout is marked by transient attacks of acute arthritis initiated by monosodium urate crystals deposited within and around joints. Whether gout is primary (90%) or secondary to another underlying disease (Table 26.6), the common feature is excessive uric acid in the tissues and body fluids. In the primary form (90% of cases), gout is the major manifestation of the disease.

Pathogenesis

Hyperuricemia (serum uric acid above 6.8 mg/dL) is necessary, but not sufficient, for development of gout. Elevated uric acid can result

TABLE 26.6 Classification of Gout

Clinical Category	Uric Acid Synthesis	Uric Acid Excretion
Primary Gout		
Unknown enzyme defects	Normal ↑	↓ Normal
Known enzyme defects (e.g., partial HGPRT deficiency)	↑	Normal
Secondary Gout		
Increased nucleic acid turnover (e.g., leukemia)	↑ ↑	↑
Chronic renal disease	Normal	↓
Congenital (e.g., Lesch-Nyhan syndrome, HGPRT deficiency)	↑ ↑	↑

HGPRT, Hypoxanthine guanine phosphoribosyl transferase.

from overproduction, reduced excretion, or both (see Table 26.6). Uric acid levels are determined by several factors:

- **Rates of synthesis.** Uric acid is the end product of purine catabolism. Increased synthesis typically reflects some abnormality in purine production. Purines are the product of two interlinked pathways: the *de novo* pathway, in which purine nucleotides are synthesized from nonpurine precursors, and the *salvage pathway*, in which nucleotides are synthesized from free purine bases in the diet and those generated by DNA or RNA catabolism.
- **Rate of excretion.** In the kidney, uric acid is filtered by the glomerulus, but is almost completely resorbed by the proximal tubule. A small fraction of resorbed uric acid is secreted by the distal nephron and excreted in the urine.

In primary gout, elevated uric acid most commonly results from reduced excretion, the basis of which is unknown in most patients. Genome-wide association studies have identified polymorphisms in genes including *URAT1*, *GLUT9*, and *KCNQ1*, which are involved in transport and homeostasis of urate and other ions. In a small minority of cases primary gout is caused by increased uric acid production due to enzymatic defects. For example, partial deficiency of hypoxanthine guanine phosphoribosyl transferase (HGPRT) interrupts the salvage pathway, so purine metabolites cannot be recycled and are instead degraded into uric acid. Complete absence of HGPRT also results in hyperuricemia, but a significant neurologic condition (*Lesch-Nyhan syndrome*) dominates the clinical picture, so this is considered a form of secondary gout. Other forms of secondary gout are associated with medications or conditions that cause hyperuricemia. Secondary gout can also be caused by increased synthesis of uric acid (e.g., following rapid cell lysis during chemotherapy for leukemia) or decreased excretion (chronic renal disease).

The arthritis in gout is triggered by precipitation of monosodium urate crystals into the joints, which results in the production of cytokines that recruit leukocytes (Fig. 26.38). Resident synovial macrophages and neutrophils phagocytose the urate crystals, which are detected by a cytosolic sensor, the inflammasome (Chapter 3). In response, the inflammasome activates caspase-1, which is involved in the production of active IL-1. IL-1 promotes accumulation of more neutrophils and macrophages within the joint. Ingested urate crystals may also rupture phagolysosomes, releasing additional proteases and inflammatory mediators. A vicious cycle develops as newly recruited inflammatory cells release cytokines, free radicals, proteases, and arachidonic acid metabolites to recruit even more leukocytes. Complement activation by the alternative pathway may further contribute to leukocyte recruitment. The result is an acute arthritis that typically remits in days to weeks. However, repeated attacks of acute arthritis lead eventually to the formation of *tophi*, aggregates of urate crystals and inflammatory tissue in the inflamed synovial membrane and peri-articular tissue, as well as severe cartilage damage that compromises joint function.

Only about 10% of individuals with hyperuricemia develop gout. Other factors linked to disease include:

- **Age of the individual and duration of the hyperuricemia.** Gout usually appears after 20 to 30 years of hyperuricemia
- **Genetic predisposition.** In addition to X-linked HGPRT abnormalities, polymorphisms in genes involved with urate or ion transport and inflammation are associated with gout
- **Alcohol consumption**
- **Obesity**
- **Drugs** (e.g., thiazides) that reduce urate excretion

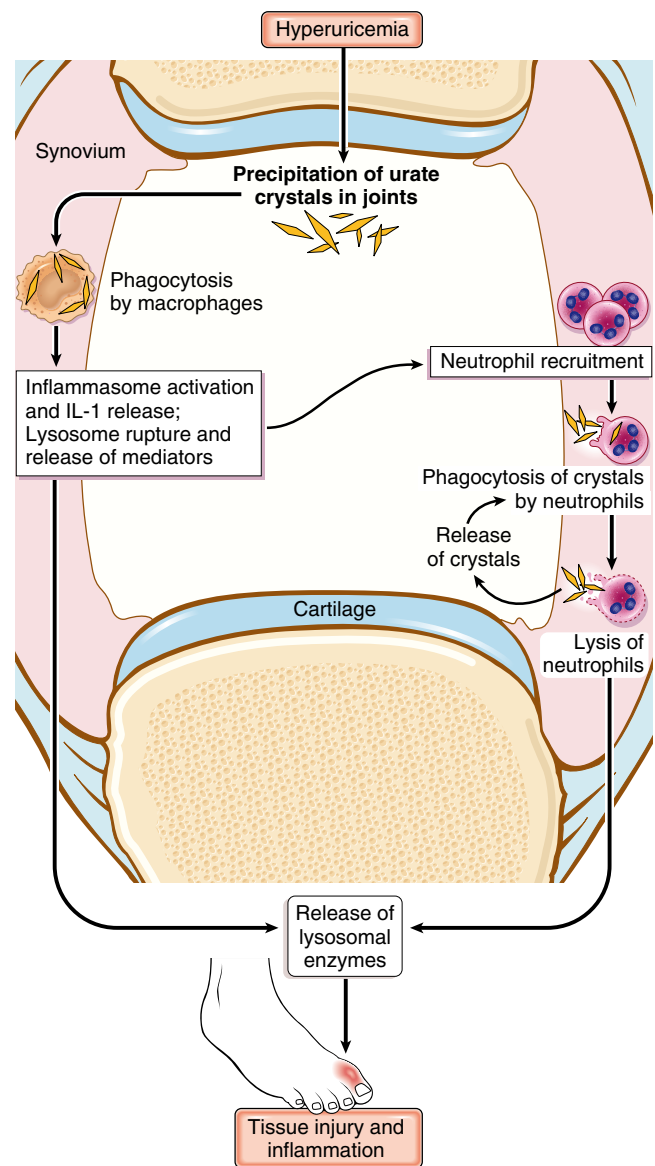


Fig. 26.38 Pathogenesis of acute gouty arthritis. Urate crystals are phagocytosed by macrophages and stimulate the production IL-1 and the rupture of lysosomes, leading to recruitment of neutrophils, tissue injury, and inflammation. *IL-1*, Interleukin 1.

MORPHOLOGY

Distinctive clinical features of gout with characteristic histologic correlates include (1) acute arthritis, (2) chronic tophaceous arthritis, (3) extraarticular tophi, and (4) urate (gouty) nephropathy. **Acute arthritis** is characterized by dense neutrophilic infiltrates that permeate the synovium and synovial fluid. Slender, needle-shaped urate crystals (best seen with polarized optics) are arranged in small clusters in the synovium and are frequently found in the cytoplasm of neutrophils within aspirated joint fluid (Fig. 26.39A). When crystals are parallel to the axis of the polarizer, they appear yellow under compensated polarized light (**negative birefringence**). The edematous and congested synovium also contains scattered lymphocytes, plasma cells, and macrophages. The acute attack remits when the episode of crystallization abates and crystals are solubilized.

Chronic tophaceous arthritis evolves from repetitive precipitation of urate crystals during acute attacks. Urate crystals encrust the articular surface and form

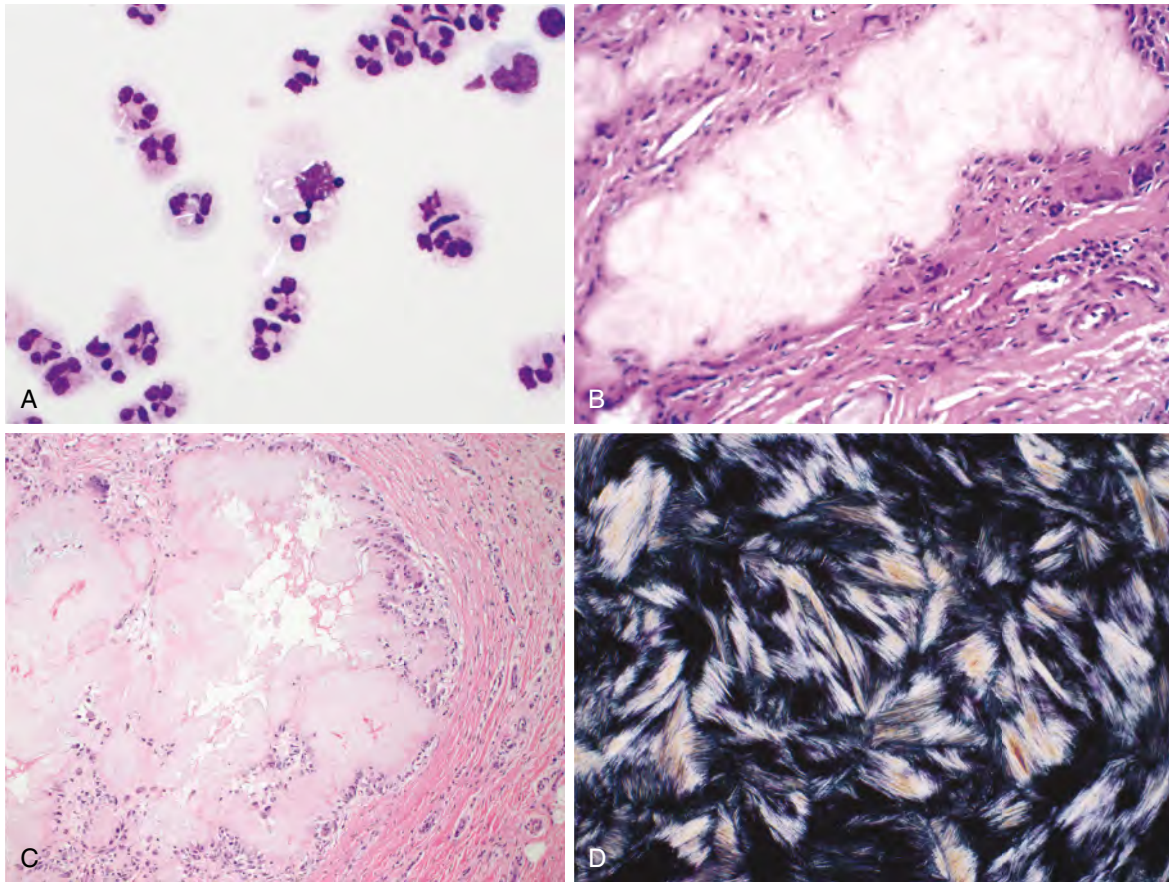


Fig. 26.39 Gout. (A) Joint aspirate from acute gout arthritis demonstrates polarizable needle-shaped urate crystals in neutrophils. (B, C) Gouty tophus—an aggregate of dissolved urate crystals is surrounded by reactive fibroblasts, mononuclear inflammatory cells, and giant cells. (D) Needle-shaped urate crystals are easily identified under polarized light.

chalky deposits in the synovium, which becomes hyperplastic, fibrotic, and thickened by inflammatory cells. The resulting pannus destroys the underlying cartilage and leads to juxtaarticular bone erosions. In severe cases, fibrous or bony ankylosis ensues, resulting in loss of joint function. **Tophi** are the pathognomonic hallmark of gout. They are formed by large aggregations of urate crystals surrounded by an intense inflammatory reaction of foreign-body giant cells. (Fig. 26.39B–C). Tophi may appear in the articular cartilage, ligaments, tendons, and bursae. Less frequently they occur in soft tissues (earlobes, fingertips) or kidneys.

Urate nephropathy refers to the renal complications caused by urate crystals or tophi in the renal medullary interstitium or tubules. Complications include uric acid nephrolithiasis (kidney stones) and pyelonephritis, particularly when urinary obstruction occurs.

Clinical Features

Four clinical stages are recognized:

- *Asymptomatic hyperuricemia* begins around puberty in males and after menopause in females.
- *Acute arthritis* presents as sudden-onset, excruciating joint pain, localized hyperemia and warmth, and, occasionally, mild fever. Most first attacks are monoarticular, with 50% occurring in the first metatarsophalangeal joint. Untreated, acute gouty arthritis may last for hours to weeks but gradually resolves completely.
- The *intercritical period* is the symptom-free interval following resolution of acute arthritis. Some individuals never have another

attack but most experience a second acute episode within months to a few years. In the absence of appropriate therapy, the attacks recur at shorter intervals and frequently become polyarticular.

- *Chronic tophaceous gout* develops about 10 years after the initial acute attack and is characterized by juxtaarticular bone erosion and loss of the joint space.

Nonpharmacologic treatments for gout include lifestyle modifications, such as weight loss in obese individuals; dietary changes to reduce purine intake; reduced alcohol and sugar-sweetened beverage consumption; and regular exercise. Pharmacologic therapies include uricosuric drugs (e.g., probenecid), xanthine oxidase inhibitors (e.g., allopurinol), urate oxidases (uricases) and drugs that reduce inflammation (e.g., NSAIDs, colchicine). Generally, gout does not shorten life span, but it does negatively impact quality of life.

Calcium Pyrophosphate Crystal Deposition Disease (Pseudogout)

Calcium pyrophosphate crystal deposition disease (CPPD), also known as pseudogout, is a variably symptomatic disorder caused by deposition of calcium pyrophosphate (CPP) crystals within and around joints. CPPD usually occurs in individuals older than 50 years and becomes more common with increasing age; up to 60% of those age 85 years or older are affected. It is divided into sporadic (idiopathic), hereditary, and secondary types. Secondary CPPD

arises in the setting of previous joint damage, hyperparathyroidism, hemochromatosis, hypomagnesemia, hypothyroidism, ochronosis, and diabetes.

Pathogenesis

Unlike urate crystals, CPP crystals form in hyaline cartilage and fibrocartilage. Studies suggest that degradation of articular cartilage proteoglycans, which normally inhibit mineralization, allows CPP crystallization around chondrocytes. As in gout, CPP crystals activate the inflammasome, causing release of IL-1 and leading to recruitment of leukocytes (see Fig. 26.38). Even in the absence of inflammation, crystal deposition in cartilage may disrupt the structure and function of the matrix. Mutations in the *ANKH* gene, which encodes an inorganic pyrophosphate transporter, and the *OPG* gene, which encodes a decoy receptor for RANKL (see Fig. 26.4), have been linked to some forms of CPPD.

MORPHOLOGY

The crystals first develop in the articular cartilage, menisci, and intervertebral disks and as the deposits enlarge they may rupture and seed the joint. The crystals form **chalky, white friable deposits**, which are seen histologically in hematoxylin and eosin-stained preparations as oval blue-purple aggregates (Fig. 26.40A). Individual **rhomboid crystals** are 0.5 to 5 μm in greatest dimension (Fig. 26.40B) and take on a blue tinge when seen under compensated polarized light when parallel to the axis of the polarizer (*positive birefringence*) (Fig. 26.40C). Inflammation is usually milder than in gout.

Clinical Features

CPPD is frequently asymptomatic. Symptomatic patients present with sudden pain in one or a few joints, which (unlike gout) may last for weeks or months. In the chronic form, persistent joint pain is present with superimposed acute flares that can be confused with osteoarthritis or rheumatoid arthritis clinically. The knees, followed by the wrists, elbows, shoulders, and ankles are affected most commonly. *Chondrocalcinosis* is defined by radiographically evident calcification of hyaline cartilage or fibrocartilage, typically by CPP crystals, but not all individuals with this finding have symptomatic CPPD. Ultimately, approximately 50% of affected individuals experience significant joint damage. Therapy is similar to that used to treat gout and includes NSAIDs, colchicine, corticosteroids, and IL-1 inhibitors. There is no known treatment that prevents or slows crystal formation.

KEY CONCEPTS

Arthritis

- Osteoarthritis (degenerative joint disease), the most common disease of joints, is a degenerative disorder of articular cartilage in which matrix breakdown exceeds synthesis. Inflammation is minimal and typically secondary, but locally produced cytokines may contribute to progression of joint degeneration.
- Rheumatoid arthritis (RA) is a chronic autoimmune inflammatory disease that affects mainly small joints but can be systemic. RA is caused by a cellular and possibly humoral immune response against self-antigens, particularly citrullinated proteins. TNF plays a central role, and anti-TNF biologics are effective therapy.
- Seronegative spondyloarthropathies are a heterogeneous group of autoimmune arthritides that preferentially involve the sacroiliac, vertebral joints and entheses and are associated with HLA-B27. Reactive arthritis is an uncommon immune-mediated disorder that develops after resolution of genitourinary or gastrointestinal infection.

- Suppurative arthritis describes direct infection of a joint space by bacterial organisms.
- Lyme disease is a systemic infection by *B. burgdorferi* that manifests, in part, as an infectious arthritis, possibly with an autoimmune component in chronic stages.
- Gout and pseudogout result from inflammatory responses to precipitated urate or calcium pyrophosphate, respectively.

JOINT TUMORS

Reactive tumor-like lesions, such as ganglion cysts, synovial cysts, and osteochondral loose bodies, commonly involve joints and tendon sheaths. They usually result from trauma or degenerative processes and are much more common than neoplasms. Primary neoplasms are rare, usually benign, and tend to recapitulate the cells and tissue types (synovial membrane, fat, blood vessels, fibrous tissue, and cartilage) native to joints and related structures. Synovial sarcoma, once thought to be related to or derived from the tissues of the joint, is now recognized as a sarcoma of uncertain origin and is discussed with soft tissue tumors.

Ganglion and Synovial Cysts

Ganglion cysts are 1 to 1.5 cm in size and nearly always located near a joint capsule or tendon sheath, especially around the wrist. They appear as firm, fluctuant, pea-sized translucent nodules. Ganglion cysts develop due to cystic or myxoid connective tissue degeneration. They are not true cysts, as the cyst wall lacks a cell lining. Lesions may be multilocular and may enlarge through coalescence and degeneration of adjacent connective tissue. The fluid that fills ganglion cysts resembles synovial fluid and may accumulate due to a one-way valve type connection to the joint space. The term ganglion derives from the Greek root meaning “knot or swelling”; it is unrelated to ganglia of the nervous system.

Herniation of synovium through a joint capsule or massive enlargement of a bursa may produce a synovial cyst. A well-recognized example are synovial cysts that develop in the setting of RA or OA (*Baker cyst*). The cysts are lined by tissue resembling synovium that may be hyperplastic and filled with fluid that often contains inflammatory cells and fibrin.

Tenosynovial Giant Cell Tumor

Tenosynovial giant cell tumor is a benign neoplasm that develops in the synovial lining of joints, tendon sheaths, and bursae. Tenosynovial giant cell tumor can be diffuse (previously known as *pigmented villonodular synovitis*) or localized. The diffuse type tends to involve large joints, while the localized type usually occurs as a discrete nodule attached to a tendon sheath, commonly in the hand. Both variants are most often diagnosed in individuals in their second to fourth decades.

Pathogenesis

Both diffuse and localized types of tenosynovial giant cell tumor harbor a reciprocal somatic chromosomal translocation, $t(1;2)(p13;q37)$, that positions the type VI collagen α -3 (*COL6A3*) promoter upstream of the coding sequence of the *CSF1* gene, which encodes macrophage colony-stimulating factor (M-CSF). As a result, the tumor cells overexpress and secrete M-CSF, which through autocrine and paracrine effects stimulates proliferation of macrophages in a manner similar to giant cell tumor of bone (described previously).

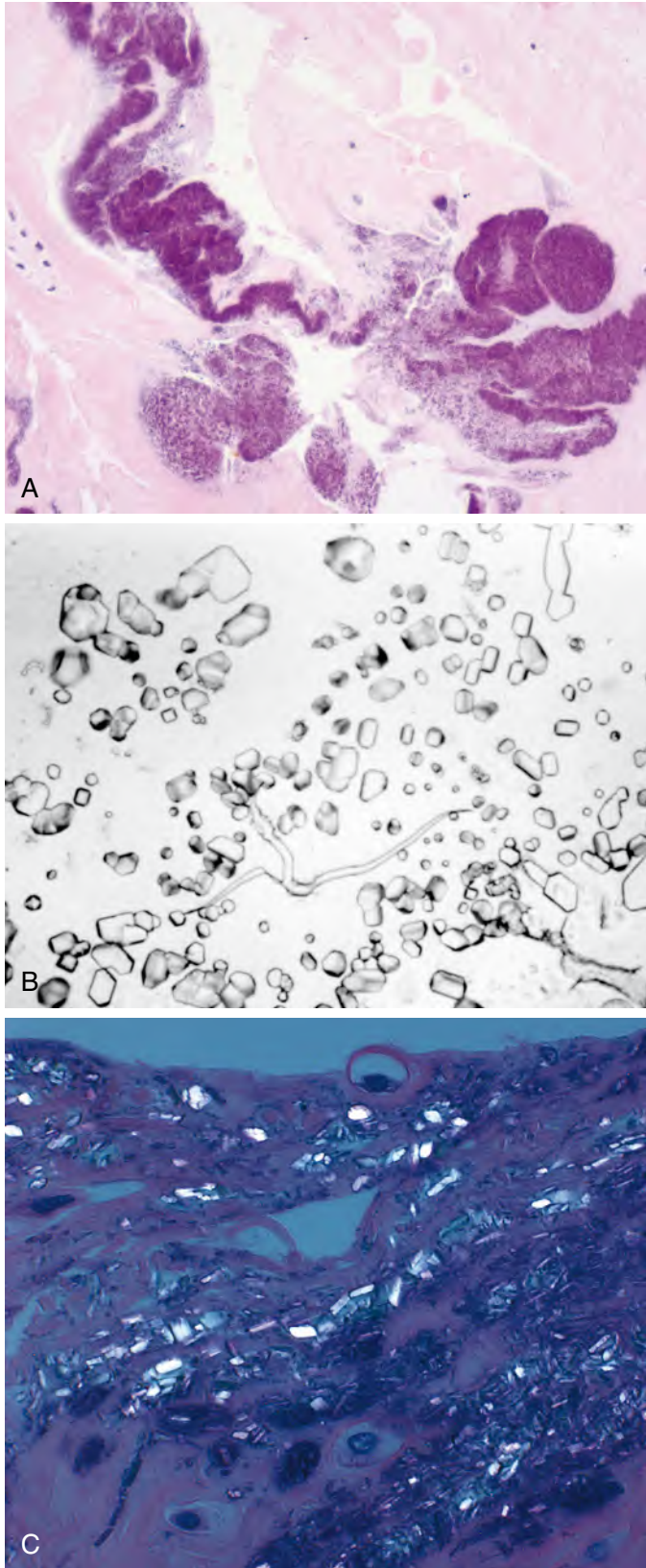


Fig. 26.40 Pseudogout. (A) Calcium pyrophosphate deposits are present in cartilage and consist of amorphous basophilic material. (B) Smear preparation showing rhomboid calcium pyrophosphate crystals. (C) The crystals are readily visualized under polarized light. (From Horvai A, Link T. *Bone and Soft Tissue Pathology*. Elsevier; 2012.)

MORPHOLOGY

Tenosynovial giant cell tumors are red-brown to orange-yellow in color. In diffuse tumors, the normally smooth joint synovium is converted into a tangled mat by red-brown folds, fingerlike projections, and nodules (Fig. 26.41A), whereas localized tumors are usually <5 cm and well circumscribed. The neoplastic cells, which are a minority of the cells in the mass, are polygonal, moderately sized, and resemble synoviocytes (Fig. 26.41B). In the diffuse variant, they spread along the surface and infiltrate the subsynovial tissue. A solid aggregate may be attached to the synovium by a pedicle. Both diffuse and localized types are densely infiltrated by macrophages that may contain hemosiderin or foamy lipids and can be multinucleated. Older lesions may become fibrotic.

Clinical Features

Diffuse tenosynovial giant cell tumors present in the knee in 80% of cases. Affected individuals complain of pain, locking, and recurrent swelling, symptoms similar to those seen with monoarticular arthritis. Sometimes a palpable mass is appreciated. Some tumors erode into adjacent bones and soft tissues, causing confusion with other types of neoplasms. Recurrence is common. Localized tenosynovial giant cell tumors present as a solitary, slow-growing, painless mass that frequently involves the tendon sheaths along the wrists and fingers. Bone erosion and recurrence are less common than in the diffuse type. Surgical excision is the mainstay of treatment. Antagonists of M-CSF signaling are effective therapies but carry a risk of severe hepatotoxicity; their use is restricted to patients in whom surgery would produce significant morbidity or functional impairment.

SOFT TISSUE TUMORS

By convention, *soft tissue* refers to nonepithelial tissue excluding the skeleton, joints, central nervous system, and hematopoietic and lymphoid tissues. Although nonneoplastic conditions can involve soft tissue, they are seldom confined to this compartment, so the area of soft tissue pathology is restricted to neoplasms. Except for skeletal muscle neoplasms, benign soft tissue tumors are 100-fold more frequent than their malignant counterparts, sarcomas. In the United States, the incidence of soft tissue sarcomas is approximately 12,000 per year, which is less than 1% of all cancers. Sarcomas, however, cause 2% of all cancer deaths, reflecting their aggressive behavior and lack of effective treatment for many types of these tumors. Most soft tissue tumors arise in the extremities, particularly the thigh. Approximately 15% arise in children, many of whom have germline mutations that place them at increased risk.

Pathogenesis

Most sarcomas are sporadic and have no known predisposing cause. A small minority are associated with germline mutations in tumor suppressor genes that produce well-described syndromes (e.g., neurofibromatosis 1, Gardner syndrome, Li-Fraumeni syndrome, Weber-Osler-Rendu syndrome). Other sarcomas are linked to known environmental exposures such as radiation, burns, or toxins, or to chronic inflammation.

Unlike many carcinomas and certain hematologic malignancies that arise from well-recognized precursor lesions (Chapter 7), sarcoma precursors are undefined. While some sarcomas recapitulate a recognizable mesenchymal lineage (e.g., skeletal muscle), all sarcomas are thought to arise from mesenchymal stem cells that acquire driver mutations in oncogenes and tumor suppressor genes. The chromosomal aberrations that are seen in sarcomas are heterogeneous but tend to fall into two categories:

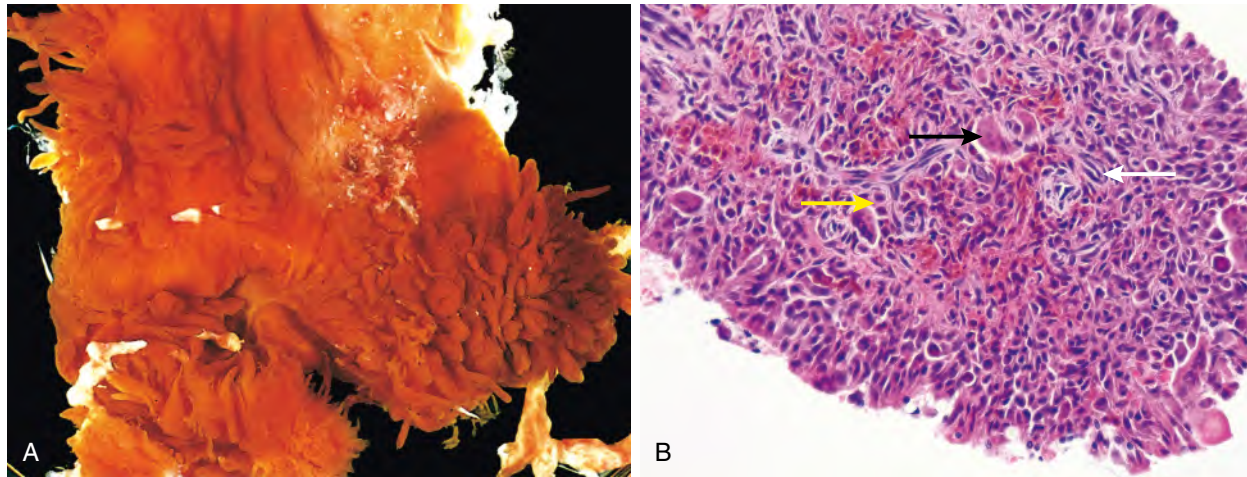


Fig. 26.41 Tenosynovial giant cell tumor, diffuse type. (A) Excised synovium with fronds and nodules typical of diffuse tenosynovial giant cell tumor. The color and texture explain the old name of pigmented villonodular synovitis. (B) Mixed population of spindled (*white arrow*), synoviocyte-like (*yellow arrow*), and osteoclast-type giant cells (*black arrow*) with abundant red-brown hemosiderin pigment in tenosynovial giant cell tumor.

TABLE 26.7 Chromosomal Abnormalities in Soft Tissue Tumors

Tumor	Cytogenetic Abnormality	Gene Fusion	Proposed Function
Ewing sarcoma	t(11;22)(q24;q12) t(21;22)(q22;q12)	<i>EWSR1::FLI1</i> <i>EWSR1::ERG</i>	Chimeric fusion protein that alters gene regulation, possibly through several mechanisms
Extraskeletal myxoid chondrosarcoma	t(9;22)(q22;q12)	<i>EWSR1::NR4A3</i>	
Desmoplastic small round cell tumor	t(11;22)(p13;q12)	<i>EWSR1::WT1</i>	
Clear-cell sarcoma	t(12;22)(q13;q12)	<i>EWSR1::ATF1</i>	
Liposarcoma—myxoid type	t(12;16)(q13;p11)	<i>FUS::DDIT3</i>	Chimeric fusion protein that arrests adipocytic differentiation
Synovial sarcoma	t(x;18)(p11;q11)	<i>SS18::SSX1</i> <i>SS18::SSX2</i> <i>SS18::SSX4</i>	Chimeric fusion proteins that dysregulate chromatin remodeling complexes
Rhabdomyosarcoma—alveolar type	t(2;13)(q35;q14) t(1;13)(p36;q14)	<i>PAX7::FOXO1</i> <i>PAX3::FOXO1</i>	Chimeric transcription factors that disrupt skeletal muscle differentiation
Dermatofibrosarcoma protuberans	t(17;22)(q22;q15)	<i>COL1A1::PDGFB</i>	Promoter driven overexpression of PDGF- β , leading to autocrine stimulation
Alveolar soft-part sarcoma	t(X;17)(p11.2;q25)	<i>ASPSCR1::TFE3</i>	Unknown
Infantile fibrosarcoma	t(12;15)(p13;q23)	<i>ETV6::NTRK3</i>	Chimeric tyrosine kinase that leads to constitutively active Ras/MAPK pathway
Nodular fasciitis	t(22;17)	<i>MYH9::USP6</i>	May increase Wnt/ β -catenin signaling

- **Simple karyotype** (~20% of sarcomas): Like many leukemias and lymphomas, some sarcomas are euploid tumors with a single or limited number of recurrent chromosomal aberrations (Table 26.7) that occur early in tumorigenesis and are specific enough to serve as diagnostic markers. Tumors with these features most commonly arise in younger individuals and show limited degrees of pleomorphism. Examples include Ewing sarcoma (described earlier) and synovial sarcoma. In some cases, the oncogenic effect of chromosomal rearrangements is reasonably well understood (see Table 26.7). In others, the pathogenic mechanisms are unknown. These recurrent events often produce a fusion gene encoding oncogenic chimeric proteins, which represent potential targets for therapy.
- **Complex karyotype** (~80% of sarcomas): These tumors are usually aneuploid or polyploid and demonstrate multiple chromosomal gains and losses, a feature that suggests underlying genomic instability. Examples include leiomyosarcoma and undifferentiated soft

tissue sarcoma. Such tumors are more common in adults and tend to be pleomorphic.

Classification of soft tissue tumors continues to evolve as new molecular genetic abnormalities are identified. Clinically, soft tissue tumors range from benign, self-limited lesions that require minimal treatment, to intermediate grade locally aggressive tumors with limited metastatic risk, to highly aggressive malignancies with high metastatic risk that are frequently fatal. All highly aggressive malignancies are classified as *sarcomas*, but this term is used less consistently among locally aggressive nonmetastasizing tumors. Pathologic classification integrates morphologic (e.g., muscle differentiation), immunohistochemical, and molecular features (Table 26.8). In addition to accurate diagnosis, tumor grade (degree of differentiation) and stage (extent of local and metastatic disease) are important prognostic indicators. The next section will consider representative soft tissue tumors (summarized in Table 26.8).

TABLE 26.8 Clinical Features of Soft Tissue Tumors

Category	Behavior	Tumor Type	Common Locations	Age (Years)	Morphology
Adipose	Benign	Lipoma	Superficial extremity, trunk	40–60	Mature adipose tissue
	Malignant	Well-differentiated liposarcoma	Deep extremity, retroperitoneum	50–60	Adipose tissue with scattered atypical spindle cells
	Malignant	Myxoid liposarcoma	Thigh, leg	30s	Myxoid matrix, “chicken wire” vessels, round cells, lipoblasts
Fibrous	Benign	Nodular fasciitis	Arm, forearm	20–30	Tissue culture–like growth, extravasated red cells
	Locally aggressive	Deep fibromatosis	Abdominal wall	30–40	Dense collagen, long, unidirectional fascicles
	Malignant	Adult fibrosarcoma	Deep extremity	40–60	Monomorphic spindle cells, fascicles
Skeletal muscle	Benign	Rhabdomyoma	Head and neck	0–60	Polygonal rhabdomyoblasts, “spider” cells
	Malignant	Alveolar rhabdomyosarcoma	Extremities, sinuses	5–15	Uniform round dyscohesive cells between septae
	Malignant	Embryonal rhabdomyosarcoma	Genitourinary tract	1–5	Primitive spindle cells, “strap” cells
Smooth muscle	Benign	Leiomyoma	Extremity	20s	Uniform, plump eosinophilic cells in fascicles
	Malignant	Leiomyosarcoma	Thigh, retroperitoneum	40–60	Pleomorphic eosinophilic cells
Vascular	Benign	Hemangioma	Head and neck	0–10	Circumscribed mass of capillary or venous channels
	Malignant	Angiosarcoma	Skin, deep lower extremity	50–80	Infiltrating capillary channels
Nerve sheath	Benign	Schwannoma	Head and neck	20–50	Encapsulated, fibrillar stroma, nuclear palisading
	Benign	Neurofibroma	Wide, cutaneous, subcutis	10–20+	Myxoid, ropy collagen, loose fascicles, mast cells
	Malignant	Malignant peripheral nerve sheath tumor	Extremities, shoulder girdle	20–50	Tight fascicles, atypia, mitotic activity, necrosis
Uncertain histotype	Benign	Solitary fibrous tumor	Pelvis, pleura	20–70	Branching ectatic vessels
	Malignant	Synovial sarcoma	Thigh, leg	15–40	Tight fascicles of uniform basophilic spindle cells, pseudoglandular structures
	Malignant	Undifferentiated soft tissue sarcoma	Thigh	40–70	High-grade anaplastic polygonal, round, or spindle cells, bizarre nuclei, atypical mitoses, necrosis

TUMORS OF ADIPOSE TISSUE

Lipoma

Lipoma, a benign tumor of fat, is the most common soft tissue tumor in adults. The conventional lipoma is the most common subtype, from which rare variants are distinguished according to characteristic morphologic and/or genetic features. Most solitary lipomas have chromosomal rearrangements involving chromosome 12q that dysregulate the expression of the *HMGA2* gene, a regulator of chromatin structure. *Lipomatosis* describes multifocal lipomas involving a limb. Most lipomas are soft, mobile, painless, and cured by simple excision.

MORPHOLOGY

The conventional lipoma is a well-encapsulated mass of mature adipocytes. It usually arises in the subcutis of the proximal extremities and trunk during middle adulthood. Infrequently, lipomas are large, intramuscular, and poorly circumscribed.

Liposarcoma

Liposarcoma, a malignant tumor of adipose tissue, is the most common sarcoma of adulthood. These tumors typically develop in deep soft tissues of the proximal extremities and retroperitoneum of individuals in their 50s and 60s.

Pathogenesis

The three major subtypes of liposarcoma (well-differentiated, myxoid, and pleomorphic) have different genetic aberrations.

- *Well-differentiated liposarcoma* harbors amplification of the *MDM2* gene, located in chromosome region 12q13–q15. *MDM2* acts as an oncoprotein by promoting the proteasomal degradation of p53.
- In *myxoid liposarcoma*, a t(12;16) translocation produces a *FUS::DDIT3* fusion gene that encodes a chimeric DNA binding protein. Through uncertain mechanism, this oncoprotein arrests adipocyte differentiation, leading to proliferation of primitive mesenchymal cells.
- *Pleomorphic liposarcomas* have complex karyotypes without consistent genetic abnormalities.

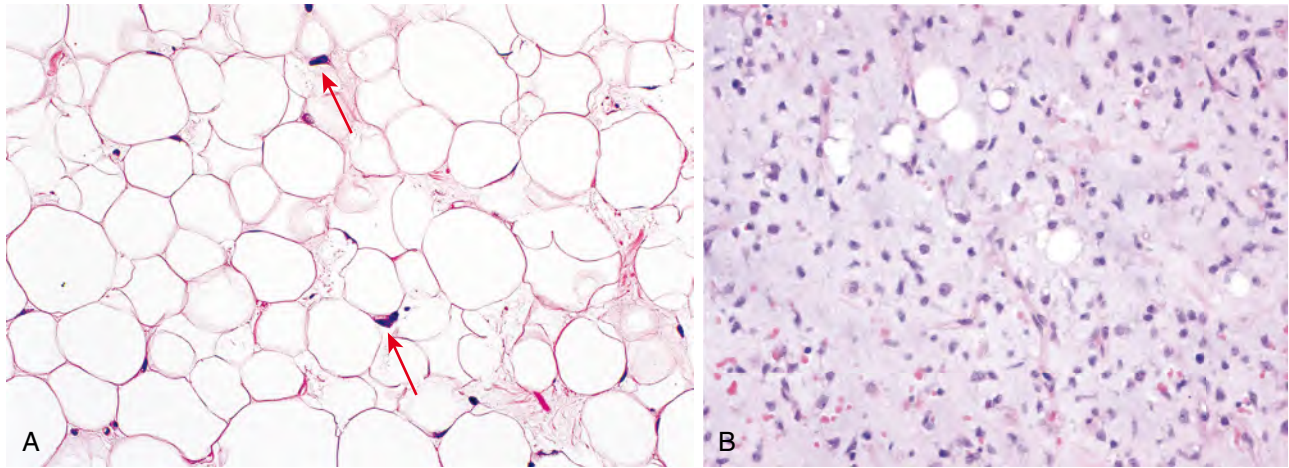


Fig. 26.42 Liposarcoma. (A) The well-differentiated subtype consists of mature adipocytes and scattered cells with hyperchromatic nuclei (arrows). (B) Myxoid liposarcoma with abundant basophilic ground substance and a rich capillary network with scattered immature adipocytes and more primitive round to stellate cells. (A, From Goldblum JR, Folpe AL, Weiss SW. *Enzinger & Weiss's Soft Tissue Tumors*. 7th ed. Elsevier; 2020.)

MORPHOLOGY

Well-differentiated liposarcoma consists of mature adipocytes with scattered atypical spindle cells (Fig. 26.42A). **Myxoid liposarcoma** displays by abundant basophilic extracellular matrix, arborizing capillaries, and primitive cells at various stages of adipocyte differentiation resembling fetal fat (Fig. 26.42B). **Pleomorphic liposarcoma** consists of sheets of anaplastic cells with bizarre nuclei admixed with variable numbers of immature adipocytes (**lipoblasts**).

Clinical Features

The three major subtypes of liposarcoma have distinctive clinical features. Well-differentiated liposarcoma may be indolent, especially in the extremities where complete surgical excision is feasible. For this reason, the diagnosis of *atypical lipomatous tumor* is used to denote extremity tumors. In the retroperitoneum, however, the diagnosis of well-differentiated liposarcoma is warranted because these tumors frequently recur, leading to significant morbidity. Myxoid liposarcoma demonstrates an intermediate local recurrence and metastatic risk (~20% and 50%, respectively). This tumor may metastasize to bone, which is an unusual pattern of spread for soft tissue sarcomas. Pleomorphic liposarcoma is aggressive and frequently metastasizes.

FIBROUS TUMORS

Nodular Fasciitis

Nodular fasciitis is a self-limited fibroblastic and myofibroblastic proliferation that occurs in young adults. The tumor occurs most frequently on the forearm, chest, and back and grows rapidly over weeks to months. A history of trauma is present in 10% to 50% of cases. Because it typically regresses spontaneously and often follows trauma, this lesion was long considered a reactive process. However, nodular fasciitis harbors a (17;22) translocation that produces a *MYH9::USP6* fusion gene, indicating that it is a clonal, self-limited proliferation. It is interesting to note that aneurysmal bone cyst (discussed earlier), a benign neoplasm previously considered reactive, is also associated with an acquired *USP6* fusion gene.

MORPHOLOGY

Nodular fasciitis arises in the deep dermis, subcutis, fascia, or muscle. It is generally <5 cm in diameter and unencapsulated, with at most slightly infiltrative borders. Histologically, nodular fasciitis is composed of plump, immature-appearing fibroblasts and myofibroblasts containing elongated nuclei with punctate nucleoli growing in a loose, “tissue culture” pattern. Mitoses are frequent but atypical forms are notably absent (Fig. 26.43). A gradient of maturation (**zonation**) from hypercellular myxoid regions to hypocellular fibrous areas is characteristic. Cystic areas, osteoclast-like giant cells, extravasated red cells, and infiltrating lymphocytes may be present.

Fibromatoses

Superficial Fibromatosis

Superficial fibromatosis is a benign fibroblastic proliferation that has an innocuous clinical course. Clinical subtypes include Dupuytren contracture, Ledderhose disease, and Peyronie disease.

- *Dupuytren contracture*, or *palmar fibromatosis*, is an irregular or nodular thickening of the palmar fascia that can be unilateral or

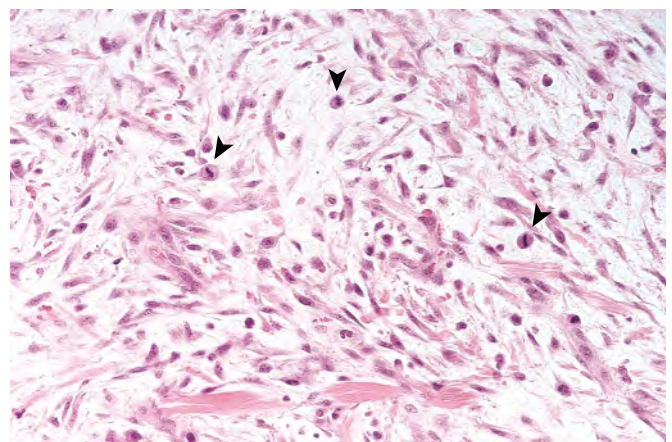


Fig. 26.43 Nodular fasciitis with plump, randomly oriented spindle cells surrounded by myxoid stroma. Note the mitotic activity (arrowheads).

bilateral. Its incidence increases with age. Attachment to overlying skin can cause puckering or dimpling and a slowly progressive flexion contracture.

- *Ledderhose disease*, or *plantar fibromatosis*, occurs in boys younger than 10 years into adolescence. It is unilateral and does not cause contractures but can be associated with palmar and penile fibromatosis.
- *Peyronie disease*, or *penile fibromatosis*, is a palpable induration or mass on the dorsolateral aspect of the penis. Eventually, it may cause abnormal curvature of the shaft and constriction of the urethra.

All three types are composed of plump spindle cells arranged in poorly defined broad bundles or long, sweeping fascicles surrounded by abundant dense collagen. Palmar and plantar fibromatoses progress in about 50% of cases. The remainder stabilize and do not progress, and some resolve spontaneously.

Deep Fibromatosis (Desmoid Tumor)

Deep fibromatoses (desmoid tumors) are large, infiltrative masses that frequently recur but do not metastasize. They arise most frequently in the teens to third decade, predominantly in females in the musculoaponeurotic structures of the anterior abdominal wall, limb girdles, or mesentery. Deep fibromatoses contain mutations in the *APC* or *CTNNB1* (β -catenin) genes, both of which lead to increased Wnt signaling. Most tumors harbor sporadic *CTNNB1* mutations. Individuals with familial adenomatous polyposis due to germline *APC* mutations are predisposed to deep fibromatosis (so-called Gardner syndrome, Chapter 17).

MORPHOLOGY

Deep fibromatoses are gray-white, firm, poorly demarcated masses up to 15 cm in size. They are rubbery and tough and infiltrate surrounding muscle, nerve, and fat. Bland fibroblasts arranged in long parallel fascicles amid dense collagen are characteristic (Fig. 26.44). The resulting histologic appearance can resemble a scar.

Clinical Features

In addition to possibly being disfiguring or disabling, deep fibromatosis can be painful. Because they are infiltrative, complete excision can be difficult and recurrence is common. Inhibitors of γ -secretase, a protease involved in Notch signaling, slow disease progression through uncertain mechanisms but are not curative.

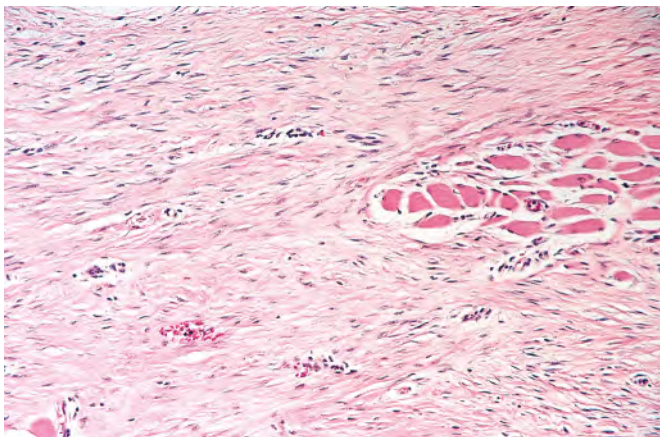


Fig. 26.44 Deep fibromatosis infiltrating skeletal muscle.

SKELETAL MUSCLE TUMORS

In contrast to tumors of other cell types, skeletal muscle neoplasms are almost all malignant. The benign variant, rhabdomyoma, is frequently associated with tuberous sclerosis (Chapter 28) and may occur in the heart or soft tissues.

Rhabdomyosarcoma

Rhabdomyosarcoma is a malignant mesenchymal tumor with skeletal muscle differentiation. Four subtypes are recognized: *alveolar* (20%), *embryonal* (50%), *pleomorphic* (20%), and *spindle cell/sclerosing* (10%). Alveolar and embryonal subtypes of rhabdomyosarcoma are the most common soft tissue sarcomas of childhood and adolescence, usually appearing before 20 years of age. Pleomorphic rhabdomyosarcoma occurs in adults, while the spindle cell/sclerosing type affects all ages. Pediatric rhabdomyosarcomas often arise in the sinuses, head and neck, and genitourinary tract; these locations do not normally contain much skeletal muscle, underscoring the notion that sarcomas arise from mesenchymal stem cells rather than differentiated cells.

As befits their distinctive clinical and histologic features, each subtype of rhabdomyosarcoma has a unique molecular pathogenesis.

- *Alveolar rhabdomyosarcoma* frequently contains fusions of the *FOXO1* gene to either *PAX3* or *PAX7* due to t(2;13) or t(1;13) translocations, respectively. *PAX3* is a transcription factor that initiates skeletal muscle differentiation; the chimeric *PAX3::FOXO1* fusion protein interferes with differentiation, a mechanism similar to many transcription factor fusion proteins found in acute leukemia.
- *Embryonal rhabdomyosarcoma* frequently harbors somatic driver mutations that lead to activation of the RAS pathway.
- A subset of *spindle cell sclerosing rhabdomyosarcoma* has mutations that involve the *MYOD1* gene, a major regulator of skeletal muscle differentiation.
- *Pleomorphic rhabdomyosarcoma* is genetically heterogeneous and associated with a constellation of genetic aberrations that are distinct from the other three types.

MORPHOLOGY

Embryonal rhabdomyosarcoma takes the appearance of a soft, gray, infiltrative mass. The tumor cells recapitulate skeletal muscle at various stages of differentiation and include sheets of primitive round and spindled cells in myxoid stroma (Fig. 26.45A). Rhabdomyoblasts with visible cross-striations may be present. **Sarcoma botryoides** (Chapter 22) is a variant of embryonal rhabdomyosarcoma that develops in the walls of hollow viscera, including the urinary bladder and vagina.

In **alveolar rhabdomyosarcoma** fibrous septa divide the cells into clusters or aggregates, resembling pulmonary alveoli. Cells in the center of the aggregates are only minimally cohesive, while those at the periphery adhere to the septa. Tumor cells are uniform and round with little cytoplasm (Fig. 26.45B).

Pleomorphic rhabdomyosarcoma is composed of large, sometimes multinucleated, bizarre eosinophilic tumor cells that may overlap histologically with other pleomorphic sarcomas. Immunohistochemistry for skeletal muscle-specific proteins such as myogenin is usually necessary to confirm rhabdomyoblastic differentiation.

Spindle cell/sclerosing rhabdomyosarcoma consist of fusiform cells with vesicular chromatin arranged in long fascicles or in a storiform pattern. Rhabdomyoblasts are occasionally present. Dense, collagenous, sclerotic stroma may be more common in adults.

Clinical Features

Rhabdomyosarcomas are aggressive and usually treated by surgical resection and chemotherapy, with or without radiation therapy. Histologic type and location are correlated with survival. The botryoid

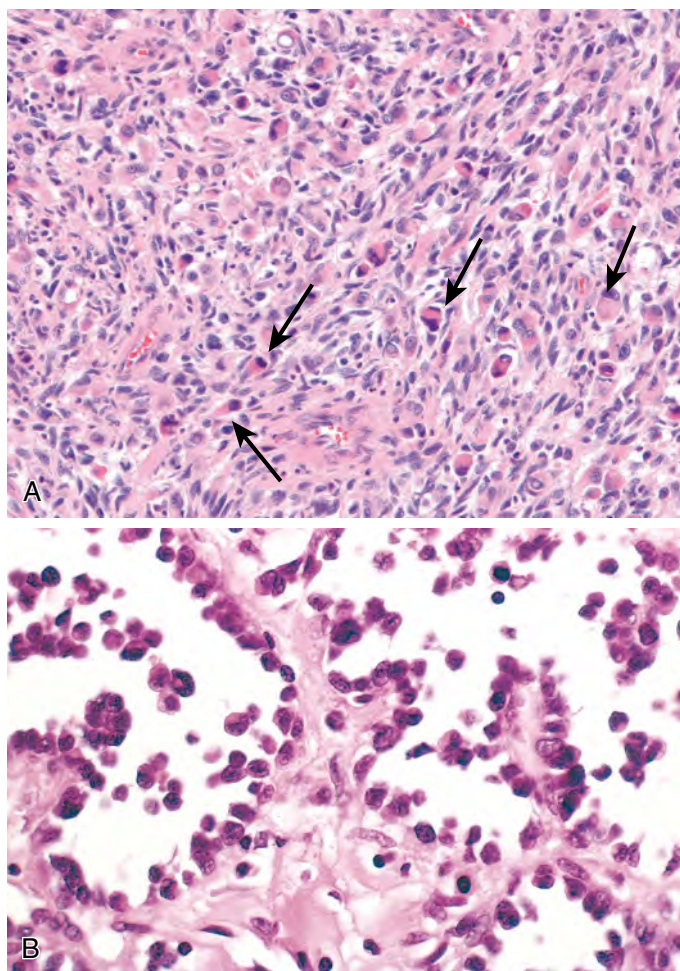


Fig. 26.45 Rhabdomyosarcoma. (A) Embryonal subtype composed of malignant cells ranging from primitive and round to densely eosinophilic with skeletal muscle differentiation (arrows). (B) Alveolar rhabdomyosarcoma with numerous spaces lined by dyscohesive, uniform round tumor cells.

variant has the most favorable prognosis while pleomorphic rhabdomyosarcoma is often fatal.

SMOOTH MUSCLE TUMORS

Leiomyoma

Leiomyoma, a benign smooth muscle tumor, most often arises in the uterus but can occur in any soft tissue site. Uterine leiomyoma (fibroid, Chapter 22) may cause a variety of symptoms including infertility and menorrhagia. Leiomyoma may also arise from cutaneous erector pili muscles (*pilar leiomyoma*) or from deep soft tissues, large vessels, or the gastrointestinal tract. Germline loss-of-function mutations in the fumarate hydratase gene (*FH*) lead to an autosomal dominant syndrome of multiple cutaneous leiomyomas, uterine leiomyomas, and renal cell carcinoma. Fumarate hydratase is a Krebs cycle enzyme, another intriguing example of the link between metabolic abnormalities and neoplasia.

Soft tissue leiomyomas are usually 1 to 2 cm in size and are composed of fascicles of densely eosinophilic spindle cells that intersect at right angles. Tumor cells have blunt-ended, elongated nuclei with minimal atypia and few mitotic figures. Solitary lesions are easily cured by excision.

Leiomyosarcoma

Leiomyosarcoma is a malignant tumor of smooth muscle. It most often develops in the deep soft tissues of the extremities and the retroperitoneum of older adults and is more frequent in females than males. Leiomyosarcomas have underlying defects in genomic stability leading to complex karyotypes.

MORPHOLOGY

Deep soft tissue leiomyosarcomas are firm, rubbery, or fleshy tumors. Histologically, they consist of eosinophilic spindle cells with blunt-ended, hyperchromatic nuclei arranged in interweaving fascicles. Brisk mitotic activity and necrosis are common. Immunohistochemical detection of smooth muscle proteins such as smooth muscle actin, desmin, and caldesmon can aid in diagnosis.

Clinical Features

Leiomyosarcomas present as painless firm masses. Retroperitoneal tumors may be large and bulky and cause abdominal symptoms. Treatment depends on tumor size, location, and grade. Superficial or cutaneous leiomyosarcomas are usually small and have a good prognosis, while those involving retroperitoneum are large, frequently unresectable, and often fatal as a result of local extension and metastatic spread, especially to the lungs. A particularly deadly form arises from the great vessels, often the inferior vena cava.

TUMORS OF UNCERTAIN ORIGIN

Although many soft tissue tumors can be assigned to recognizable histologic types, a large proportion do not recapitulate any known mesenchymal lineage. This group includes tumors with simple or complex karyotypes; an example of each is described here.

Synovial Sarcoma

Synovial sarcoma was so named because in the first described cases, the tumor involved soft tissues near the knee joint leading to speculation that it arose from synovium. However, synovial sarcoma shows variable epithelial differentiation and can present in locations that lack synovium. Synovial sarcomas account for approximately 10% of all soft tissue sarcomas and typically occur in adolescents and young adults. Individuals often present with a deep soft tissue mass that has been present for several years. Most synovial sarcomas demonstrate a unique chromosomal translocation $t(X;18)(p11;q11)$ composed of the *SS18* gene fused to one of three *SSX* genes. The fusions encode chimeric proteins that interfere with normal chromatin remodeling and dysregulate gene expression.

MORPHOLOGY

Synovial sarcomas can be monophasic or biphasic. Monophasic synovial sarcoma consists of uniform spindle cells with scant cytoplasm and dense chromatin arranged in short, tightly packed fascicles. The biphasic type demonstrates a second component of glandlike structures composed of cuboidal to columnar epithelioid cells (Fig. 26.46). Immunohistochemistry is helpful in the diagnosis of these tumors because tumor cells, especially in the biphasic type, are positive for epithelial antigens (e.g., keratins), thereby differentiating them from many other sarcomas.

Clinical Features

Synovial sarcomas are treated with limb-sparing surgery and chemotherapy. The 5-year disease-specific survival varies from 62% to 83% and is related to tumor stage and patient age. Common sites of metastases are

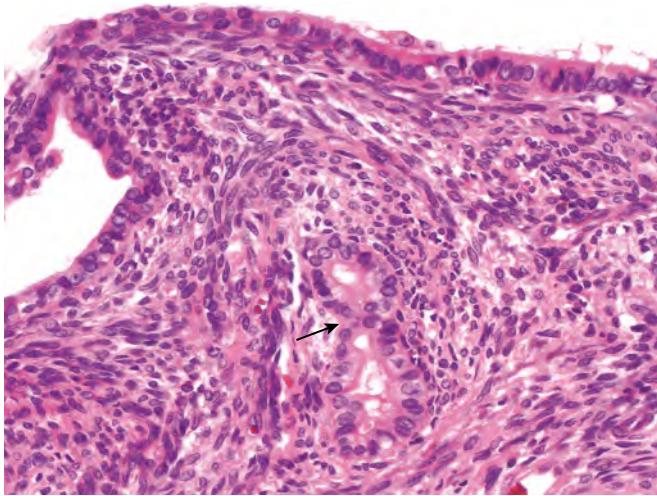


Fig. 26.46 Synovial sarcoma with classic biphasic histologic appearance consisting of spindle cells and gland-like components (arrow).

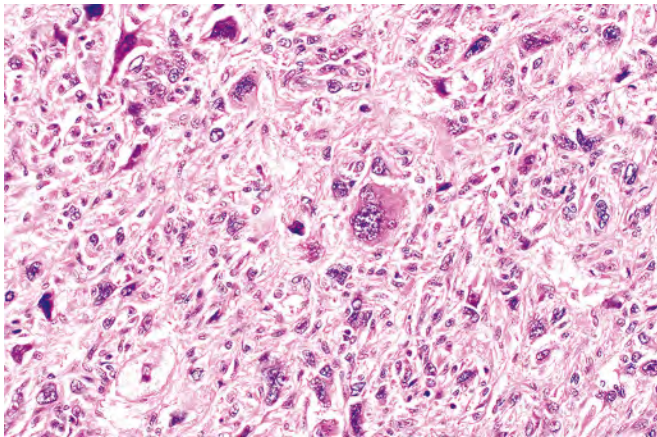


Fig. 26.47 Undifferentiated pleomorphic soft tissue sarcoma revealing anaplastic spindled to polygonal cells. (A, From Goldblum JR, Folpe AL, Weiss SW. *Enzinger & Weiss's Soft Tissue Tumors*. 7th ed. Elsevier; 2020.)

the lungs and regional lymph nodes, the latter an unusual location for the spread of sarcoma.

Undifferentiated Soft Tissue Sarcoma

Undifferentiated soft tissue sarcoma includes malignant mesenchymal tumors that cannot be classified into another category by morphology, immunophenotype, or genetics. The major subtype, *undifferentiated pleomorphic sarcoma*, arises in the deep soft tissues of the extremities, especially the thigh of middle-aged or older adults. It is an anaplastic tumor that typically has a complex karyotype due to genomic instability.

MORPHOLOGY

Undifferentiated pleomorphic sarcomas are usually gray-white fleshy masses that may be quite large (10 to 20 cm in size). Necrosis and hemorrhage are common. Microscopically, they are most commonly composed of sheets of large, anaplastic polygonal cells with hyperchromatic, irregular, sometimes bizarre nuclei (Fig. 26.47). Other undifferentiated soft tissue sarcomas may contain predominantly spindled or round cells. Mitotic figures, including atypical nonsymmetric forms, are abundant. By definition tumor cells lack both differentiation along recognized lineages or diagnostic genetic abnormalities.

Clinical Features

Undifferentiated pleomorphic sarcomas are aggressive malignancies that are treated with surgery, adjuvant chemotherapy, and/or radiation. The prognosis is generally poor. Metastases occur in 17% to 50% of cases.

KEY CONCEPTS

Soft Tissue Tumors

- Soft tissue sarcomas are malignant mesenchymal lesions that are distinct from tumors of epithelial, central nervous system, hematopoietic, or lymphoid tissues.
- Soft tissue tumors likely arise from pluripotent mesenchymal stem cells rather than mature cells.
- Tumors that recapitulate a mature mesenchymal tissue (e.g., smooth muscle) can be subdivided into benign and malignant forms.
- Some tumors are composed of cells for which there is no normal counterpart (e.g., synovial sarcoma and undifferentiated pleomorphic sarcoma).
- Sarcomas with simple karyotypes demonstrate reproducible chromosomal and molecular abnormalities that contribute to pathogenesis and have diagnostic utility.
- Most adult sarcomas are genetically heterogeneous with complex karyotypes, histologically pleomorphic, and associated with a poor prognosis.

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